

A DISTRIBUTED DOUBLET-BASED METHOD FOR UNSTEADY AERODYNAMIC ANALYSIS WITH RELAXED WAKES

by

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Abstract

A novel potential flow-based method for unsteady aerodynamic analysis with relaxed wakes is presented. This method models aerodynamic surfaces and wakes using distributed doublet elements (DDEs) of parabolic strength in both the spanwise and streamwise directions. Due to the nature of the method, the velocities induced by a DDE-based system are bounded everywhere, meaning there are no non-physical or infinite induced velocities in the velocity field, as opposed to lower-order filament-based potential flow methods. In addition to providing a robust velocity field, the use of DDEs provides a degree of stability in predicting unsteady forces that is unachievable with conventional lower-order methods.

The DDE-based method was validated through a series of four studies. The first study predicted the span efficiency of various planar elliptical wings, which agreed with the classical lifting line solution to within 3%. The second study predicted the unsteady lift response to an airfoil passing through a sharp-edged gust, which matched the analytical solution provided by the Küssner function, illustrating the stability of the unsteady force calculations within the DDE-based method. The third study predicted the unsteady time-averaged thrust force produced by a small, rigid rotor at various tip-path plane angles and advance ratios, matching experimentally-obtained data to within 10%. The final study compared the transient sectional thrust of a small, rigid rotor blade operating under unsteady conditions to those obtained via a CFD-based analysis. The comparison agreed well in magnitude and azimuthal location.

Two exploratory studies focusing on small, rigid rotors were performed to demonstrate potential applications of the DDE-based method. The first study quantified the impacts of rolling maneuvers

on the time-averaged thrust and moments of a rigid rotor, with the results indicating that a rolling maneuver can have a significant effect on the thrust and roll moment coefficients. The second study predicted the transient thrust of two rotors operating under multiple tip-path plane angles and advance ratios, showing transient thrust oscillations of up to 60% of the mean thrust output in some cases. The thrust output of individual rotor blades oscillated by up to 90% of the mean total rotor thrust under certain conditions.

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List of Symbols

A_n, B_n, C_n	Strength coefficients of a distributed doublet element
AR	Aspect ratio
C_{D_i}	Induced drag coefficient
C_L	Lift coefficient
C_P	Pressure coefficient
C_T	Thrust coefficient
$\bar{C}_T$	Mean thrust coefficient
$\hat{C}_T$	Normalized thrust coefficient
c	Chord length, m
D	Diameter, m
$[D_i]$	Influence coefficients of a DDE
DDE	Distributed doublet element
e	Span efficiency factor
e_{ext}	Span efficiency factor obtained through Richardson extrapolation
F	Force, N
GCI	Grid convergence index obtained through Richardson extrapolation
J	Propeller advance ratio
M	Number of chordwise elements
N	Number of spanwise elements
n	Rotational speed, rev/s

R	Radius, m
U_∞	Freestream velocity, m/s
$\hat{U}_a(\xi, \eta)$	Approximation of the freestream velocity vector field across a DDE, m/s
V	Velocity, m/s
$\hat{v}_T$	Trailing edge unit vector
$\hat{W}_a(\xi, \eta)$	Approximation of the induced velocity vector field across a DDE, m/s
$\hat{W}_T(\xi)$	Approximation of the wake induced velocity along the trailing edge of a DDE, m/s
x_t/c_r	Ratio of the wing tip leading edge location to the wing root chord length
α	Angle of attack, °
α_{tpp}	Rotor tip-path plane angle, °
Γ	Circulation, m^2/s
γ	Vorticity, $1/s$
Δx_c	Chord length of a DDE element, m
Δx_w	Distance traveled by lifting surface in one timestep, m
$\hat{\lambda}$	Normalized thrust oscillation amplitude
μ	Rotor advance ratio
μ_∞	Normalized freestream velocity
$\mu(\xi, \eta)$	Doublet strength
Ω	Rotational speed, rad/s

Chapter 1

Introduction

Computational methods for aerodynamic analysis are vital for investigating and quantifying the performance of aerodynamic bodies. They provide a valuable supplement to, or substitute for, experimental testing which can be prohibitively expensive. There are various computational methods available, each with their own strengths and weaknesses due to their governing methods and the assumptions applied to the analysis. As a general rule, the higher the degree of exactness with which a flow is represented, the higher the computational expense. Due to this, balancing the capability and expense of a computational method is an important consideration during the selection process.

Most computational methods for aerodynamic analysis approximate solutions to the Navier-Stokes equations in order to predict flow field properties. The implementation and the fidelity of this approximation is one way with which methods can be categorized. Computational fluid dynamics (CFD)-based methods typically discretize the flow field into cells of finite volume over which the Navier-Stokes equations can be numerically solved. The approximations and assumptions made to the system can vary widely, with one example being the inclusion or exclusion of viscous effects. The breadth to the applicability and capability of computational fluid dynamics makes it a valuable tool with which aerodynamic systems can be analyzed. However, these methods typically require extensive computational power and time to prepare and solve, often limiting their applications to more in-depth analyses.

An example of the complexity of implementing a high-fidelity CFD-based analysis is in the control volume discretization, or grid generation process. This is a resource-intensive process that requires specialized skills and is critically important to the solution stability and accuracy (1). The grid charac-

teristics and sizing needed for an accurate analysis are dependent on the geometry, analysis parameters, and the assumptions made regarding the governing equations. These factors drive requirements such as the grid resolution necessary near the surface, which determines the effectiveness of the boundary layer modelling. As a result of this grid dependence, a solution’s sensitivity to the grid sizing must typically be quantified through grid independence studies (2).

In addition to the complexity of the grid generation, the solution time for a three-dimensional CFD-based analysis is often measured in days or weeks, and may require the use of supercomputing clusters. In order to reduce the computational expense of a CFD-based approach, simplifications can be made to the governing Navier-Stokes equations. The assumption of adiabatic and inviscid flow, for example, reduces the Navier-Stokes equations to the Euler equations (3), reducing the required grid complexity and computational expense of the solution. Euler methods are typically applied to analyses where boundary layer effects are considered insignificant, such as supersonic flows (1).

Potential flow-based methods are derived from even further simplifications to the Navier-Stokes equations, making assumptions that the flow field is irrotational, inviscid, and typically incompressible. Aerodynamic systems are represented using singularity elements placed along the surfaces, with the superpositioning of their effects forming the flow field. Due to the simplified governing equations and the surface-based singularity representation rather than the volume-based approach of higher-fidelity methods, potential flow-based analyses require substantially less time and computational power to run when compared to CFD-based methods (4). A trade-off of this decreased computational expense is that potential flow-based methods are often unsuitable for analyses with significant viscous contributions (1). Though the simplifications made to their governing equations limit the applicability of potential flow-based methods, they remain the basis of many widely-used computationally-inexpensive prediction methods for aircraft and aerodynamic surfaces.

1.1 Motivation

Potential flow-based methods provide aircraft designers with a relatively fast way of quantifying and qualifying a design’s performance early on in the design process (4). The runtime of these methods is typically counted in minutes, and the low resource requirements lend it to parallelization, meaning multiple analyses can be run simultaneously. In addition to this, the integrated nature of the aircraft

design process has driven interest in optimization-based approaches. Capable and fast aerodynamic performance prediction methods can be teamed with structural models and multi-objective optimization algorithms to create efficient designs based on specific criteria, with examples given by Refs. (5; 6; 7; 8). However, the applicability of these optimization-based approaches are constrained by the suitability of the aerodynamic performance prediction method used. Optimizations typically require hundreds of thousands of design evaluations, meaning that using high-fidelity CFD-based methods within an optimization-based approach is often not feasible due to the time and computational power required. For these reasons, potential flow-based methods still play an important role in aircraft conceptual design and analysis.

Recent decades have seen substantial growth in the adoption of unconventional aircraft configurations such as multirotor and vertical takeoff and landing (VTOL) aircraft, due mainly to the emergence of both the drone and urban air mobility industries. The electrification of aircraft propulsion has also spurred interest in distributed propulsion for fixed-wing aircraft, as is the case with the X-57 Maxwell pictured in Fig. 1.1 and discussed in detail by Borer et al. (9). Due largely to the X-57 Maxwell project, the AIAA introduced the Workshop for Integrated Propeller Prediction (WIPP) in 2019, which is dedicated to analyzing integrated propeller-wing systems using CFD-based methods (10). The unconventional nature of these modern aircraft configurations poses interesting design and analysis challenges for aerodynamicists, as they incorporate a multitude of rotors and/or propellers in close proximity to one another, and to other aerodynamic surfaces such as wings.



Figure 1.1: The X-57 Maxwell (11)

The complexity of analyzing these aircraft configurations exists even at the component level. Many remotely-piloted aircraft systems and urban air mobility vehicles use rigid rotors, which must operate under a wide-range of inflow conditions. During the ascending and descending stages of flight, these rotors have largely axial flow, meaning that the majority of the inflow into the rotors is in-line with their

axes of rotation. Under these conditions, the rotors are acting in a predominantly steady-state manner, similar to a conventional forward-facing propeller. While the vehicle is in forward flight, however, the rotor planes are typically acute to the incoming flow. Due to this, advancing and retreating blade effects act on the individual rotor blades, and the overall forces and moments generated by the rotors are transient. The transient nature of these rotor loads necessitates an unsteady analysis framework, whether analyzing individual isolated rotors or the aircraft configuration as a whole.

Existing unsteady potential flow-based methods are limited in their applicability when analyzing these unconventional systems. Strong interactions between aerodynamic surfaces and shed wakes, as is the case with rotors in forward flight or integrated propeller-wing systems, are a source of instability for these methods. These instabilities are discussed in further detail in Chapter 2. As a result of these limitations, systems of this nature are typically analyzed using CFD-based methods. In a notable recent study, Misiorowski et al. (12) quantified the transient thrust oscillations of a single isolated rigid rotor at one operating condition, using an unsteady Navier-Stokes solver. This was done in the course of quantifying rotor interactions on quadrotors. In a similar vein, McQuaid et al. (13) analyzed both an isolated rotor and a quadcopter as a means of investigating the onset of vortex ring state. Both of these studies required the use of supercomputing clusters. Cerny and Breitsamter (14) used an unsteady Navier-Stokes solver to analyze a propeller operating at seven different inflow angles, comparing the results to experimentally-obtained data. Outside of these three studies, the unsteady aerodynamic analysis of small rotors is notably lacking from the literature.

Regarding the analysis of integrated propeller-wing systems, the AIAA Workshop for Integrated Propeller Prediction has spurred interest in the research of propeller-wing interactions using CFD-based methods, with examples given by Refs. (15; 16; 17; 18). This research area is still emerging, with the WIPP representatives noting that the “effort represents a critical first step in the ability to accurately predict propeller effects with confidence as it culminates in an open powered test database that can be used for CFD validation” (19). While the application of CFD-based methods to these unconventional and unsteady systems is maturing, potential flow-based methods could potentially provide insights into the effects of airfoil camber, thickness, or wing planform shape (4), which can help to identify broader trends in the design space. Without computationally-inexpensive and stable unsteady performance prediction methods, the design and analysis of unconventional aircraft configurations is hampered.

1.2 Research Objectives

The purpose of this research is to introduce a new potential flow-based method for the transient analysis of complex and unsteady aerodynamic systems with relaxed wakes. As such, this method aims to support the analysis and design of unconventional and unsteady systems, including rotors with advancing and retreating blade effects, multirotor vehicles, and integrated propeller-wing systems. The method introduced herein is straightforward to implement when compared to the geometry and grid definition processes required of CFD-based analysis. In addition to this, the computationally-inexpensive nature of the method enables it to be used within optimization-based processes.

Chapter 2

Potential Flow

Potential flow theory forms the basis of many computationally-inexpensive aerodynamic performance prediction methods. Potential flow theory predated digital computers and was founded on the assumption of an irrotational velocity field as a necessary simplification to the Navier-Stokes equations at the time (20). Coupled with the assumption that the flow is inviscid and incompressible, potential flow methods were initially intended to be computationally-inexpensive enough for human computers to use. As digital computers have grown more powerful, the scope and capabilities of potential flow-based methods have increased while their runtimes have decreased (21). The capabilities and the computationally-inexpensive nature of their design have kept them an appropriate choice for many aerodynamic performance prediction applications. The increasing use of optimization-based design approaches has shown that potential flow-based methods are still indispensable.

2.1 Characteristics of Potential Flow-Based Methods

There can be significant differences in the capability and computational expense of two different approaches among potential flow-based methods, depending on how the systems are modelled and which assumptions are made. Simpler and faster methods typically use lumped vorticity in the form of vortex filaments (22), while more comprehensive methods distribute vorticity over a surface (23). Some approaches model both the upper and lower surfaces of the wing or rotor to account for thickness (24), while others only model the camber line or zero-lift line (25). Wake representations may also vary be-

tween methods, with some using fixed or prescribed wake geometries and others using a relaxed wake model, which evolves the wake shape over time. Lastly, how aerodynamic loads are calculated is an important distinction, with some methods relying on pressure integration and others relying directly on the singularity strength.

2.1.1 Singularity Elements

Potential flow-based methods use singularity elements to model the flow field. These elements are placed along the aerodynamic surfaces and other boundaries, such as the wake shear layer. The superpositioning of their effects forms a vector field approximation of the flow, which can then be used to approximate aerodynamic loads. Depending on the element type, the strength of a singularity element can be lumped into a point or line, or distributed over a surface. The various types of singularity elements have their own strengths and weaknesses, which can define the applicability of the overall method. The following section will give an overview of several commonly-used singularity elements along with their advantages and disadvantages, which provides important context for this research.

Vortex Filaments

Vortex filaments form the basis of many commonly-used potential flow-based methods. They are often represented as straight line segments with constant circulation strengths, which make them computationally-inexpensive and simple to implement. An example is shown in Fig. 2.1, which illustrates a filament of strength Γ spanning from points P_1 to P_2 , inducing a tangential velocity q_i at point P .

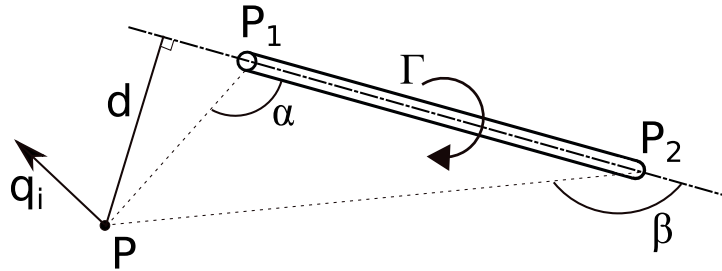


Figure 2.1: A vortex filament inducing a velocity at a point. Recreated from (20).

The tangentially induced velocity q_i is given by:

$$q_i = \frac{\Gamma}{4\pi d}(\cos \alpha - \cos \beta) \quad (2.1)$$

where d represents the shortest distance from point P to the axis of the vortex filament. One concern with the use of vortex filaments is that the tangential velocity induced by the filament becomes infinite as the distance to the vortex filament, d , approaches zero and is singular at zero. In a numerical environment, the singularity at the center of a vortex filament is often managed through the use of a solid-core filament model (26)(27), which prescribes a finite radius to the vortex filament rather than modelling it as infinitely thin. Within the solid core, the induced velocity is corrected such that it varies linearly with radial location, approaching zero at the centre of the core (26).

Vortex filaments are often used to construct vortex rings, which are planar quadrilateral elements with vortex filaments of equal circulation strength around the perimeter. An illustration of a vortex ring is given in Fig. 2.2. These elements are used to model aerodynamic surfaces and wakes, often oriented such that two sides are aligned with the streamwise direction, modelling the shed circulation between neighbouring elements in what is known as a vortex lattice method. Examples of vortex ring systems are illustrated in Figs. 2.3 and 2.4.

As the filaments around the perimeter of a vortex ring are all of equal strength and form a closed loop, individual vortex rings follow Helmholtz's first and second laws, which state that the strength of a filament must be constant along its length and a vortex filament cannot start or end in a fluid (27). Thus, through constructing a system using vortex rings, it will inherently follow Helmholtz's laws and remain irrotational, which is necessary for potential flow analysis.

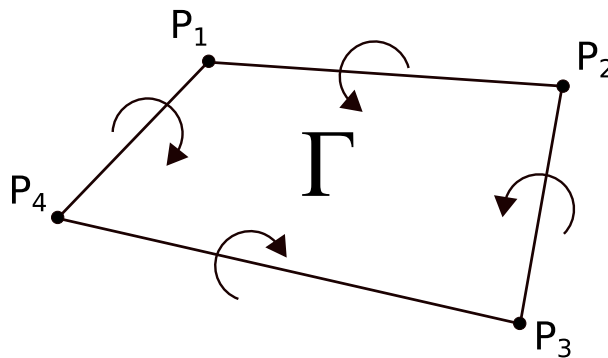


Figure 2.2: A diagram of a generic vortex ring element.

Vortex Sheets and Distributed Normal-Doublets

The lumped vorticity contained within a vortex filament, represented by the circulation strength Γ , can be distributed linearly across an infinitely-thin planar surface as vorticity, $\gamma(\xi, \eta)$, forming what is known as a vortex sheet.¹ The distributed nature of a vortex sheet results in a velocity field that is bounded everywhere, meaning that vortex sheets do not induce infinite velocities as long as the sheet is continuous (28). However, the induced velocities can become singular at the perimeter of a sheet with non-zero strength, in the same way that the velocity induced by a vortex filament at its core is singular (28). Additionally, the tangentially-induced velocity of a vortex sheet is discontinuous across the plane of the sheet (27)(28).

The use of vortex sheets rather than vortex filaments removes the potential for infinite induced velocities, which is desirable for applications where a solid-core correction factor for the vortex filaments is inadequate. However, implementing a completely vortex sheet-based system in three-dimensions is complicated by the fact that the vorticity distribution must be divergence-free, as it is defined as the divergence of the curl of the velocity field, which is by definition zero for potential flow fields (27). Prescribing a divergence-free vorticity distribution directly to a vortex sheet is nontrivial, which is why purely vortex sheet-based solutions are typically not used in three-dimensional analyses (28).

Drela (28) outlines one method by which to achieve a divergence-free vorticity distribution, through first prescribing a higher-order scalar function $\mu(\xi, \eta)$ which acts as a stream function for the vorticity. To achieve a linearly-varying vorticity distribution, then, the scalar function $\mu(\xi, \eta)$ must be second-order. The surface gradient of this scalar function $\mu(\xi, \eta)$, rotated 90° about the surface unit normal, defines the vorticity:

$$\gamma(\xi, \eta) = \hat{n} \times \tilde{\nabla} \mu(\xi, \eta) \quad (2.2)$$

By defining the vorticity from the scalar function $\mu(\xi, \eta)$, the resultant vorticity distribution is divergence-free. However, Drela (28) also notes that the scalar function $\mu(\xi, \eta)$ can be interpreted as the strength of a distributed normal-doublet sheet, referred to hereafter as simply a distributed doublet sheet. That is to say, a vortex sheet with a divergence-free vorticity distribution $\gamma(\xi, \eta)$ is functionally equivalent to a distributed doublet sheet of strength $\mu(\xi, \eta)$. Due to this equivalence, and due to the complications

¹When discussing vortex sheets and similar planar singularity elements, a local (ξ, η) reference frame is used as shown in Figs. 2.6 and 3.2.

that arise from prescribing a divergence-free vorticity distribution directly, three-dimensional analyses using higher-order distributed elements often use a framework that relies on the scalar function $\mu(\xi, \eta)$, meaning that the elements are treated as distributed doublet sheets.

Equivalence of Circulation and Doublet Strength Distributions

An important takeaway from the relationship between vortex sheets and distributed doublet sheets is the conditional equivalence of the doublet strength distribution, $\mu(\xi, \eta)$, to the circulation distribution, $\Gamma(\xi, \eta)$. A distributed doublet sheet can be viewed as an infinite series of infinitesimally small constant-strength doublet sheets of varying strength represented by the distribution $\mu(\xi, \eta)$. As stated previously, constant-strength doublet sheets and constant-strength vortex rings are functionally identical, meaning that the terms $\mu(\xi, \eta)$ and $\Gamma(\xi, \eta)$ may be used interchangeably under certain conditions.

If the higher-order doublet strength distribution $\mu(\xi, \eta)$ is forced to be continuous and never permitted to have a step-change to zero, the integration of the resulting vorticity distribution defined by Eq. 2.2 would yield an expression representing both $\mu(\xi, \eta)$ and $\Gamma(\xi, \eta)$. The relationship between the doublet strength and vorticity distributions under these conditions is equivalent to the relationship between the circulation and vorticity distributions. That is to say, it is conditionally acceptable for the definition of the vorticity introduced in Eq. 2.2 to be based on either the doublet strength or circulation distribution:

$$\gamma(\xi, \eta) = \hat{n} \times \tilde{\nabla} \mu(\xi, \eta) = \hat{n} \times \tilde{\nabla} \Gamma(\xi, \eta) \quad (2.3)$$

As mentioned with the introduction of Eq. 2.2, the relationship between the doublet strength and the vorticity involves a 90° rotation about the surface-normal direction. The explanation for this is related to the equivalence of the distributed doublet and vortex sheets. Figure 2.3 illustrates a system of four vortex rings, each of constant and unique strength Γ_N . Imagine these vortex rings representing an infinitesimally small area of a larger, higher-order distributed doublet sheet. According to Helmholtz's laws, circulation cannot be created or destroyed within a system, meaning that changes in circulation in one direction must be taken up in the other direction. As an example, the changes in circulation between Γ_1 and Γ_2 from filaments A-B and B-C in Fig. 2.3 must be accounted for through the filament B-E, which is the case as the strength of filament B-E is defined as the difference between Γ_1 and Γ_2 . Likewise, changes in the strength of filaments C-F and F-I must be accounted for in E-F, and so forth.

At any point where the circulation changes magnitude, this change must be accounted for in order to obey Helmholtz's second law. When dealing with distributed elements such as vortex or doublet sheets, this is often referred to as the condition of irrotationality (29).

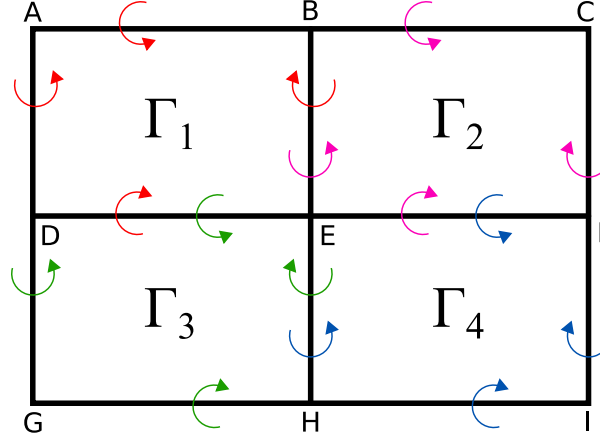


Figure 2.3: A generic constant-strength vortex ring system.

The circulation distribution $\Gamma(\xi, \eta)$ over a distributed doublet sheet, under the condition that the circulation and doublet strengths are equivalent, represents the strengths of these infinitesimally small vortex rings. The rate of change of $\Gamma(\xi, \eta)$, then, is actually representative of the shed circulation acting in the direction perpendicular to the rate of change. This is the reason for the 90° rotation between the circulation distribution $\Gamma(\xi, \eta)$ and the vorticity distribution $\gamma(\xi, \eta)$ of the equivalent vortex sheet. Through defining the vorticity distribution from the doublet strength $\mu(\xi, \eta)$, and conditionally through the circulation distribution $\Gamma(\xi, \eta)$, the resulting vorticity distribution is divergence-free and satisfies the condition of irrotationality.

2.1.2 Representations of Aerodynamic Surfaces

There are two ways in which singularity elements are commonly arranged to represent aerodynamic surfaces. The first is known as a panel method, which models the aerodynamic surfaces with thickness. The singularity elements are placed along both the upper and lower portions of the aerodynamic surfaces, forming a closed-body (27). Panel methods aim to discretize the geometry with high fidelity. The second commonly-used method is known as a lifting surface method, which models the aerodynamic surfaces as infinitely thin. Singularity elements are typically placed along the zero-lift line or camber line of the

airfoil (1)(27), as shown in Fig. 2.4 which illustrates a wing and wake modelled with vortex rings using a lifting surface method.

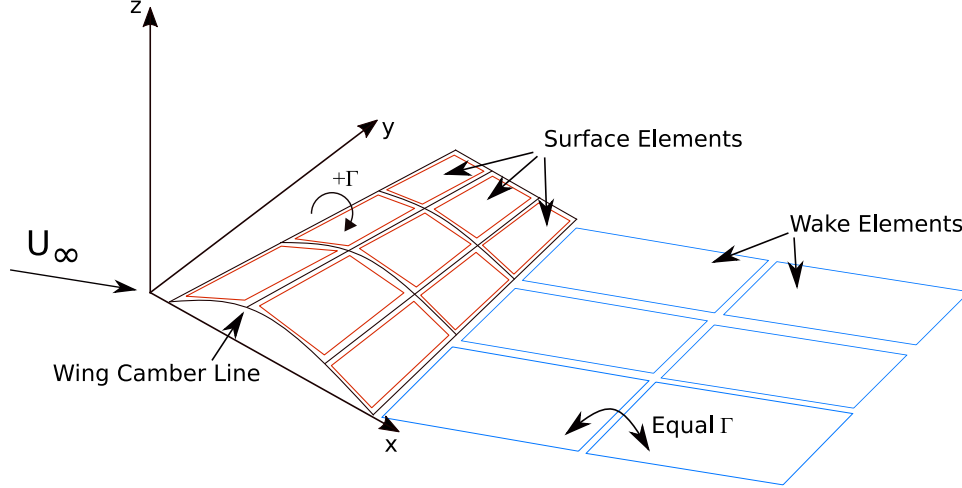


Figure 2.4: A representation of a wing and wake with vortex rings. Recreated from (27).

Lifting surface methods are typically less computationally-expensive than panel methods, as they require fewer surface elements (1). In addition to this, using a zero-thickness model avoids any issues that could arise from wake piercing, where wake elements enter into the interior of a closed body, as can be observed with panel codes and strong aerodynamic interactions (30). Using a relaxed wake model within a panel method can be problematic, as the aerodynamic surfaces' internal regions have a constant velocity potential and, therefore, stagnant flow. As a result, any relaxed wake elements that pierce the body will indefinitely remain within the interior region, leading to a non-physical flow field. As the purpose of this research is to develop a method for the analysis of complex systems with strong surface-wake interactions, this research will focus primarily on lifting surface based representations.

2.1.3 Representations of Wakes

The manner in which the strength and shape of the wake are prescribed and how these characteristics change with time are important areas of distinction. As a solution progresses, wake elements are shed from the trailing edges of the surfaces. The strengths assigned to newly-created wake elements are equal to the strength of the trailing edge surface elements from which they are shed (27). For a conventional steady-state analysis in which the flow conditions are unchanging, the strength of the surface elements

do not change with time once the singularity element strengths have converged. As a result of this, the strengths of the corresponding wake elements are unchanging in the streamwise direction. As a simplification for steady-state analysis, the trailing edge surface element strength will typically be propagated down the entire wake every timestep, resulting in a wake of constant strength in the streamwise direction (27)(31). This is illustrated in Fig. 2.4, where the circulation strength of the wake elements, Γ_W , are set to be equal in the streamwise direction. This type of steady-state analysis and simplified wake-strength representation are typically applied to time-independent systems such as conventional wings, or rotors in purely axial flow.

For time-dependent solutions, the strengths of the surface elements change as the flow conditions change. To properly capture unsteady wake effects, the strengths of the individual wake elements must remain constant once they are shed and the solution progresses. This results in a wake that has streamwise changes in strength, representative of the changes in strength of the surface elements as they contend with changing flow conditions. The upper portion of Fig. 2.5 illustrates an unsteady two-dimensional airfoil and wake, with the streamwise wake vorticity given as $\gamma_w(t)$. Represented by discrete vortices in this example, a discretized wake is illustrated in the lower portion of Fig. 2.5, with vortices of varying strength Γ_W spaced at intervals equivalent to the distance travelled by the airfoil over a timestep. The strengths assigned to these vortices are based on the integrated wake vorticity across that timestep, resulting from the flow conditions acting on the airfoil during that timestep.

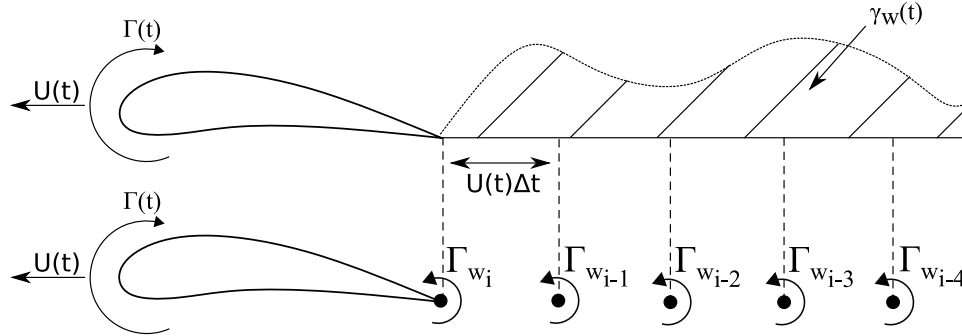


Figure 2.5: An example of the discretization of an unsteady two-dimensional wake. Recreated from (27).

Modelling streamwise changes in wake strength is a necessary component of an unsteady analysis, as the strength of the wake is a crucial factor in quantifying the influence that it holds over the surface elements. The wake strength also impacts its shape when using a relaxed wake model, which moves

the wake elements with the velocity field. For steady-state analyses, the relaxation procedure often involves modifying an existing prescribed wake geometry iteratively until the wake shape converges (27). Alternatively, this relaxation procedure can be incorporated into the timestepping loop so that the wake elements are moved with their local velocities as the solution progresses, which is a necessary approach for time-dependent unsteady analyses (27).

2.1.4 Force Calculations

There are several ways in which the steady and unsteady aerodynamic forces can be approximated within a potential flow-based system. A common method for all analysis types is pressure integration, which relates the surface velocities acting on the system to the surface pressures and, subsequently, the forces. Also applicable to both steady and unsteady analyses is the Kutta-Joukowski theorem, which relates the singularity strength of the elements and the magnitude and direction of the flow field to the resulting force vector. Unsteady effects can also be approximated using indicial response methods, which take the unsteady response of a two-dimensional airfoil undergoing a change in aerodynamic boundary conditions, and apply it to a three-dimensional section of an aerodynamic surface. A brief description of these three methods follows.

Pressure Integration

The forces acting on an aerodynamic surface can be determined via pressure integration, which is frequently incorporated into panel methods. Pressure integration involves determining the surface velocities acting on the singularity elements, which is then related to the surface pressures and, in turn, the force acting across the element. As outlined by Katz and Plotkin (27), implementing a pressure integration-based approach typically involves relating the velocity at an element control point, Q_k , to the pressure coefficient of the element, C_{P_k} , via the expression:

$$C_{P_k} = 1 - \frac{Q_k^2}{U_\infty^2} \quad (2.4)$$

where U_∞ represents the freestream velocity. This simplified method assumes a common surface velocity and by extension, pressure distribution, across the element when determining C_{P_k} . With the pressure

coefficient of an element having been approximated, it is possible to determine the force acting on the element via:

$$\Delta F_k = -C_{P_k} \left(\frac{1}{2} \rho U_\infty^2 \right) \Delta S_k \mathbf{n}_k \quad (2.5)$$

in which ΔF_K represents the force acting in the direction of the unit normal vector of the element $\mathbf{n}_k$, and ΔS_k being the area of the element.

This pressure integration-based approach is used extensively in panel methods, where the discretized geometry of the system closely resembles the actual aerodynamic surfaces. As noted by Cole (32), the forces approximated through pressure integration rely on the accuracy of the geometric discretization of the system, specifically in the fidelity of the modelling around the leading edge. The forces predicted via Eq. 2.5 are oriented normal to the element surfaces and thus require an adequate number of elements near the leading edge to model the leading edge suction force. As a result, pressure integration is generally not used within a lifting surface-based representation, which discretizes the camber line or zero-lift line.

The Kutta-Joukowski Theorem

For lifting surface-based methods, the Kutta-Joukowski theorem is typically applied to determine the forces acting on an element. The primary formulation of this theorem relates the circulation strength of a vortex filament, Γ , to a resulting freestream lift force vector, F , via the magnitude and direction of the freestream velocity vector, U_∞ (27):

$$F = \rho U_\infty \times \Gamma \quad (2.6)$$

This theorem was generalized by Prandtl (33) as:

$$dF = \rho v \times g \quad (2.7)$$

where dF is the resultant force due to the velocity, v , acting on vortex element, g , in which $g = \Gamma dS$ for a vortex filament or $g = \gamma(\xi, \eta) dA$ for a vortex sheet (33). With this generalized formulation, the Kutta-Joukowski theorem can be applied not only to the freestream force as is the case with Eq. 2.6,

but also the induced forces, in which the induced velocities of the system form the velocity v in Eq. 2.7. This adaptation of the Kutta-Joukowski theorem was introduced by Eppler (34), and used to determine induced drag by Mortara et al. (35), as well as induced lift and drag by Bramesfeld (36), and Cole (32).

The reliance of the Kutta-Joukowski theorem on singularity strength rather than a surface velocity distribution makes it a favourable choice over pressure integration for lifting surface methods. In addition to this, the resultant force from the Kutta-Joukowski approach implicitly accounts for the leading-edge suction force (32), meaning that the discretization of the leading edge curvature is not as influential as with a pressure integration-based approach.

The Unsteady Kutta-Joukowski Formulation

Force calculations within an unsteady framework require the approximation of a transient non-circulatory component, commonly referred to as apparent mass. This term is associated with the instantaneous acceleration of the fluid immediately adjacent to the airfoil (28). The unsteady Kutta-Joukowski formulation for a filament-based method is given by Drela (37) as:

$$\bar{F}'_L = \rho \Gamma \bar{V} \times \hat{s} + \rho \frac{\partial \Gamma}{\partial t} \frac{c}{|\bar{V}_\perp|} \bar{V} \times \hat{s} \quad (2.8)$$

where $\bar{F}'_L$ is the lift per unit span, $\bar{V}$ is the relative velocity on the element surface, $\hat{s}$ is a unit vector of the circulation orientation, and c is the chord length. The value $\bar{V}_\perp$ is defined as the component of the velocity perpendicular to the bound vortex. (37):

$$\bar{V}_\perp = \bar{V} - (\bar{V} \cdot \hat{s}) \hat{s} \quad (2.9)$$

The first term to the right side of the equality in Eq. 2.8 represents the instantaneous force acting on the airfoil, which is equivalent to the expressions outlined in Eqs 2.6 - 2.7. The second term on the right side of the equality in Eq. 2.8 represents the apparent mass, a non-circulatory term that acts as an inertial force opposing the acceleration of the surface due to the surrounding air mass which must be accelerated as well (38). The time derivative of the circulation within the apparent mass term in Eq. 2.8 requires the use of a finite differencing scheme across timesteps. Together with the modelling of the shed circulation in the wake, the addition of the apparent mass approximation constitutes a fully unsteady framework for a potential flow-based method (28).

Numerical Instabilities in Unsteady Force Calculations

The prediction of unsteady forces can be challenging when using conventional vortex filament-based methods. One such example is the ratio of timestep size to distance travelled by the surface elements. Simpson and Palacios (39) used an unsteady vortex lattice method (UVLM) outlined by Katz and Plotkin (27), and found that newly shed wake panels should be of approximately equal area compared to the trailing-edge surface elements from which they inherit their circulation strength. That is to say, the distance travelled by the surface elements each timestep, denoted by Δx_w , should be approximately equal to the chord length of the trailing edge surface element, denoted by Δx_c . The reference to this length-ratio constraint appears to be related to the work of Rusak et al. (40), who note that vortex-lattice methods tend to become unstable when the wake-element length is less than that of the surface element. A likely reason for this instability is the finite-differencing methods typically used within UVLMs to determine the apparent mass term, introduced in Eq. 2.8 (27). However, enforcing a timestep size constraint such as $\Delta x_w/\Delta x_c \approx 1$ while analyzing rotors is problematic since the distance travelled by the surface elements of rotors depends on their distance from the axis of rotation. Forward flight, with edgewise flight components, compounds this problem as the distance travelled by surface elements in these cases also depends on the rotor blades' azimuthal location.

Indicial Response Methods

Some unsteady potential flow methods, such as RCAS (41), use an indicial response method to approximate unsteady forces. These methods model the response of a two-dimensional airfoil to a step-change in aerodynamic boundary conditions, such as a change in angle of attack or a plunging motion, and incorporate the effects of shed circulation and apparent mass (42). This response is superimposed over time to model the effects of a series of sequential changes to the boundary conditions. The resulting two-dimensional unsteady response is then used to approximate the unsteady response of a three-dimensional wing section within a strip method (43). Because indicial response methods incorporate the effects of shed circulation, the wake strength at one spanwise location does not influence the unsteady forces acting on the surface at another spanwise location (43). Both pressure integration and the unsteady Kutta-Joukowski formulation do not have this limitation, as they model the effects of the shed circulation in three-dimensions. Further information on indicial response methods are given by (43; 44; 42).

2.2 Existing Potential Flow Methods for Unsteady Analysis

The following sections give an overview of the most relevant potential flow-based methods for unsteady aerodynamic analysis, which form the basis for many specialized codes used to analyze unsteady wings and rotors. Understanding the advantages and disadvantages of these methods provides important context for the motivation behind this research.

2.2.1 The Unsteady Vortex Lattice Method

One of the more common potential flow methods for unsteady analysis is the unsteady vortex lattice method (UVLM), outlined by Katz and Plotkin (27). This method uses constant-strength vortex rings to model lifting surfaces and wakes, similar to the system illustrated in Fig. 2.4. Individual wake element strengths are held constant over time, modelling streamwise changes in wake strength through a method similar to that shown in Fig. 2.5. Due to the UVLM modelling the system using vortex rings, it is typically the faster option when compared to other unsteady aerodynamic performance prediction methods, including other potential flow-based methods.

One drawback to UVLM-based methods is their reliance on vortex filaments, which necessitate correction factors to reduce the potential for infinite induced velocities. As these correction factors are only utilized when determining the velocities near vortex filaments, they typically do not impact the solution (27). Velocities in potential flow-based methods are commonly calculated at element control points, usually placed at the center of the vortex rings and, therefore, an adequate distance from the vortex filaments at the perimeter of the element. However, issues can arise when the edge of one element is in close proximity to another element's control point, which can happen with strong wake-on-wake or surface-wake interactions, as is the case with complex systems such as rotors in forward flight or propeller-wing systems.

Further information is provided by Refs. (45; 46; 47), which include examples of unsteady applications of the UVLM method. The unsteady vortex lattice method has also been extended and improved upon in various ways, including using vortex particle wake models (48; 49; 50), and modelling leading-edge vortex shedding (51).

2.2.2 The Distributed Vorticity Element Method

Based on the multiple-lifting line method developed by Horstmann (52), the distributed vorticity element (DVE) method introduced by Bramesfeld (36) and Bramesfeld and Maughmer (25) is a variation on the conventional vortex-lattice method. In essence, the DVE method uses vortex sheets as a replacement for the discrete trailing vortices present in traditional vortex lattice methods. A typical element from the DVE method is shown in Fig. 2.6. These elements consist of a leading and trailing edge vortex filament of equal and opposite quadratic circulation strength along the element span, with a vortex sheet in between that varies linearly in the spanwise direction, modelling the shed vorticity from the quadratic filaments. Conceptually, a single DVE can be thought of as a vortex ring-based system with infinite spanwise elements. Using these vortex sheets reduces the number of filaments in the system and reduces the potential for infinite or singular induced velocities. In addition to this, the second-order spanwise circulation distribution of a DVE requires fewer elements to model spanwise changes in circulation when compared to constant-strength vortex rings.

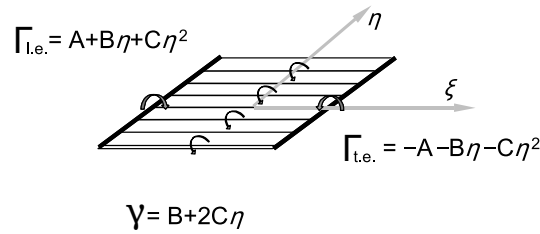


Figure 2.6: A distributed vorticity element (36).

For steady-state analyses where streamwise changes in the wake strength are irrelevant, the leading and trailing edge vortex filaments are omitted from the wake since they are of equal magnitudes yet opposite orientations, and thus their influences cancel. The wake of a steady aerodynamic system is represented entirely with vortex sheets, which have constant strength in the streamwise direction and change in a first-order continuous manner in the spanwise direction. With the wake free of vortex filaments, strong wake-on-wake interactions can occur with the wake elements well-behaved. An example of a steady-state relaxed wake analysis is illustrated in Fig. 2.7, which models a wing half-span using triangular surface elements and quadrilateral wake elements. As the wake rolls up near the wingtip, the elements intersect one another without singular or large induced velocities. The wake of a steady-state

analysis using the DVE method can be modelled for long distances downstream due to the numerical robustness that the vortex sheets provide within the relaxation procedure (25).

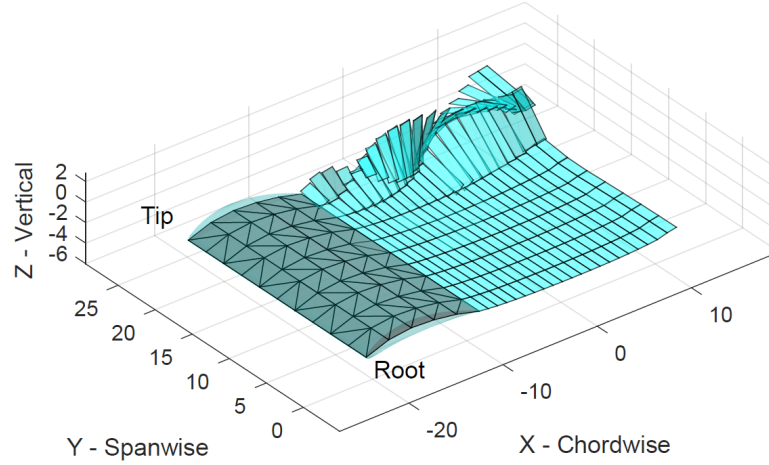


Figure 2.7: A steady-state analysis of a wing half-span with a relaxed wake, modelled using DVEs (53).

As surface elements still have bound vortices at their leading and trailing edges, steady-state analyses with DVE-based methods are subject to certain instabilities. For example, a relaxed wake analysis can become unstable if a wake element passes near a surface element, as it can encounter a large induced velocity and deform unpredictably. Independent of the wake model, if two surface elements are arranged closely together, the induced velocities at these control points can be influenced heavily by the vortex filaments. Due to these bound vortices, steady-state analyses with DVE-based methods are not entirely free of the instabilities present in VLM-based methods.

Further issues arise with DVE-based methods during unsteady analysis, where the bound vortices are needed in the wake to model streamwise changes in circulation. Unsteady analysis with the DVE method requires a fixed wake model, as relaxed wake interactions are erratic when the bound vortices are included in the wake. Even with a fixed wake model, unsteady systems with strong surface-wake interactions can result in issues as the vortex filaments in the wake may pass near the surface elements' control points and cause large induced velocities that impact the solution. As a result, unsteady aerodynamic analyses with DVE-based methods face similar issues to UVLM-based methods.

The DVE method has been applied to both fixed-wing and rotorcraft analysis. Barcelos et al. (54) used the DVE method with a relaxed wake model to analyze quadrotor flight configurations as a way

of quantifying the impact of rotor-on-rotor interactions. This analysis was done under a steady-state framework, as the relaxed wake model is unstable when modelling streamwise changes in circulation, as previously discussed. Cole (32) used the DVE method to analyze propeller-wing interactions, using a steady-state relaxed wake for the propeller and a fully-unsteady fixed wake for the wing. This choice of wake models was a necessary compromise, as a fully-unsteady propeller wake would introduce potential instabilities when interacting with the wing surface, while a steady-state relaxed wake model for the wing would neglect the unsteady nature of the propeller-wing interaction. More detail on using the distributed vorticity element method in unsteady analysis is provided by Cole et al. (43) (55).

2.2.3 Higher-Order Distributed-Strength Methods

A well-known higher-order distributed element-based potential flow method is PANAIR. It uses first-order source and second-order doublet distributions across quadrilateral surface elements (23), meaning that the system is represented entirely with higher-order elements and does not require the use of vortex filaments. The higher-order distributed nature of the elements used in PANAIR are inherently more computationally expensive than the vortex ring counterparts from methods such as the UVLM. PANAIR can be used as both a panel method or lifting surface method, and the forces and moments in PANAIR-based methods are typically determined using pressure integration. However, PANAIR assumes a quasi-steady framework (56), limiting its applicability for the analysis of complex unsteady systems.

The induction model introduced by Johnson (23) was used by Pate (57), and Pate and German (24), who applied it to a panel method for unsteady analysis. The method models the surface and wake solely with linearly-varying vortex sheets, meaning that the strength distribution relies on directly determining a divergence-free vorticity distribution as discussed in Section 2.1.1. The purpose of this method is to model accurate surface velocities for an unsteady analysis (57), and as such it is strictly a panel method.

2.3 Purpose of this Research

The purpose of this research is to develop a robust potential flow-based method for the analysis of unsteady systems with relaxed wakes. This is accomplished through the development of an entirely new distributed doublet-based method that removes all reliance on vortex filaments, which form the basis of

many instabilities in the UVLM and DVE-based approaches. With the surface and wake represented using distributed doublet sheets of parabolic strength distribution in both the spanwise and streamwise directions, systems modelled with the distributed doublet element (DDE)-based method contain no lumped vorticity, even when modelling unsteady wakes. In addition to this, the DDE-based method includes the modelling of both the shed circulation and the non-circulatory apparent mass, constituting a fully-unsteady framework. As a result of this framework and the entirely distributed nature of the system, stable unsteady relaxed wake analyses are possible even when the systems contain strong surface-wake or surface-on-surface interactions.

2.3.1 Research Contributions

The DDE-based method introduced in the herein study re-purposes the second-order distributed doublet induction model outlined by Johnson (23) into a fully-unsteady lifting surface method, as a natural progression to the UVLM, DVE, and PANAIR methods. Combining the stability and distributed nature of these second-order planar doublet elements into a lifting surface method framework allows it to be used for fully-unsteady relaxed wake analysis, while still retaining some of the benefits of the lifting surface method in terms of computational expense and complexity.

Chapter 3

The Distributed Doublet Element Method

The Distributed Doublet Element (DDE) method was developed to analyze complex and unsteady systems with relaxed wakes, which is challenging or unfeasible with existing filament-based methods. The DDE-based approach improves upon lower-fidelity methods by distributing the vorticity over planar surfaces in both the streamwise and spanwise directions, rather than lumping the strengths into vortex filaments. As a result of this distribution and unlike lower-fidelity methods, the velocity field induced by a DDE-based system is bounded everywhere. This gives the DDE method a distinct advantage in modelling these unsteady systems.

An overview of the DDE-based algorithm is shown in Fig. 3.1. To begin, the wing or rotor is discretized into thin surfaces using planar triangular elements that are placed along the camber line. The initial doublet strength distributions for these elements are determined by imposing flow tangency conditions along the surface, and necessitating the strength distribution be continuous across all elements on a surface. A timestepping procedure then commences, and a row of triangular wake elements are created at the trailing edge of the surface. The strength distribution for the surface and newly-created wake elements are determined, and the wake is relaxed. The strength distribution for the entire surface and wake is updated once again to account for the movement and stretching of the wake elements. The

forces acting on the surface elements are then calculated. The next timestep begins, and this process continues until a user-specified timestep limit is reached.

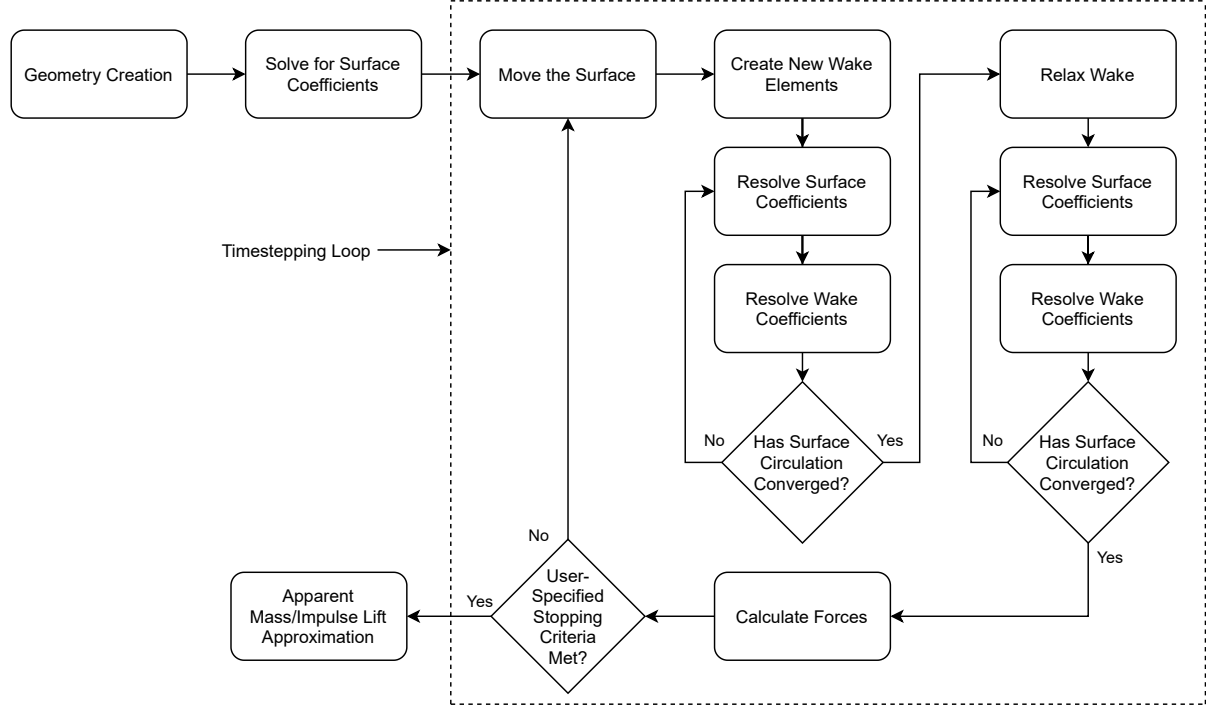


Figure 3.1: An overview of the DDE method.

3.1 The Distributed Doublet Element

At the heart of the method introduced in this thesis are distributed doublet elements, which take the form of planar triangles. As shown in Fig. 3.2, they have a local orthogonal (ξ, η, ζ) -reference system. The 1-2 edge of the triangle is always parallel with the local η -axis, while the 1-3 edge is not necessarily parallel with the ξ -axis as the triangles are not required to be right-angled. The local coordinate system origin is located at the centroid of the element.

The circulation strength across an element is described by the expression:

$$\Gamma(\xi, \eta) = \frac{1}{2}A_1\eta^2 + A_2\eta + \frac{1}{2}B_1\xi^2 + B_2\xi + C_2\eta\xi + C_3 \quad (3.1)$$

with A , B and C representing coefficients of the circulation distribution, and (ξ, η) representing the

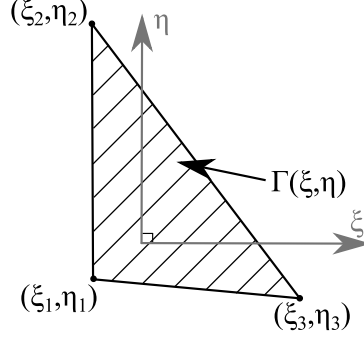


Figure 3.2: A top-down view of a DDE in the local coordinate system.

location on the surface of the element. As outlined by Drela (28), the velocity induced by a distributed doublet element with strength $\Gamma(\xi, \eta)$ on an arbitrary field point is given by:

$$V = \iint_S \Gamma(\xi, \eta) \left[\frac{\hat{n}}{|r-s|^3} - 3\hat{n} \cdot (r-s) \frac{r-s}{|r-s|^5} \right] d\eta d\xi \quad (3.2)$$

with the normal vector of the element represented by $\hat{n}$, the location of the field point represented by r , and the location of the differential element on the surface of the DDE given by s . Equation 3.2 is solved in terms of the element strength coefficients through the method outlined by Johnson (23) for the evaluation of the doublet integrals for an arbitrary field point. The method for solving Eq. 3.2 is discussed in detail in Appendix A. The subsequent velocities that a DDE induces are:

$$\begin{bmatrix} q_\xi \\ q_\eta \\ q_\zeta \end{bmatrix} = [D_i] \begin{bmatrix} A_1 \\ A_2 \\ B_1 \\ B_2 \\ C_2 \\ C_3 \end{bmatrix} \quad (3.3)$$

where the q -vector is the induced velocity acting on the field point in the local element reference frame as determined by the influence coefficients within the matrix $[D_i]$ and the element strength coefficients. The influence coefficients within $[D_i]$ are dependent only on the DDE geometry and the location of the field point relative to the element. The element strength coefficients are determined based on the required boundary conditions prescribed for the system, discussed in further detail in Section 3.3.

The velocity induced by a distributed doublet sheet is bounded everywhere, with the possible exception of the edge of the sheet (23). If the circulation strength is non-zero along the edge of a DDE, that edge will behave equivalent to a vortex filament, capable of inducing infinite and singular velocities (23)(28). However, if there is a neighbouring DDE with which the circulation strength is continuous, then the effects of these edges cancel out as they act in opposite directions, and the velocity is bounded and continuous across that junction. As a result, the DDE method relies on a continuous and uninterrupted circulation distribution, with the strength being zero where there are no neighbouring surface or wake elements.

3.2 Representation of Lifting Surfaces and Wakes

All surface and wakes in the DDE method are represented using distributed doublet elements, meaning all elements in the system follow the relationship outlined in Eq. 3.3. All surfaces and wakes are modelled with zero-thickness, with surface elements placed along the wing, rotor, or propeller camber line. An example of a system modelled with DDEs is displayed in Fig. 3.3, which shows a wing half-span with a 4-bladed wingtip-mounted propeller. As the DDEs are triangular, non-planar surfaces such as camber lines of wings or rotors are modelled as gapless surfaces. Although the DDE method can technically model thickness effects similar to a panel method, it is intended to model complex systems with strong surface-wake interactions. Thus, for the purposes of this research, the lifting surfaces are modelled with zero thickness to avoid any wake piercing.

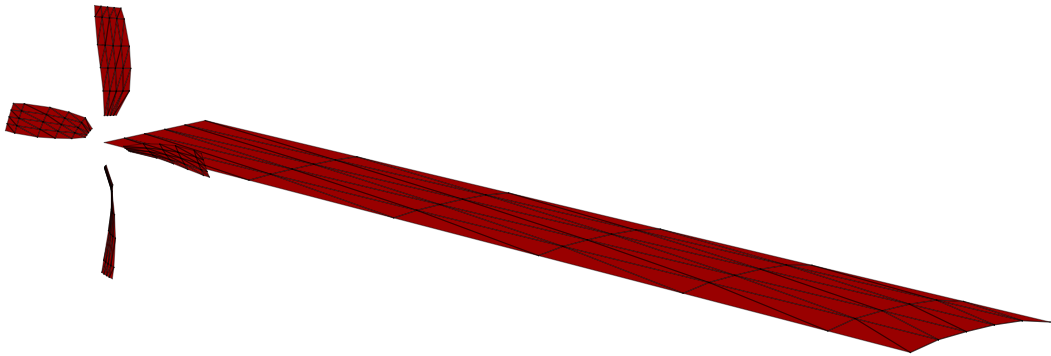


Figure 3.3: A wing half-span with a wingtip-mounted 4-bladed propeller modeled with DDEs.

The wake is developed using a timestepping method. The lifting surfaces are moved every timestep,

and new wake elements are formed in the region bounded by the previous and current trailing edge locations. In the case of a rotor, the wake elements generated have different chordal lengths along the radius of the rotor, as the location from the center of rotation determines the distance travelled by the trailing edge in a timestep. As an illustration of this point, a top-down view of a rotor in fully-edgewise flow with a relaxed wake is displayed in Fig. 3.4, in which the wake element colours differentiate between the blades from which they were shed. The wake relaxation procedure moves the wake vertices with their local velocities once per timestep, meaning the wake shape evolves as the solution progresses. As with the modelling of the surfaces, the wake geometry is modelled as a gapless sheet of elements.

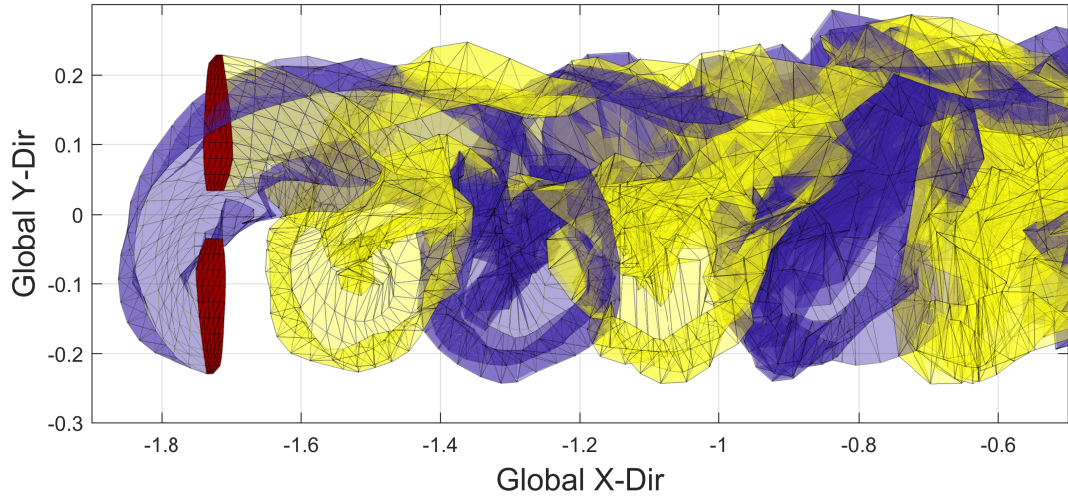


Figure 3.4: Top-down view of a rotor in fully-edgewise flow using a relaxed-wake model.

3.3 Determination of the Circulation Strength Distribution

As previously stated, due to the nature of distributed doublet elements, any stepwise change in the circulation along the edge of an element or between two neighbouring elements is equivalent to a vortex filament with a strength equal to that stepwise change (28). However, if the triangular elements are joined together with a continuous circulation distribution across neighbouring elements, there are no stepwise changes, and therefore the velocity induced by that system will be bounded everywhere. Due to this fact, the circulation distribution across the system must be continuous. This distribution is determined via the boundary conditions imposed on the system, which are, as a result, focused on removing any

stepwise change in strength. At the leading edge and free ends of lifting surfaces, the circulation strength is prescribed to zero. Between neighbouring elements, the circulation and vorticity are prescribed to be continuous to facilitate a smooth transition in strength across the junction. At the trailing edge of lifting surfaces, the circulation strength is carried into the wake. As an example, a circulation distribution fulfilling these boundary conditions is illustrated in Fig. 3.5, in which a graphical representation of the circulation distribution is superimposed over the corresponding planar elliptical wing.

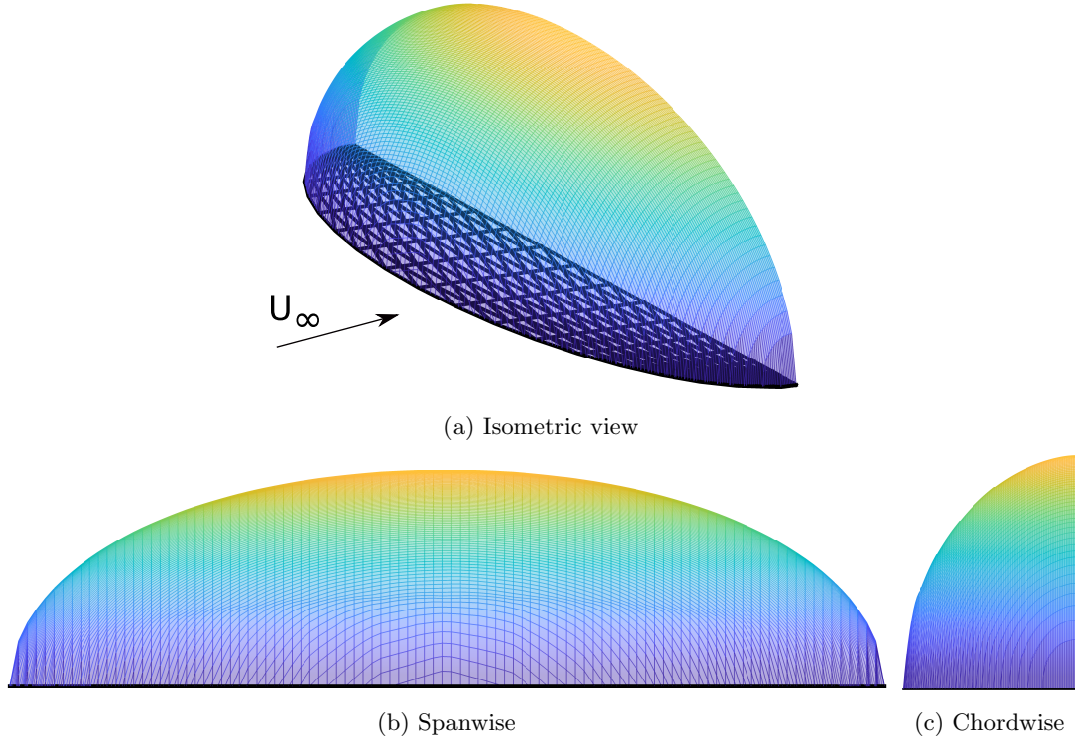


Figure 3.5: A graphical representation of the circulation distribution across a planar elliptical wing.

In addition to these prescribed circulation and vorticity conditions, flow tangency boundary conditions are imposed at four collocation points for every surface element. These points are located at the centroid of the element and the 90% location from the centroid to each of the three vertices, as shown in Fig. 3.6. These flow tangency conditions imposed near the element vertices ensure that at junctions of multiple elements, such as a single vertex connected to multiple DDEs, flow tangency is enforced in the immediate area. An example of this is illustrated in Fig. 3.7, which shows a top-down view of a simple rectangular wing consisting of eight elements. Eight collocation points surround the central vertex of the wing. While the tangential velocities at these points may differ, the normal velocity components are

forced to zero, which helps define a smooth circulation distribution across this junction, as the circulation distribution is closely tied to the self-induced normal velocity of an element.

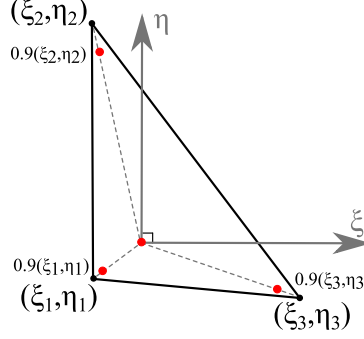


Figure 3.6: The locations of collocation points on a DDE.

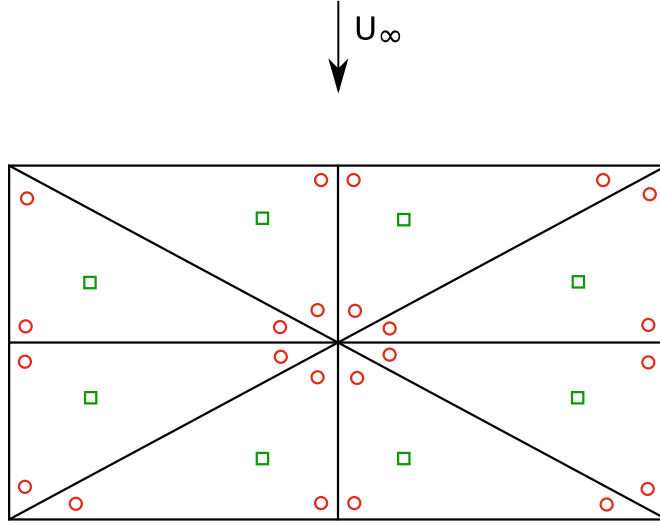


Figure 3.7: A top-down view of a rectangular wing with the collocation points highlighted.

The increased number of collocation points per element also reduces the number of elements required to model a system. The higher-order distributed nature of the DDE method means that, in theory, fewer elements should be required to model a surface when compared to conventional vortex-lattice methods. However, the accuracy of a solution is tied to the number of flow tangency conditions on the surface. Thus, by increasing the number of collocation points per element, the total number of elements required to model a system can be reduced. An example of this is illustrated in Fig. 3.7, in which the green squares represent the traditional location of collocation points at the element centre, and the red circles represent the additional collocation points used in the DDE method. This increase in the number of flow

tangency conditions provides comprehensive coverage of the wing while taking advantage of the reduced number of elements needed due to the higher-order circulation distribution.

As there are six circulation-strength coefficients per DDE, the number of boundary conditions for a perfectly-constrained system is $6N_E$, where N_E represents the number of surface elements in the system. This ideally-constrained system would produce a circulation distribution that satisfies all boundary conditions completely. In actuality, the DDE-method is over-constrained. The excess number of boundary conditions is a result of prescribing a continuous higher-order circulation distribution across an entire surface. Prescribing a gapless second-order circulation distribution between two DDEs requires three boundary conditions. In addition to this, the gradient of the circulation distribution is set equal at the edge mid-point between neighbouring elements in both the ξ and η -directions, resulting in a further two boundary conditions. As such, a typical single-edge boundary between two DDEs can require a total of five boundary conditions, or 2.5 boundary conditions per edge per DDE. With the system being over-constrained, the element strengths are solved by finding the minimum norm least-squares solution. Due to this approximation, the element strengths are not guaranteed to satisfy every boundary condition completely, and the network may feature discontinuities in the circulation distribution. These discontinuities must be corrected; otherwise, singularities remain in the system.

As noted by Johnson (23), it is virtually impossible to define a second-order doublet distribution (analogous to the circulation distribution for this application) that is continuous across all panel edges of a network. To remove discontinuities, any gaps in the circulation distribution between elements are removed through averaging, and the leading edge and free-end strengths are prescribed to zero. This results in a continuous strength distribution. These corrections, however, impact the system's adherence to the flow tangency conditions. To mitigate this effect, the error introduced by this correction is quantified through re-evaluating the flow tangency conditions based on the updated strength distribution. This process iterates as the prediction-correction algorithm outlined in Fig. 3.8. The prediction is the calculation of the least-squares solution, and the correction is the removal of discontinuities. This algorithm continues iterating until the solution converges to a continuous circulation distribution, determined through monitoring the maximum change in circulation occurring across the surface elements every iteration. With the convergence of the continuous circulation distribution, any error within the system is then reflected in deviation from the flow tangency conditions, which is quantified with a coefficient of determination at the conclusion of the procedure outlined in Fig. 3.8.

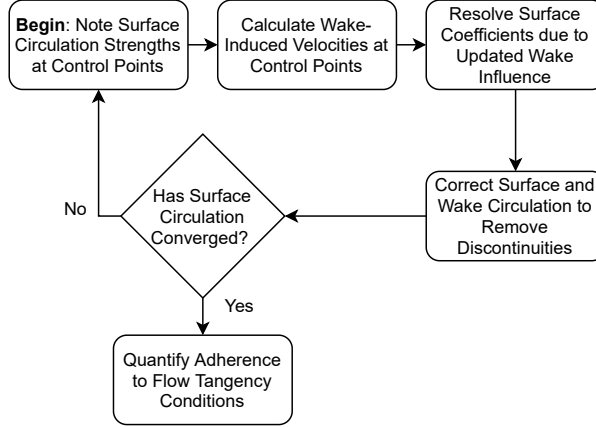


Figure 3.8: The prediction-correction algorithm used within the timestepping procedure.

It should be noted that the purpose of the iterative procedure outlined in Fig. 3.8 is two-fold. Aside from ensuring a continuous circulation distribution, this process also ensures that flow tangency is maintained at all collocation points while analyzing unsteady systems. As shown in Fig. 3.1, the DDE method uses this iterative process twice within each timestep, once before and once after the wake relaxation process. Apart from mitigating the error introduced by the least-squares solution, this process is also used to determine the updated strengths of the surface elements and newly-created wake elements. As the strength of the surface elements impact the strength of the newly-created wake elements, and as the newly-created wake elements impact flow tangency through their induced velocities, the strengths of the surface and wake elements are closely intertwined.

The iterative procedure outlined in Fig. 3.8 is necessary in order to ensure flow tangency and the Kutta condition are adhered to within each timestep, especially in unsteady analyses in which the flow tangency conditions may vary significantly between timesteps due to changing inflow conditions. A single iteration of this process is often not enough to reach convergence when the freestream velocities acting on an aerodynamic surface are changing rapidly between timesteps, which leads to a lack of flow tangency and a solution that is dependent on the timestep size. Katz and Plotkin describe this iteration-based technique when discussing unsteady motion of a two-dimensional thin airfoil (27). Murua (58) mentions the possibility of implementing such a scheme within the UVLM, though they opted not to due to the added computational expense.

3.4 Force Calculations

This section discusses how the aerodynamic forces that act on a surface are determined within the DDE method. The total force acting on a surface is determined by summing the contributions of the freestream, induced, and non-circulatory forces. The freestream and induced lift and side forces, as well as the non-circulatory forces, are determined for every surface DDE. Induced drag is calculated along the trailing edge of the aerodynamic surface. The freestream and induced forces are found by integrating the Kutta-Joukowski theorem across elements, while the non-circulatory forces are approximated through a finite-differencing scheme outside of the timestepping loop.

3.4.1 Freestream Forces

The freestream force acting on an element is found through integrating the Kutta-Joukowski theorem across the surface of the element:

$$\hat{F} = \rho \iint_S \hat{U}_a(\xi, \eta) \times \hat{\omega}(\xi, \eta) \, d\eta \, d\xi \quad (3.4)$$

where $\hat{F}$ is the resultant force vector acting on the element, $\hat{U}_a(\xi, \eta)$ is a vector field approximation of the freestream velocities across the surface of an element, with the subscript a denoting that this is an approximation, and $\hat{\omega}(\xi, \eta)$ is a vector field of the surface vorticity of the element. The analytical solution to Eq. 3.4 is covered in detail in Appendix B.1.

As discussed in Section 2.1.1, the surface vorticity vector field is determined via the gradient of the circulation distribution given by Eq. 3.1, rotated 90° about the $\hat{\zeta}$ -axis in the local reference frame:

$$\hat{\omega}(\xi, \eta) = \frac{\partial \Gamma}{\partial \eta} \hat{\xi} - \frac{\partial \Gamma}{\partial \xi} \hat{\eta} = (A_1 \eta + C_2 \xi + A_2) \hat{\xi} + (-B_1 \xi - C_2 \eta - B_2) \hat{\eta} \quad (3.5)$$

The approximation of the freestream velocity vector field across the element surface, $\hat{U}_a(\xi, \eta)$, is determined by finding the linear variations in the three components of the freestream velocity across an element in the local reference frame, as a function of location on the surface:

$$\hat{U}_a(\xi, \eta) = (k_{\xi a} \xi + k_{\xi b} \eta + k_{\xi c}) \hat{\xi} + (k_{\eta a} \xi + k_{\eta b} \eta + k_{\eta c}) \hat{\eta} + (k_{\zeta a} \xi + k_{\zeta b} \eta + k_{\zeta c}) \hat{\zeta} \quad (3.6)$$

where k represents a coefficient of this linear approximation. This first-order method of representing the freestream velocity across an element was chosen rather than assuming a constant velocity distribution for Eq. 3.4, as the DDE method is meant to be used predominantly in rotor analysis which does not feature constant freestream velocity distributions.

It should be noted that Eq. 3.4 is integrated across the surface of the element to produce a single resultant force vector for every element, which is assumed to act on the centroid of each element. This force can then be broken down into its lift and side force components for wing analysis, or axial and in-plane components for rotor analysis. If the integration in Eq. 3.4 is ignored and only the integrand is evaluated, the result is a vector field of the freestream force acting across the element surface. This would be a beneficial approach for analyses where higher-fidelity force distributions are required.

3.4.2 Induced Forces

The induced lift and side forces, which arise from velocities that the flow induces along the lifting surfaces, are determined for every surface element in a manner similar to the freestream force calculations above. The induced drag is calculated along the trailing edge of the lifting surfaces for every chordwise row of surface elements.

Induced Lift and Side Force

Following a similar methodology to the freestream force calculation, the induced lift and side force calculations begin through integrating the Kutta-Joukowski theorem across the element surface:

$$\hat{F} = \rho \iint_S \hat{W}_a(\xi, \eta) \times \hat{\omega}(\xi, \eta) \, d\eta \, d\xi \quad (3.7)$$

where $\hat{W}_a(\xi, \eta)$ represents the induced velocity vector field across the surface of an element, found by summing the induced velocity contributions from all surface and wake elements. This vector field expression is determined using the same first-order approximation introduced in Eq. 3.6, with the subscript a again denoting that this is an approximation. As with Eq. 3.4, the analytical solution to Eq. 3.7 is discussed in detail in Appendix B.1. The lift and side force (or axial and radial components for rotor analysis) can then be found by dotting the resultant force $\hat{F}$ with the unit vectors of the lift

and side force directions. This is done to negate the component of $\hat{F}$ in the drag direction (or tangential direction for rotors), as this is determined separately as outlined in the following section.

Induced Drag

Unlike the other force calculations, the induced drag force is found along the trailing edges of the lifting surfaces. The method introduced here is an extension of the method introduced by Eppler (34), and subsequently used by Mortara et al. (35), Bramesfeld (36), and Cole (32). Unlike the leading edge and free ends of lifting surfaces in the DDE method, the trailing edge may have a non-zero circulation strength across its span, which is continued into the wake to form a continuous strength distribution. Based on the Kutta condition and the Kutta-Joukowski theorem, the strength of this circulation along the trailing edge is proportional to the force generated by the surface. In fact, the freestream and induced lift and side forces could be calculated solely along the trailing edge based on this circulation value, if a chordwise force distribution was not desired.

To calculate the induced drag force along the trailing edge, the Kutta-Joukowski theorem is integrated across the span of the trailing edge vector of the surface elements:

$$\hat{F} = \rho \int_{\xi} \hat{W}_T(\xi) \times \hat{v}_T \Gamma_T(\xi) d\xi \quad (3.8)$$

where $\hat{W}_T(\xi)$ is a second-order approximation of the wake-induced velocity vector field along the trailing edge as a function of the element span, $\hat{v}_T$ is a unit vector of the trailing edge direction, and $\Gamma_T(\xi)$ represents the circulation along the trailing edge vector as a function of the element span.

The wake-induced velocity vector field $\hat{W}_T(\xi)$ is represented in a similar manner as the the freestream velocity vector field introduced in Eq. 3.6. However, the wake-induced velocity vector field is expressed solely as a function of element span along the trailing edge, and is a second-order approximation:

$$\hat{W}_T(\xi) = (u_1\xi^2 + u_2\xi + u_3)\hat{\xi} + (v_1\xi^2 + v_2\xi + v_3)\hat{\eta} + (w_1\xi^2 + w_2\xi + w_3)\hat{\zeta} \quad (3.9)$$

where u , v , and w represent coefficients of the vector field approximation. A second-order approximation was used in order to match the second-order nature of the spanwise circulation distribution. The analytical solution to Eq. 3.8 is discussed in further detail in Appendix B.2.

3.4.3 Unsteady Forces

The unsteady or non-circulatory forces in the DDE method are calculated using a finite-differencing scheme, which takes place after the timestepping loop and marks the final step in the method, as shown in Fig. 3.1. Recall from Section 2.1.4 that the unsteady Kutta-Joukowski formulation for a filament-based method is given by Drela (37) as:

$$\bar{F}'_L = \rho \Gamma \bar{V} \times \hat{s} + \rho \frac{\partial \Gamma}{\partial t} \frac{c}{|\bar{V}_\perp|} \bar{V} \times \hat{s} \quad (2.8)$$

The first term to the right side of the equality in Eq. 2.8 represents the instantaneous force acting on the lifting surface section. Within the DDE method, the equivalent of this term is calculated via the surface integrations of the Kutta-Joukowski theorem represented by Eqs. 3.4 and 3.7, as well as the integration along the trailing edge represented by Eq. 3.8. The second term on the right side of the equality in Eq. 2.8 represents the apparent mass, which in the DDE method is determined with the expression:

$$\hat{F} = \rho \frac{\partial}{\partial t} \left[\iint_S \Gamma(\xi, \eta) \, d\eta \, d\xi \right] \hat{l} \quad (3.10)$$

with the time-derivative approximated across timesteps using a finite-differencing scheme. The resulting force $\hat{F}$ acts in the lift direction, denoted by $\hat{l}$ and determined using the local freestream velocity vector at the element control point, $\hat{U}$, and the spanwise direction vector, $\hat{s}_w$, in a similar manner to Eqs. 2.8 and 2.9:

$$\hat{l} = \frac{\hat{U} \times \hat{s}_w}{|\hat{U} - (\hat{U} \cdot \hat{s}_w) \hat{s}_w|} \quad (3.11)$$

The integrated circulation across every element according to Eq. 3.10 is determined each timestep. This equation follows closely to the time-dependent non-circulatory term in Eq. 2.8, though the chord term c is not present due to the integration of the circulation in Eq. 3.10 being performed in both the spanwise and chordwise directions. Further details on the approximation of Eq. 3.10 are given in Appendix B.3, including the analytical solution to the integrated circulation of an element.

Stability of the Unsteady Force Calculation

As discussed in Section 2.1.4, the numerical stability of unsteady filament-based methods requires the enforcement of a timestep constraint of $\Delta x_w / \Delta x_c \geq 1$, where Δx_w is the distance travelled by the surface elements each timestep, and Δx_c is the chord length of the trailing edge surface elements. Enforcement of this timestep size constraint becomes problematic for surfaces with changing velocities along the span, such as rotor blades. Since its second-order distributed doublet sheets can be imagined as a system of infinitely small vortex rings, the timestepping portion of the DDE method inherently satisfies this constraint of $\Delta x_w / \Delta x_c \approx 1$. In contrast, using a conventional vortex-lattice method would require an infinite number of elements, leading to both Δx_w and Δx_c being infinitesimally small, and therefore satisfying $\Delta x_w / \Delta x_c \approx 1$. As the apparent mass term represented by Eq. 3.10 is approximated after the timestepping loop, this approximation can be made with a timestep size different than the one used within the timestepping loop to ensure that the constraint $\Delta x_w / \Delta x_c \geq 1$ is maintained. This guarantees the stability of the apparent mass approximation, independent of the velocity or distance travelled by a surface element inside of a timestep.

3.5 Summary

The DDE method uses second-order distributed doublet sheets to model aerodynamic systems, eliminating the singularities and instabilities which are inherent in vortex filament-based potential flow methods. Aerodynamic surfaces are modelled using planar triangular DDEs placed along the camber line. The wake is represented as a continuous sheet of DDEs, and is grown and relaxed with the timestepping process. Freestream and induced forces are calculated by integrating the Kutta-Joukowski theorem across the elements, and non-circulatory forces are approximated based on the time rate of change of the circulation. Due to the distributed nature of the DDEs and the method used for approximating non-circulatory forces, the DDE method does not have constraints on its stability, as opposed to UVLM or DVE-based methods.

Chapter 4

Validation

The DDE method was evaluated through a series of four validation studies, with the methodology and results of these studies presented within this chapter. The first study predicted the span efficiency factors of several planar elliptical wings and quantified the impact of the discretization scheme on the results. The second study modelled a flat plate passing through a sharp-edged gust as a means of validating the unsteady force calculations by comparing the lift response to the analytical solution, which is given by the Küssner function. Following this, the time-averaged thrust coefficients produced by a rotor under various operating conditions were predicted using the DDE method and compared to experimentally-obtained data from Ryerson University’s large subsonic wind-tunnel and rotor-test stand. Lastly, predictions of a rotor’s transient thrust coefficient in near-edgewise flight were compared to high-fidelity CFD-based results, constituting the validation for fully-unsteady transient analysis of rotors.

4.1 Elliptical Wing Study

In a panel density study, the predicted span efficiencies of various planar elliptical wings were compared to classical analytical results. This study aimed to determine the effect of the discretization scheme on the resulting force predictions. Two distinct planar elliptical wing shapes were evaluated, with both illustrated in Fig. 4.1. The ratio x_t/c_r relates the chordwise location of the wingtip, x_t , to the chord length at the wing root, c_r , with $x_t/c_r = 0$ denoting a straight leading edge and $x_t/c_r = 1$ denoting a straight trailing edge. Both wings have the same planform area, an aspect ratio of 7, and were modelled

as infinitely thin and without camber. The chord at the wingtip was set equal to $1/10^{th}$ of the root chord to avoid the chordwise elements coming to a point.

The point of comparison for this panel density study is the span efficiency factor, which relates the actual induced drag coefficient of a wing, C_{D_i} , to that produced by an ideal elliptical lift distribution:

$$e = \frac{C_L^2}{(\pi AR)C_{D_i}} \quad (4.1)$$

where AR is the aspect ratio of the wing. According to lifting line theory, an elliptical lift distribution produces the lowest induced drag for a planar wing with a given aspect ratio (27), with a resulting span efficiency factor of one representing the theoretical maximum for a planar wing. This classical approach assumes a fixed drag-free wake with a planar wake trace in the Trefftz plane. For the purposes of this study, an elliptical lift distribution is achieved by using an elliptical chord distribution with zero twist.

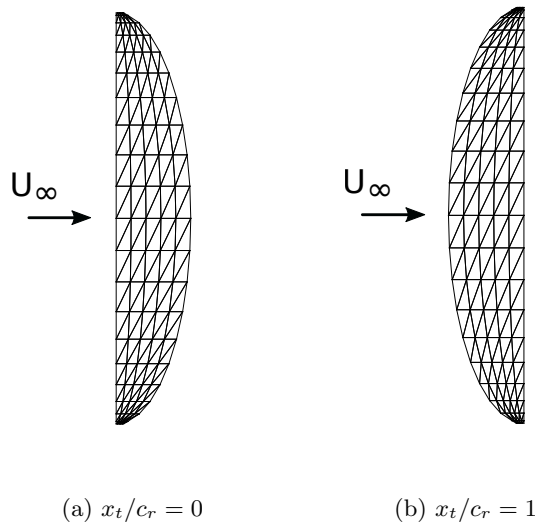


Figure 4.1: The two planar elliptical wing shapes considered in this study.

When using a fixed wake model, the sweep of the trailing edge impacts the shape of the wake, with a forward-swept trailing edge resulting in a non-planar wake trace in the Trefftz plane due to the angle of attack of the wing (36). As a result, the elliptical wing with a straight trailing edge is expected to match best the analytical solution of $e = 1$ from classical lifting line theory, which assumes a planar wake trace. The wing with a forward-swept trailing edge, in contrast, is expected to have a span efficiency factor greater than one due to the non-planar wake trace.

The results of this study are presented in two sections, with the first focusing on the effect of the spanwise element count and spacing, and the second focusing on the effect of the chordwise element count. These studies are performed independently as a means of isolating the effects of the spanwise and chordwise distributions on the solution. Within these panel density studies, both planar elliptical wing shapes are evaluated using a varying number of distributed doublet elements. All analyses were performed at a fixed angle of attack of four degrees, resulting in a lift coefficient of $C_L \approx 0.32$. A fixed, drag-free wake model was used to facilitate a direct comparison to the classical lifting line solution. The timestep size was set equal to the root chord of the wing, and the solutions were evaluated until the span efficiency factor converged to within 1E-4.

4.1.1 Effect of Spanwise Element Count

The planar elliptical wing configurations illustrated in Fig. 4.1 were evaluated with a number of different spanwise elements, denoted by N , and with both constant and cosine spanwise spacings. A cosine spacing scheme distributes the elements across the span such that the resolution near the wingtips is increased relative to the wing root. The chordwise element count for these analyses, denoted by M , was fixed at 10. An example of increasing spanwise resolution for the $x_t/c_r = 0$ wing is presented in Fig. 4.2 for both the constant and cosine spanwise spacing schemes.

The results of the spanwise element density study are presented in Fig. 4.3, with Fig. 4.3a illustrating the impact of a changing number of spanwise elements on the resulting span efficiency factors for both the constant and cosine spacing schemes. In the case where $x_t/c_r = 1$, the span efficiency factor predicted using the DDE method falls within 2% of the classical lifting line solution of $e = 1$ for the majority of panel densities. This slight discrepancy is due to the fixed number of chordwise elements used, as shown later in Section 4.1.2. The predicted span efficiency of $e \approx 0.89$ for $N = 2$ is due to the poorly discretized elliptical planform provided with this low spanwise element count, as illustrated in Fig. 4.2a. In the case of $x_t/c_r = 0$, the span efficiency factor converges to $e \approx 1.05$. As previously noted, a wing with a forward-swept trailing edge operating at a positive angle of attack will produce a wake trace with dihedral in the Trefftz plane, meaning span efficiency factors greater than one are expected.

The time required for each of these cases are provided in Fig. 4.3b on a semi-log plot. These analyses were performed on a desktop computer with 16 GB of system memory, using a single core of a 3.7 GHz Intel i7-8700k processor. For the data presented in Fig. 4.3b, the computational cost grows in the

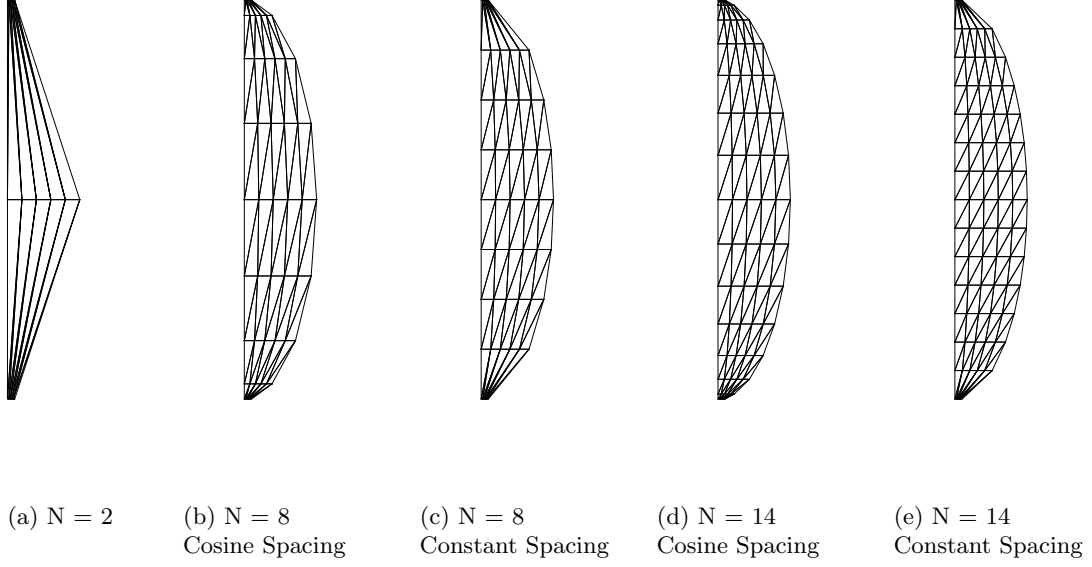
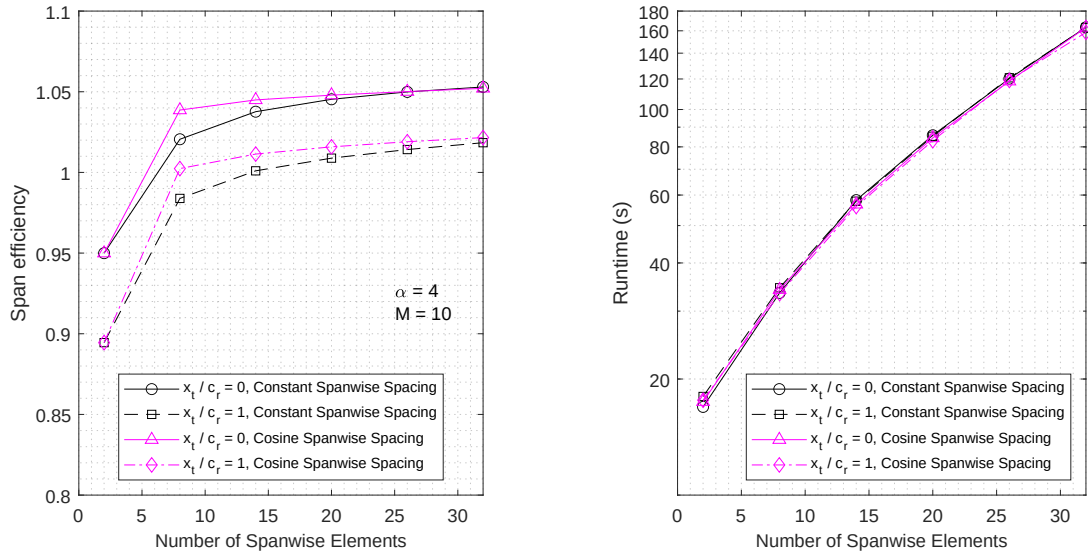


Figure 4.2: Effect of spanwise element count and spacing on an elliptical wing shape ($x_t/c_r = 0$).



(a) Spanwise element count versus span efficiency.

(b) Spanwise element count versus runtime.

Figure 4.3: Effect on spanwise element count on span efficiency and runtime for elliptical wings.

order of $N^{3/2}$. This cost relationship is driven primarily by the number of iterations performed within each timestep, as outlined in Section 3.3, which itself is highly-dependent on the nature of the analysis. Therefore, the order of the computational cost of the DDE method cannot be generalized as a whole.

The effect of the spanwise discretization on the circulation distribution is illustrated in Fig. 4.4, which shows the spanwise circulation distribution at the trailing edge, Γ_{TE} , plotted against the spanwise location, y , for the elliptical wing with $x_t/c_r = 1$. These values were non-dimensionalized to vary from zero to one, using the trailing edge circulation at the wing root, $\Gamma_{TE,0}$, and the wing half-span, $b/2$, respectively. The analytical equation for an ellipse is plotted for reference, with both the $N = 8$ and $N = 20$ cases matching this elliptical shape closely. The $N = 2$ case significantly deviates from an elliptical distribution, resulting in the low span efficiency factors reported in Fig. 4.3. These results indicate that eight spanwise elements with a cosine spacing scheme is adequate for representing both the elliptical chord distribution and the resulting circulation distribution in the spanwise direction.

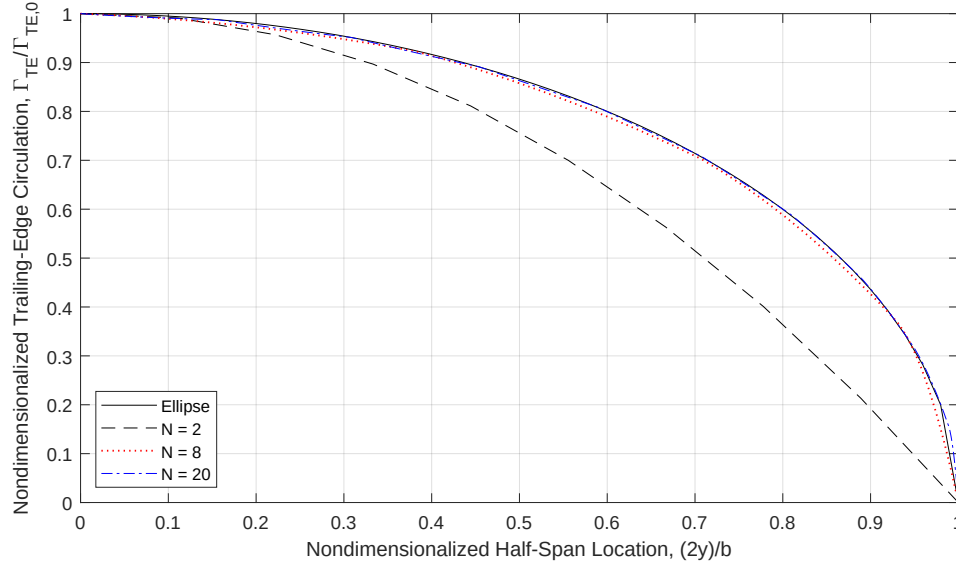


Figure 4.4: Trailing edge circulation versus spanwise location for elliptical wings with varying spanwise elements ($x_t/c_r = 1$, $M = 10$, cosine spacing).

The span efficiencies from Fig. 4.3a corresponding to 8, 20, and 32 spanwise elements were used within the Richardson extrapolation procedure provided by Celik et al. (59). The resulting extrapolated span efficiency factors, e_{ext} , and the corresponding grid convergence indices, GCI, are presented in Table

4.1. The grid convergence index is a measure of the error between a computed value and the asymptotic solution, given as a percentage (59). The extrapolated span efficiency factors presented in Table 4.1 are greater than expected, notably with the $x_t/c_r = 1$ cases where the analytical solution is $e = 1$. This overestimation of the span efficiency factor is due to the fixed number of chordwise elements used for this study, meaning that the extrapolated results presented in Table 4.1 represent the solution as the spanwise element counts are increased to infinity while the chordwise element counts remain fixed at $M = 10$. The impact of chordwise element count is discussed in Section 4.1.2.

Table 4.1: Results of spanwise grid convergence study

x_t/c_r	Distribution	e_{ext}	GCI (%)
0	Constant	1.062	1.05
0	Cosine	1.055	0.33
1	Constant	1.030	1.43
1	Cosine	1.026	0.51

The grid convergence indices for the wings with a cosine distribution are significantly less than those with a constant spacing, as a result of the cosine spacing schemes converging faster as the number of spanwise elements is increased. This faster convergence results from the increased resolution near the wingtips, where the circulation and vorticity change rapidly. These results indicate that a cosine spanwise spacing scheme is significantly more effective than constant spacing, with only eight spanwise elements needed to achieve similar accuracy to a constant spacing scheme with approximately 16 spanwise elements. This comparison highlights the importance of the element resolution near the wingtips, while demonstrating that only a few elements are necessary to model the majority of the inboard portion of the wing where the vorticity is low.

A time-history of lift and induced drag coefficients, as well as span efficiency factor, are provided in Fig. 4.5 and correspond to the analysis of an elliptical wing with $x_t/c_r = 0$, $N = 20$ and cosine spacing. Both the lift and drag coefficients are within 1% of their final value by the tenth timestep, with each timestep being equivalent to the chordal length of the wing as previously discussed. The wing must move far enough upstream of the starting vortex to gain an accurate steady-state force prediction, which is why the lift coefficient rises and the induced drag coefficient falls as this distance is increased and the influence of the starting vortex is diminished.

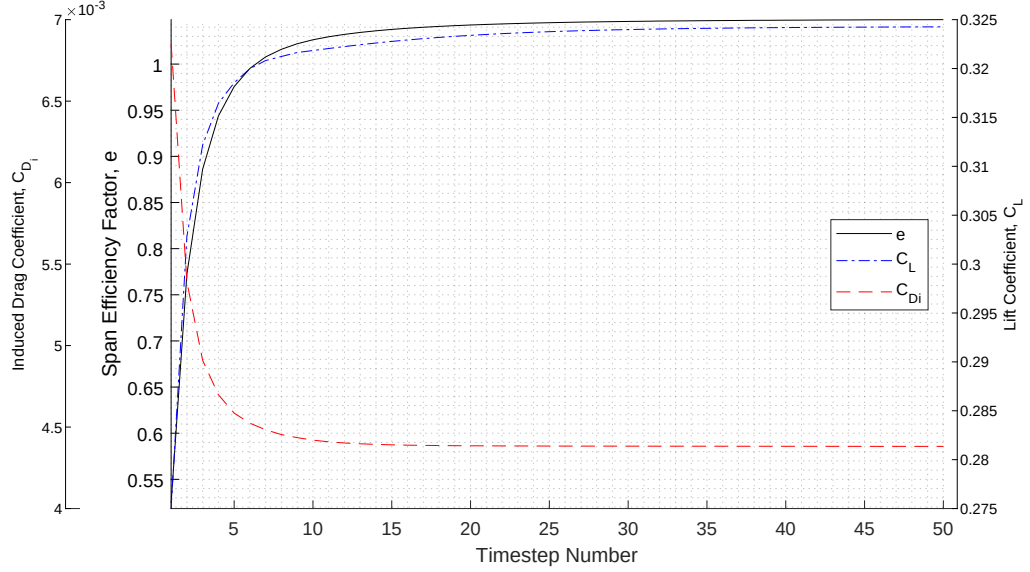


Figure 4.5: Span efficiency factor, and lift and drag coefficient histories for an elliptical wing ($x_t/c_r = 0$, $N = 20$, $M = 10$).

4.1.2 Effect of Chordwise Element Count

In a similar vein to the previous analysis, both planar elliptical wing configurations illustrated in Fig. 4.1 were evaluated with a varying number of chordwise elements, with the spanwise element count fixed at 20 with constant spacing. Figure 4.6 illustrates the impact of increasing M on the panel density of an elliptical wing with $x_t/c_r = 0$. Unlike the previous analysis, an increasing number of chordwise elements does not alter the overall geometric representation of this uncambered wing in any way; it simply increases the potential fidelity of the chordwise circulation distribution.

The results of this portion of the study are presented in Fig. 4.7, with Fig. 4.7a illustrating the impact of the chordwise element count on the span efficiency factor. The results follow a similar trend for both the $x_t/c_r = 0$ and $x_t/c_r = 1$ planform shapes. Past a chordwise element count of two, increasing the chordwise element density results in the span efficiency asymptotically approaching a solution, with the extrapolated span efficiencies presented in Table 4.2 along with the corresponding grid convergence indices. These extrapolations were performed using the results corresponding with 6, 14, and 22 chordwise elements. In the case of $x_t/c_r = 1$, the span efficiency converges to $e = 1$,

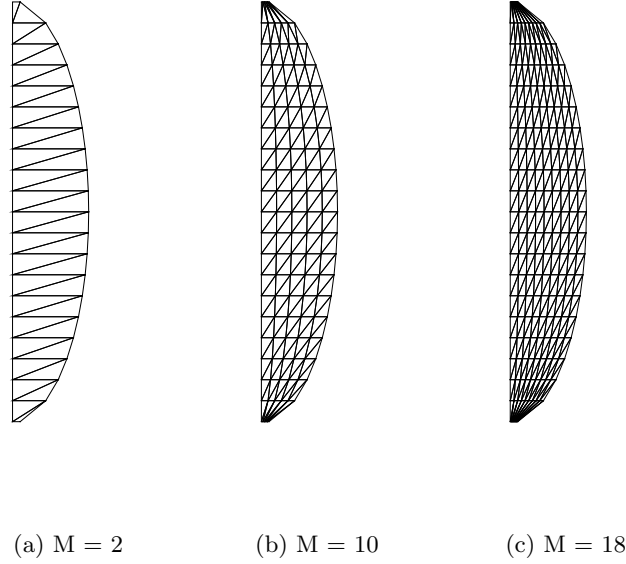


Figure 4.6: Effect of increasing chordwise element count on an elliptical wing shape.

as expected from the classical solution. The case of $x_t/c_r = 0$ converges to a span efficiency factor that is approximately 2% higher than that of $x_t/c_r = 1$ due to the non-planar wake produced by the forward-swept trailing edge.

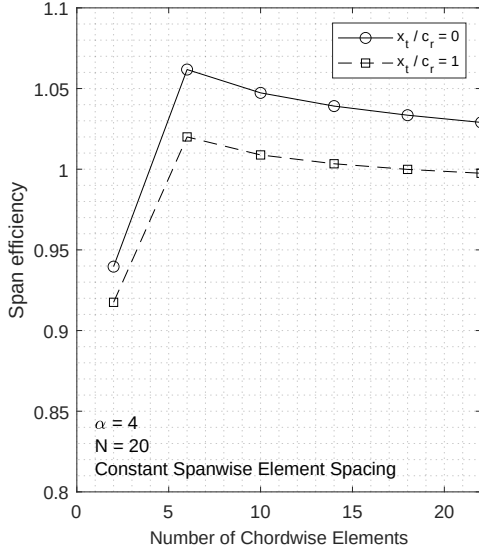
Table 4.2: Results of chordwise grid convergence study

x_t/c_r	e_{ext}	GCI (%)
0	1.022	0.87
1	0.996	0.18

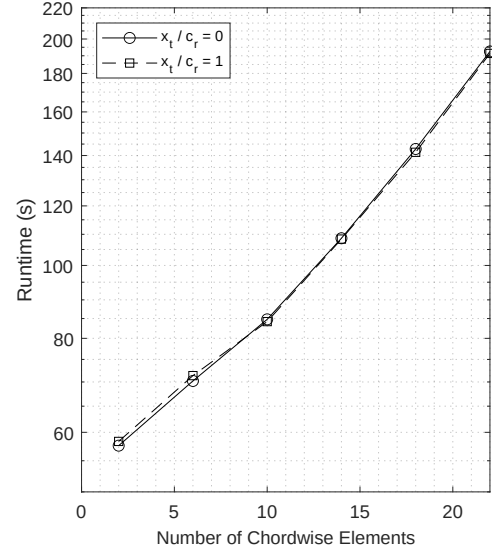
Figure 4.7b illustrates the impact of increasing M on the solution time, with the computational cost varying in the order of M^2 for these analyses. When comparing the total number of surface elements represented in Figs. 4.3b and 4.7b, increasing the chordwise element count is often less computationally expensive. This is due to the fact that an increase in the number of chordwise elements does not change the number of wake elements produced each timestep.

4.1.3 Summary

A panel density study was performed using two different elliptical wing shapes with a varying number of spanwise and chordwise elements used to represent them. For both wings, the results of the studies



(a) Chordwise element count versus span efficiency.



(b) Chordwise element count versus runtime.

Figure 4.7: Effect on chordwise element count on span efficiency and runtime for elliptical wings.

converged to within three percentage points of the expected span efficiency factor from the classical lifting line solution. The wing with the forward-swept trailing edge resulted in a higher span efficiency factor than the wing with the straight trailing edge, as expected from the theory. This validation study indicates that the spanwise resolution near the wingtip is far more influential on the solution than near the wing root, meaning that cosine spacing is a more efficient spanwise discretization scheme.

4.2 Transient Lift Response to Gusts

Apparent mass and shed circulation can contribute significantly to the resulting forces produced by an aerodynamic surface under unsteady conditions. The DDE method uses a finite-differencing scheme discussed in Section 3.4.3 to approximate the apparent mass term in the unsteady Kutta-Joukowski formulation, while the shed circulation is modelled through streamwise changes in the wake doublet strength. The purpose of this study is to validate the modelling of both of these effects, as well as to quantify the impact of the timestep size and panel density on the resulting force predictions.

To demonstrate the effectiveness of the DDE method for unsteady analysis, a two-dimensional flat plate was modelled passing through a sharp-edged gust. Figure 4.8 illustrates this system, with an infinitely-thin flat plate at zero angle of attack entering into a sharp-edged gust. The resulting transient lift response was then compared to the analytical solution given by the Küssner function (42)(60). For the purposes of this validation study, the gust had a magnitude equal to 5% of the freestream magnitude.

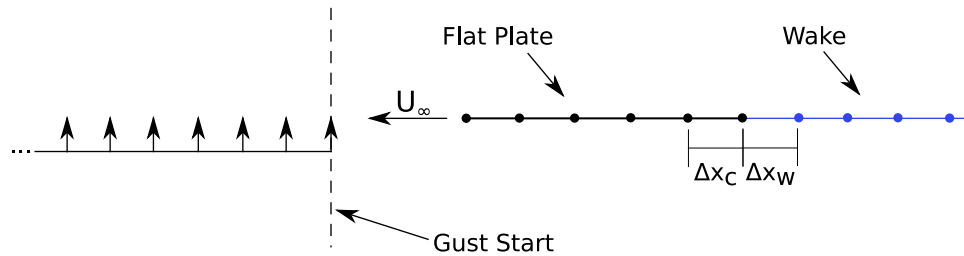
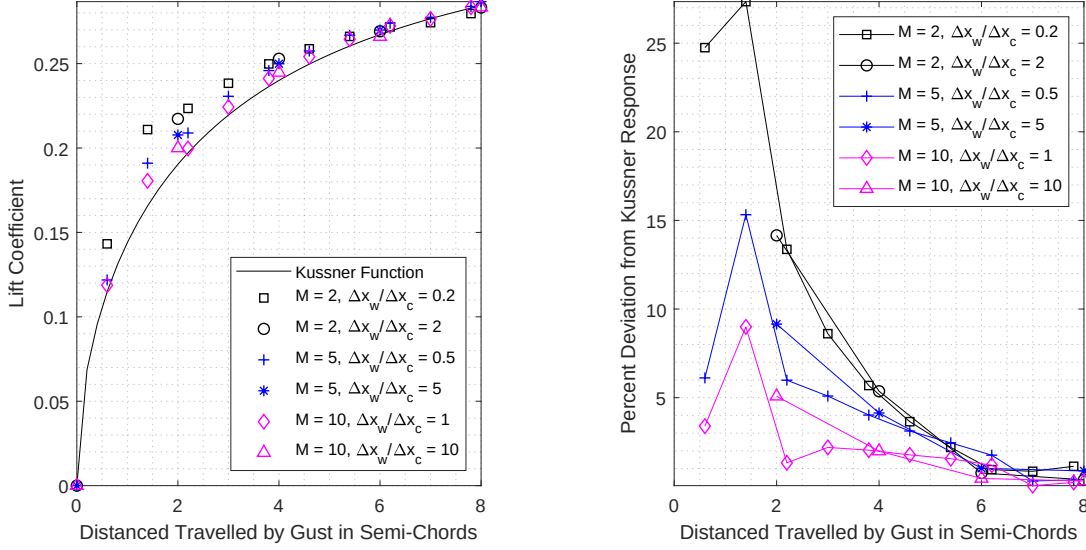


Figure 4.8: A two-dimensional flat plate approaching a sharp-edged gust.

Several different $\Delta x_w/\Delta x_c$ ratios were evaluated within this study. As outlined in Section 2.1.4 and illustrated in Fig. 4.8, this ratio represents the distance travelled by the flat plate relative to the average length of a single chordwise element. These various $\Delta x_w/\Delta x_c$ ratios were obtained using different timestep sizes and chordal element counts, with the number of chordwise elements indicated by the parameter M .

The results from this study are presented in Fig. 4.9, plotted alongside the analytical response given by the Küssner function. Figure 4.9a indicates that, unlike conventional UVLM-based methods, there is independence from the $\Delta x_w/\Delta x_c$ ratio and timestep size with the DDE method for a given number of chordwise elements. Though the ratios for each chordwise element count in Fig. 4.9a vary by a factor of ten, there is very little difference in the lift coefficient across the gust response as predicted by the DDE method. For a given number of chordwise elements, the solution is largely independent of timestep size,

meaning that the surface and wake elements can be sized separately. This capability is important when modelling rotating wings that have changing ratios along the radius of rotation.



(a) Lift coefficient response to sharp-edged gust.

(b) Percent error relative to Küssner Function.

Figure 4.9: DDE prediction of lift response of a 2D wing section as it encounters a sharp-edged gust.

The solution does, however, have a dependency on the number of chordwise elements. The results in Figure 4.9b indicate that an increase in the number of chordwise elements, M , results in a reduced error when compared to the analytical solution regardless of the timestep size or $\Delta x_w/\Delta x_c$ ratio. An increase in the chordwise resolution results in an increase in flow tangency conditions in the streamwise direction, meaning a sufficient resolution is necessary for good agreement with the Küssner function. As the chordwise strength across a single DDE is represented with a second-order polynomial, there is a limit to the complexity possible in the chordwise circulation when using a low chordwise element count.

As a final means of verifying the unsteady force predictions due to gusts, the transient lift coefficient of a finite wing passing through a gust was predicted and compared to results presented by Wang et al. (61). The rectangular wing has an aspect ratio of 6.68 and begins at a lift coefficient of zero, passing through a 1-cosine gust with an amplitude of 0.1% of the freestream velocity and a length of four chords (61). The resulting transient lift response predicted by the DDE method is plotted in Fig. 4.10 for

$\Delta x_w/\Delta x_c$ ratios of 0.96 and 3.8, alongside predictions from Wang et al. (61) using a UVLM-based method and the commercial aeroelastic code, ZAERO. The results produced by the DDE method agree well with those of ZAERO across the entire response, agreeing to within 2% and 6% for the $\Delta x_w/\Delta x_c$ ratios of 0.96 and 3.8 respectively. The maximum lift coefficient predicted by the UVLM-based method is approximately 4% higher than that of ZAERO and the DDE method, though all methods agree well across the majority of the response. As with the sharp-edged gust validation, the $\Delta x_w/\Delta x_c$ ratio has little impact on the DDE results, given that the two ratios in Fig. 4.10 vary by a factor of approximately four.

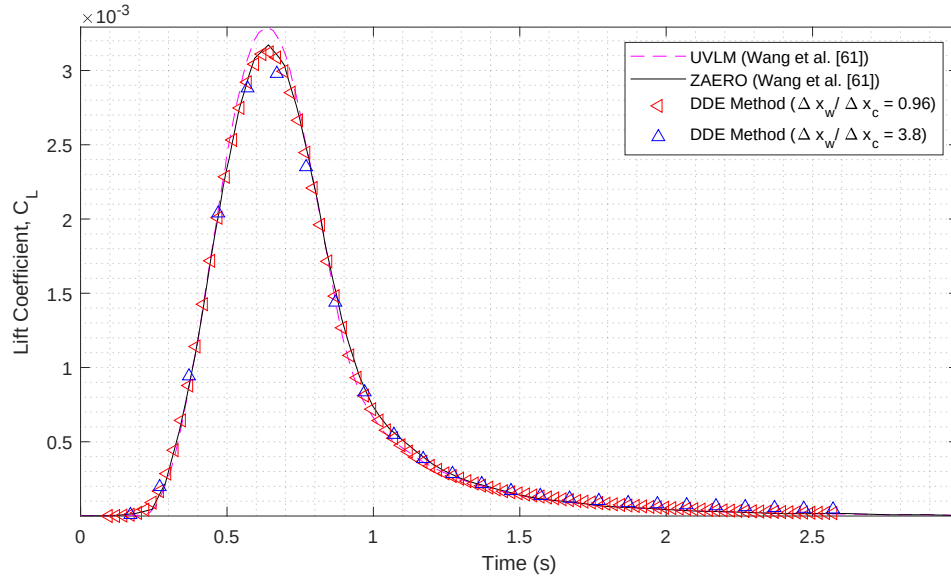


Figure 4.10: Lift coefficient response to a 1-cosine gust (61).

4.3 Time-Averaged Rotor Thrust Predictions

For this validation study, experimental data was obtained and used to validate the time-averaged unsteady thrust predictions of the DDE method. Time-averaged rotor thrust loads were determined for several different forward speeds and tip-path plane angles, both experimentally and through using the DDE method. The experimental data was obtained through Ryerson University's (RU) subsonic wind-tunnel and its rotor-test stand.

Description of Experimental Setup and Conventions

A planform of the wind-tunnel is shown in Fig. 4.11. It is a closed-return tunnel with a test section of length 1.5 m and a cross-sectional size of 0.914 m x 0.914 m. For the purposes of this study, airspeeds ranging from 5 – 25 m/s were reached in the test section. The corresponding turbulence intensities at these speeds are approximately 0.2-0.3% as reported by Kolaei et al. (62). This reference also lists further details on the wind-tunnel, experimental test procedures, and the analysis used to determine the experimental uncertainty associated with the thrust measurements reported in this study.

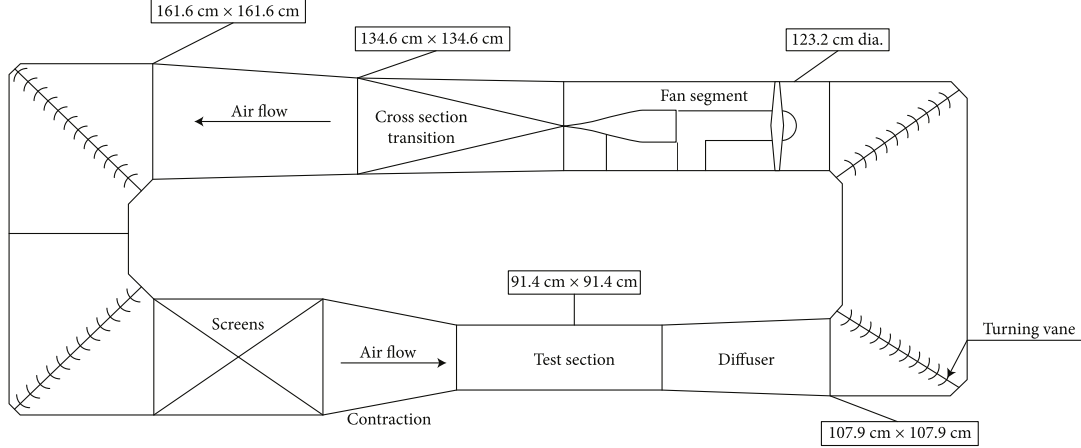


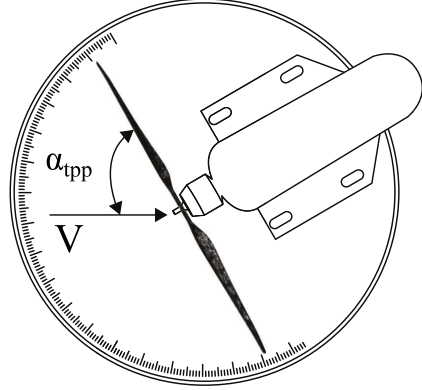
Figure 4.11: Ryerson University's subsonic wind-tunnel (62).

The rotor-test stand was mounted on a turntable in the test section of the wind-tunnel as shown in Fig. 4.12a, thus enabling tip-path plane angles ranging from 90° to -90° . The definition of the tip-path plane angle, α_{tpp} , is shown in Fig. 4.12b. For example, a tip-path plane angle of 0° represents inflow parallel to the rotor plane.

This study distinguishes between propeller and rotor conventions for advance ratio and thrust co-



(a) The rotor-test stand.



(b) Definition of tip-path plane angle (62).

Figure 4.12: The rotor-test stand configuration and angle of attack convention.

efficient. Propeller convention is used only in this section, for any comparison with data listed in the propeller-performance database published by the Applied Aerodynamics Lab of the University of Illinois at Urbana Champaign (63). Propeller advance ratio (J) is defined as:

$$J = \frac{U_\infty}{nD} \quad (4.2)$$

where U_∞ is the freestream velocity, n is the rotational speed in revolutions per second, and D is the propeller diameter. Propeller thrust coefficient (C_{T_p}) is defined as:

$$C_{T_p} = \frac{T}{\rho n^2 D^4} \quad (4.3)$$

where T is the thrust force and ρ is the air density. In later sections of this study, results are given in terms of normalized freestream velocity, μ_∞ , rotor thrust coefficient, C_T , and tip-path plane angle. Normalized freestream velocity is defined as the ratio of a rotor's freestream velocity magnitude to its blade tip speed:

$$\mu_\infty = \frac{U_\infty}{\Omega R} \quad (4.4)$$

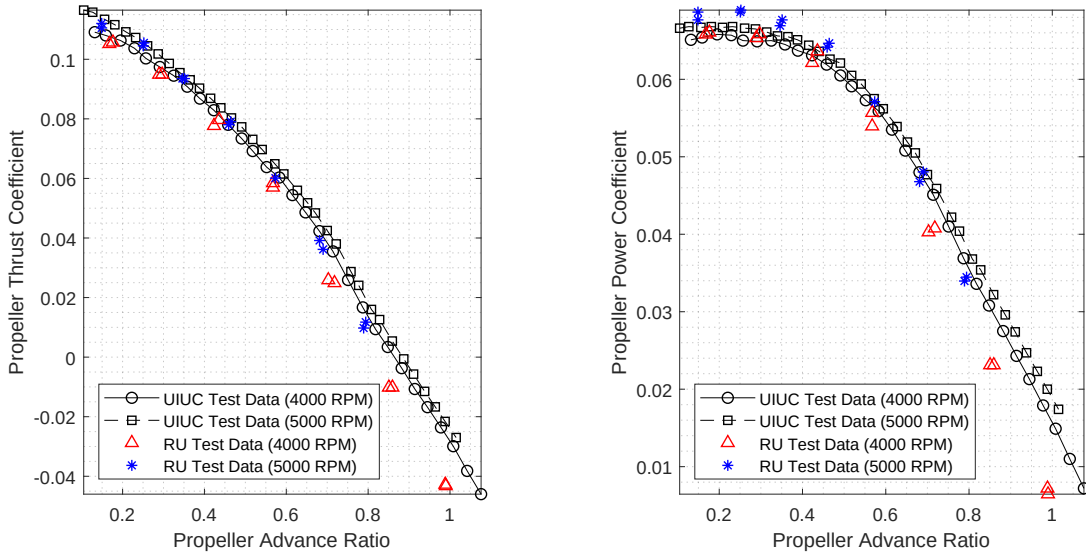
where R is the rotor radius and Ω is the rotational speed in radians per second. Rotor thrust coefficient is defined as:

$$C_T = \frac{T}{\rho A (\Omega R)^2} \quad (4.5)$$

where A is the rotor disk area.

Comparison of RU and UIUC Experimental Rotor Loads

In order to substantiate the experimental results from the RU wind-tunnel and rotor-test stand, a Master Airscrew (MA) 11x7E propeller was evaluated at various propeller advance ratios, and the results were compared with published data taken in the wind-tunnel of the University of Illinois at Urbana Champaign (UIUC) (64; 63). A comparison of thrust and power coefficients at various advance ratios are shown in Fig. 4.13. The RU and UIUC experimental results for both thrust and power coefficients differ less than 3% for advance ratios of approximately 0.5 or less. At higher advance ratios, these deviations increase as the propeller approaches a windmilling state.



(a) Thrust coefficient versus advance ratio.

(b) Power coefficient versus advance ratio.

Figure 4.13: Comparison of UIUC and RU wind-tunnel test data for Master Airscrew 11x7E propeller.

T-MOTOR 18x6.1 Rotor Testing

A T-MOTOR 18x6.1 carbon fibre rotor was evaluated at angles of attack of 0° , 15° , and 90° at various normalized freestream velocities with a fixed rotational speed of 3000 RPM. The geometry of the T-MOTOR 18x6.1 was obtained from a three-dimensional scan presented by Kolaei et al. (62), who provide the chord and twist distributions, shown in Fig. 4.14, as well as section data.

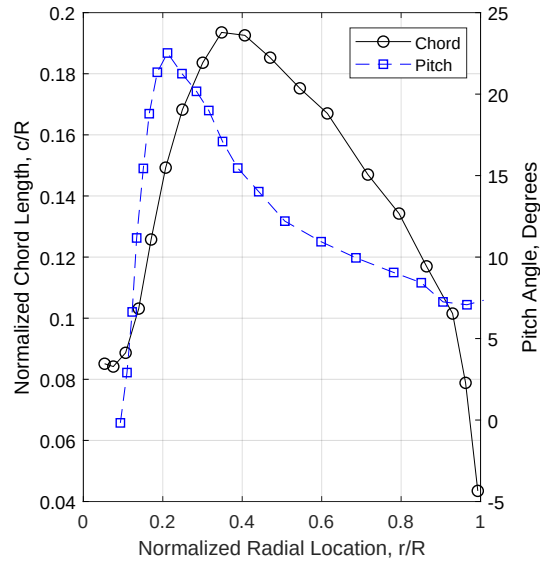


Figure 4.14: T-MOTOR 18x6.1 rotor geometry (62).

4.3.1 Comparison of Results

The T-MOTOR 18x6.1 rotor was evaluated with the DDE method using a relaxed wake model. An example of such an analysis is shown in Fig. 4.15, which depicts a top-down view of the T-MOTOR 18x6.1 in fully-edgewise flow, meaning the inflow is parallel to the rotor plane. The rotor is traversing from right to left while rotating counter-clockwise. The blades are coloured in red, with each blade shedding wake elements of a specified colour, purple or yellow. Though the forces predicted by the DDE method are relatively independent of timestep size, within reason, this analysis was performed using 40 azimuth locations per rotation in order to ensure that the wake was represented at an adequate resolution for the relaxation procedure. The total number of timesteps used within an analysis of this

nature varies, as the convergence of the thrust force requires the starting vortex to convect far enough downstream that its influence on the rotor blades is diminished.

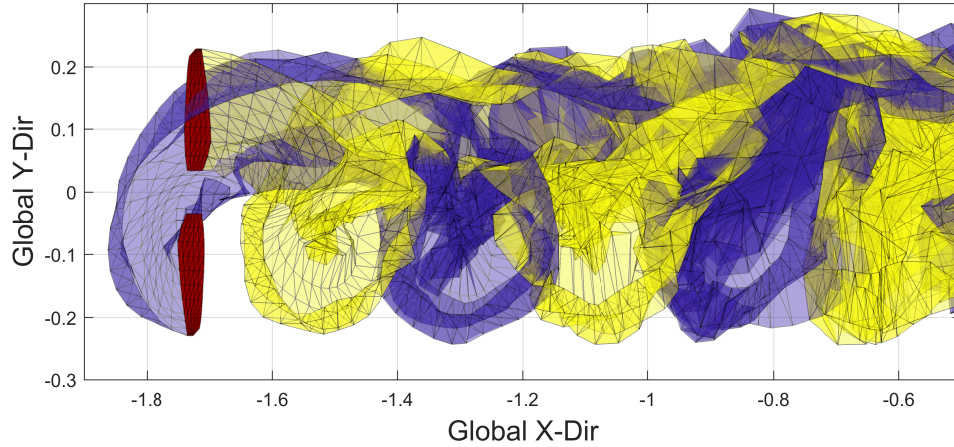


Figure 4.15: Top-down view of the T-MOTOR 18x6.1 in fully-edgewise flow.

Time-averaged thrust coefficients for the T-MOTOR 18x6.1 rotor, as predicted by the DDE method, are provided in Fig. 4.16 for tip-path plane angles $\alpha_{tpp} = 0^\circ$, 15° , and 90° . These are plotted alongside the experimental measurements obtained using the RU rotor-test stand, with the error bars defined according to the uncertainty analysis conducted by Kolaei et al. (62). The percent deviation in thrust between the DDE method and the experimental data is less than 10% for the forward flight cases at $\alpha_{tpp} = 0^\circ$ and 15° across the range of normalized freestream velocities. The largest percent deviation in reported thrust between the DDE method and the experimental data comes as the rotor approaches a windmilling state in fully axial flow at $\alpha_{tpp} = 90^\circ$, where the thrust loads become relatively small. In general, the thrust loads predicted by the DDE method agree reasonably well with experimental results, even for highly unsteady cases of fully edgewise flight with significant surface-wake interaction.

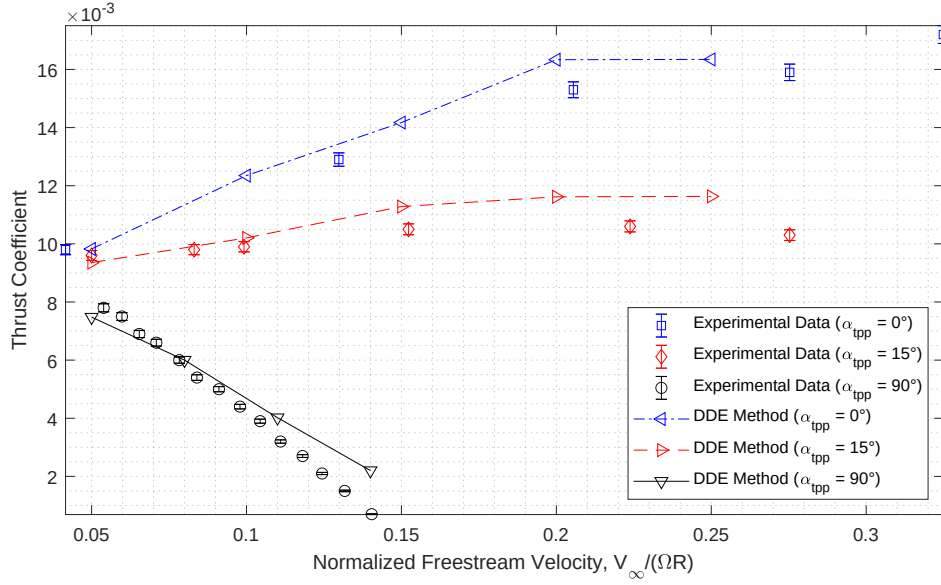


Figure 4.16: Comparison of time-averaged thrust forces for various advance ratios and angles of attack.

4.3.2 Summary

This validation study used the DDE method to predict the time-averaged thrust produced by a rotor operating under various tip-path plane angles and freestream advance ratios. The thrust predictions from the DDE method compare favourably to experimental data obtained through wind-tunnel testing. The predictions fall within 10% of the experimental values, except in the case of the rotor as it approaches a windmilling state in purely axial flow. Furthermore, this study illustrates the DDE method repeatedly being applied to fully-unsteady relaxed-wake analyses with strong surface-wake interactions without suffering numerical instabilities.

4.4 Transient Rotor Thrust Predictions

This validation study uses the DDE method to predict the transient blade sectional thrust produced by a rotor in near-edgewise flight. These predictions are compared to a CFD simulation presented by Misiorowski et al. (12), who analyzed the system using a Navier-Stokes solver with a detached-eddy simulation model. The purpose of this study is to validate the DDE method for the prediction of unsteady transient rotor thrust loads, which forms the basis of a sample application presented in Section 5.2.

For this validation study, an APC 12x5.5MR rotor was modelled at a tip-path plane angle of $\alpha_{tpp} = 5^\circ$ and a normalized freestream velocity of $\mu_\infty = 0.162$, following the geometry and operating conditions specified by Misiorowski et al. (12). A relaxed wake model was used, with a timestep size corresponding to 40 azimuth locations per revolution. The rotor was modelled for five full rotations, which allowed the starting vortex to convect far enough downstream to minimize its influence on the resulting thrust loads, which were taken from the final full rotation. The point of comparison for this validation study is the sectional thrust coefficient of a single blade of the rotor, $dC_T/d(r/R)$, which was determined by non-dimensionalizing the sectional blade thrust values using Eq. 4.5, and differentiating with the non-dimensionalized spans of the sections.

The results from both Misiorowski et al. (12) and those derived using the DDE-based method are shown in Fig. 4.17. For this study and subsequent discussions, azimuth angles $\psi = 0^\circ$ and $\psi = 180^\circ$ correspond to the rotor blades being oriented parallel to the projection of the freestream vector onto the rotor plane, as shown in Fig. 4.17. The blade beginning in the downstream-oriented position at $\psi = 0^\circ$ is advancing through azimuthal locations $\psi = 0^\circ - 180^\circ$, and retreating through $\psi = 180^\circ - 360^\circ$. The opposite is true of the other blade.

The time required for the CFD prediction presented in Fig. 4.17a is unknown, though Misiorowski et al. (12) noted that 512 2.6 GHz Intel Xeon E5-2650 processors were utilized, along with 256 GB of system memory for the grid generation. The DDE prediction was performed in approximately 25 minutes, utilizing a single 3.7 GHz core from an Intel i7-8700k processor and 16 GB of system memory.

The CFD and DDE method predictions show the equivalent minimum and maximum sectional thrust coefficients of approximately 0 and 0.021, respectively. The maximum sectional thrusts occur near $0.75 r/R$ in approximately the same azimuthal location of $\psi = 120^\circ$. On the retreating side, near-zero

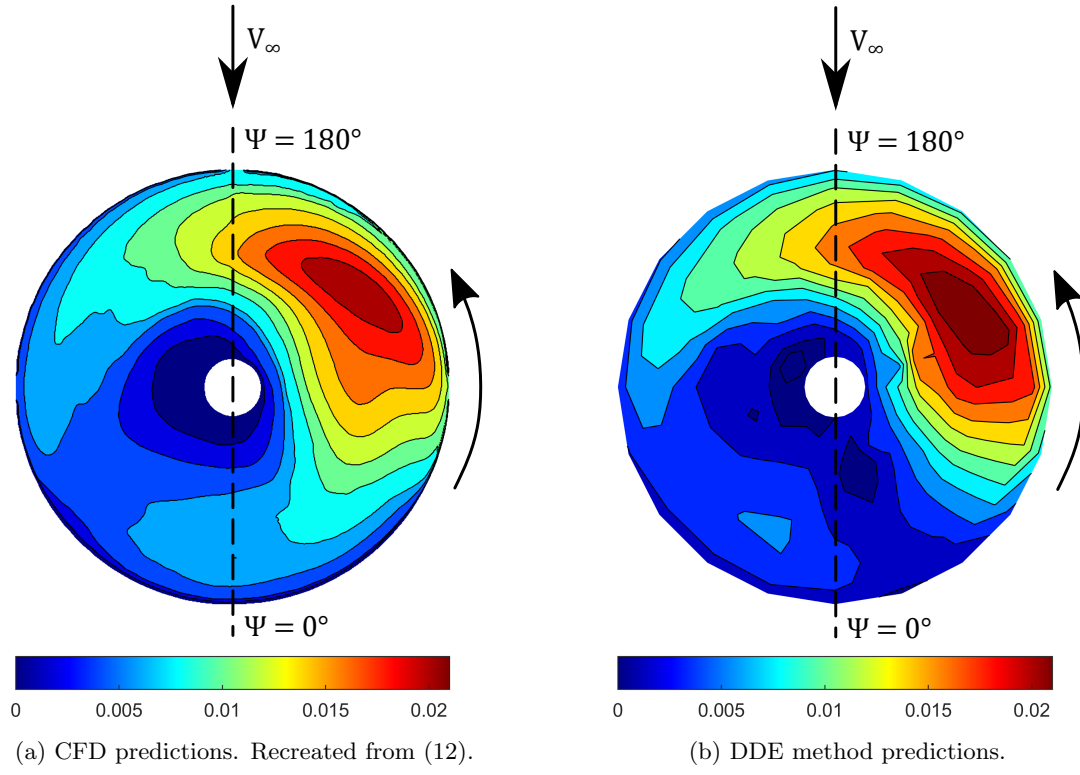


Figure 4.17: Sectional thrust coefficient $dC_T/d(r/R)$ predictions for $\alpha_{tpp} = 5^\circ$, $\mu_\infty = 0.162$.

sectional thrust coefficients at the blade root are predicted by both methods, with the predictions agreeing quite well up to $\psi = 320^\circ$. Appreciable differences exist near $\psi = 0^\circ$, where the blade is pointed downstream. At this azimuth location, the blade is engulfed by the rotor wake while producing comparatively low amounts of thrust. The absence of viscous effects within the DDE-based prediction likely contributed to the discrepancies in this region. As a whole, however, the DDE and CFD-based sectional thrust predictions presented in Fig. 4.17 agree well in both magnitude and trend.

4.5 Summary

The DDE method was applied to four different validation studies that ultimately demonstrated its suitability for the transient analysis of rotors operating under highly-unsteady conditions. The first validation study evaluated planar elliptical wings, where the impacts of panel density were quantified by observing changes in the predicted span efficiency factor. The panel density study results agreed with the analytical solution, showing asymptotic convergence to $e = 1$ for an increasing number of chordwise and spanwise elements for an elliptical wing with a straight trailing edge. A higher span efficiency factor was predicted for an elliptical wing with a forward-swept trailing edge, as expected from the theory.

The second validation study used the DDE method to predict the transient lift due to gusts for both a flat plate airfoil and a finite wing, with the purpose of evaluating the apparent mass approximation and the modelling of the shed circulation in the wake. The first analysis modelled a two-dimensional flat plate encountering a sharp-edged gust. The results were compared with the analytical results given by the Küssner function, and showed independence of timestep size. The error relative to the analytical solution was inversely proportional to the number of chordwise elements used, with ten chordwise elements resulting in a maximum percent error of 9%. This study indicates that the apparent mass approximation can be applied to rotors in edgewise flight without numerical instabilities, and with reasonable accuracy provided an adequate number of chordwise elements are used. These results were further supported by the analysis of a finite rectangular wing passing through a 1-cosine gust, with the DDE-based predictions agreeing with those found in the literature.

The third validation study compared time-averaged rotor thrust predictions from the DDE method to wind-tunnel experiments of a rotor operating under a range of different tip-path plane angles. Good agreement was shown between the predicted and experimentally-obtained rotor thrust loads across most

cases. The maximum deviation occurred as the rotor approached a windmilling state in purely axial flow, where the thrust produced by the rotor approaches zero.

In the fourth study, the transient sectional thrust coefficients of a rotor in near-edgewise flight were found using the DDE method and compared to CFD results. Both predictions agreed well in magnitude and azimuth location, predicting the same maximum and minimum sectional thrust values and their respective radial and azimuthal locations. The largest deviation between the methods occurred near an azimuth location of $\psi = 0$, where the blade is pointed downstream. Overall, this comparison to a high-fidelity CFD-based simulation indicates that the DDE method is well-suited for modelling rotors operating under unsteady conditions.

Chapter 5

Sample Applications

Two separate studies are presented in this chapter that focus on the performance of small, rigid rotors operating under unsteady conditions. These studies offer examples of well-suited applications for the DDE method. The first study models the impact of rolling maneuvers on the time-averaged thrust and moments produced by a rigid rotor, as a way of quantifying the effect of rolling maneuvers on a rotor during the flight of a quadcopter. The second study uses the DDE method to provide a detailed analysis of the transient thrust response of two different rotors in partial and fully edgewise flow, modelling the unsteadiness of the advancing and retreating blades and observing the trends that occur every rotation. Both of the sample applications presented in this chapter represent analyses that are challenging or infeasible using existing UVLM-based or DVE-based potential flow methods.

5.1 Rotor Response to Rolling

A fast, low-fidelity simulation of a rotor may simplify the flow conditions by assuming that the velocity across the rotor disk is constant and equal to the value at the rotor hub. This assumption is reasonable for multirotor vehicles in steady-level flight, with examples provided by Tsaltas (65) and Tsaltas and Bramesfeld (66). However, a rotor is subjected to highly-varying kinematic velocities as the vehicle maneuvers in-flight. These maneuvers change the magnitude and alter the distribution of the kinematic velocity across the rotor disk, impacting the thrust and moments it produces compared to steady-level

flight conditions. As a result, the validity of a constant kinematic velocity assumption across the rotor disk is questionable when analyzing a maneuvering rotor.

This study investigates the extent to which the non-uniform kinematic velocity distribution affects the predicted time-averaged rotor loads, as a means of testing the validity of the constant kinematic velocity assumption. This is accomplished by analyzing a T-MOTOR 18x6.1 rotor, introduced in Section 4.3, as it rolls at various rates and forward speeds. The time-averaged rotor loads during the roll maneuver are compared to the steady-level flight loads, with the differences indicating the extent to which the change in kinematic velocity across the rotor disk impacts the predicted loads.

Analyzing the change in the thrust and moments produced by a rotor as a result of a flight maneuver is a highly unsteady problem, and thus a well-suited application of the DDE method. Though a rotor disk has a constant kinematic velocity distribution in steady-level flight, the rotor blades themselves are operating under unsteady conditions due to advancing and retreating blade effects. With the addition of a roll maneuver, this unsteadiness is compounded by changing kinematic velocities across the rotor disk. The DDE method can quantify the extent to which the roll maneuvers impact the resulting rotor loads due to this additional complexity. This is done by simulating the rotor in steady-level edgewise flight until the solution converges, at which point a rolling maneuver commences at a constant roll rate. The analysis concludes when the solution has converged again. Due to the low computational expense of the DDE method, several operating conditions and roll rates were analyzed, and the results are compared in this study.

Quadrotor vehicles have two primary configurations, illustrated in Fig. 5.1. The first is known as a square configuration, featuring two leading rotors and two trailing rotors. The second is known as a diamond configuration, with a single leading and trailing rotor and two outboard rotors. A detailed aerodynamic analysis of these two configurations is given by Barcelos et al. (54), which indicates highly complex aerodynamic effects primarily due to the proximity of the rotors.

Two variations of rolling maneuvers were analyzed for this study and discussed herein. The first variation features the rolling motion centred about the longitudinal axis of the rotor, as encountered by the leading and trailing rotors of the diamond configuration during a rolling maneuver. In this case, the roll axis passes through the hubs of these leading and trailing rotors. The second variation of the rolling maneuver is centred on an axis offset from the longitudinal axis of the rotor, as encountered by all rotors of the square configuration and the outboard rotors on the diamond configuration, during a

rolling maneuver. In this case, with the roll axis offset from the rotor hub, this maneuver features both a rotation and translation of the rotor.

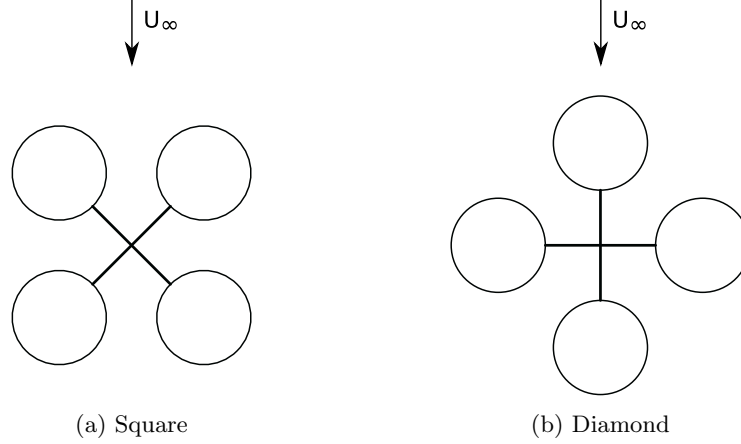


Figure 5.1: The two primary configurations for quadrotor vehicles in forward flight.

5.1.1 Rolling About the Longitudinal Axis of the Rotor

The leading and trailing rotors of the diamond configuration illustrated in Fig. 5.1b roll about their longitudinal axis as the vehicle rolls because the rotor hubs lay on the vehicle centreline. To investigate the effects on a rotor performing such a roll maneuver, a single T-MOTOR 18x6.1 rotor was analyzed in fully edgewise flow while rolling about its longitudinal axis, subjected to several different roll rates and freestream advance ratios. A top-down view of the rotor is shown in Figure 5.2, rotating counterclockwise and at a tip-path plane angle of $\alpha_{tpp} = 0$. The rolling maneuvers were conducted about the x-axis, and the direction of a positive roll rate is given by the right-hand rule.

An example of the kinematic velocity distribution resulting from a positive roll maneuver, as seen from the $y - z$ plane, is illustrated in Fig. 5.3. Rolling about the longitudinal axis of the rotor results in one half of the rotor disk experiencing an effective increase in angle of attack, and the other experiencing an effective reduction in angle of attack. For the purposes of this study, as the rotor is rotating counterclockwise and the freestream velocity is aligned with the positive x -direction, a positive roll rate results in an effective reduction in angle of attack for the advancing blade and an effective increase for the retreating blade. The opposite is true of a negative roll rate.

The results of this study are presented in Fig. 5.4, which plot the results of seven different roll rates

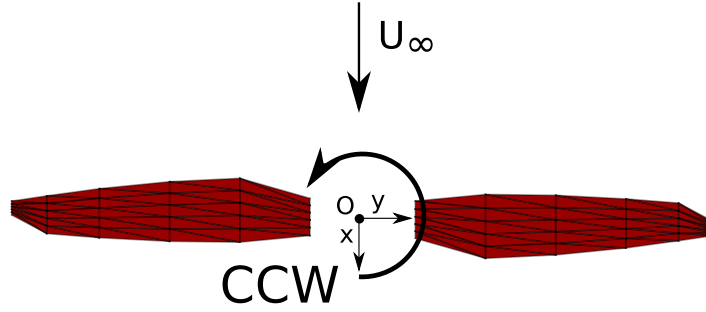


Figure 5.2: A top-down view of a rotor in edgewise flight with the roll axis coincident with the rotor hub.

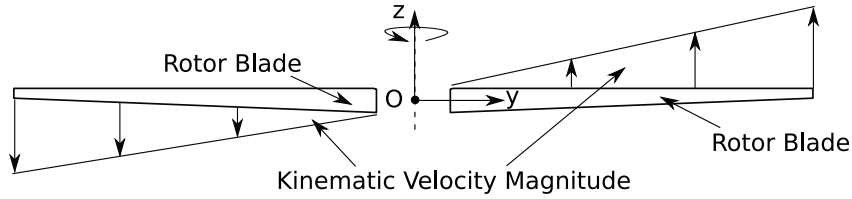
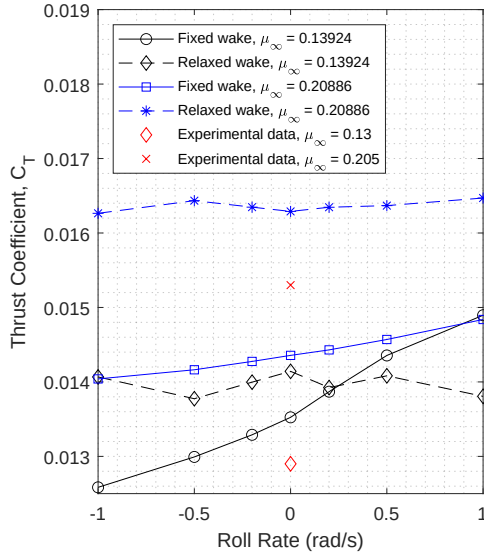
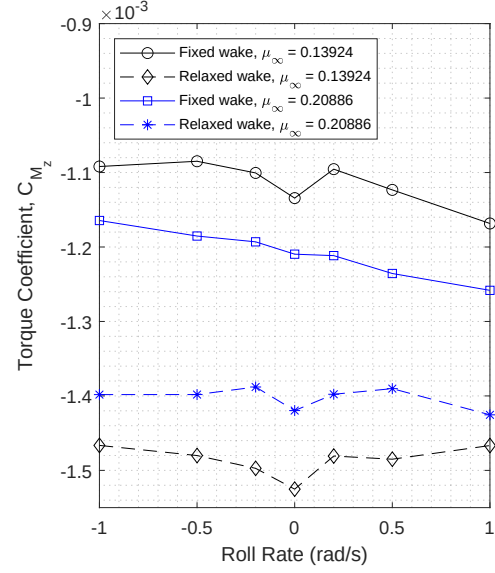


Figure 5.3: The kinematic velocity distribution across a rotor disk due to the roll axis being coincident with the rotor hub.

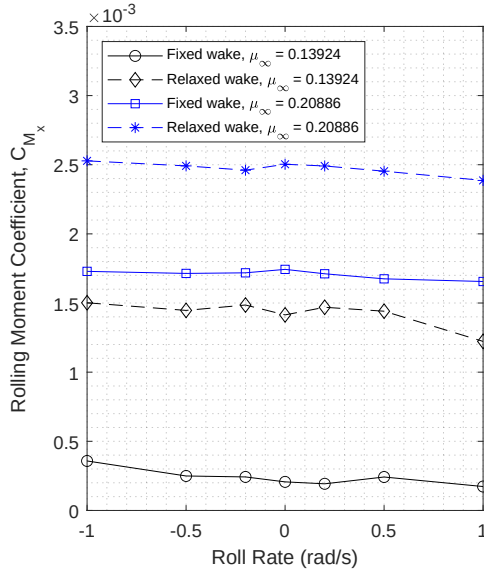
from -1 to 1 rad/s against the resulting time-averaged thrust, torque, and rolling and pitching moment coefficients for two different freestream advance ratios. For additional context, two experimental thrust data points taken from Fig. 4.16 are included in Fig. 5.4a at a roll rate of zero. The differences between the predicted forces and moments produced by the DDE method based on the wake model are immediately apparent. The fixed and relaxed wake thrust coefficients differ by 15-20% across the range of roll rates at $\mu_\infty = 0.209$. The fixed wake thrust coefficients trend with the roll rate at both freestream advance ratios, whereas the relaxed wake thrust coefficients are relatively constant across the range of roll rates. The differences between the fixed and relaxed wake results are present in the moment coefficients as well, such as with the rolling moment coefficients which differ by up to 75% relative to the relaxed wake results. These large differences are due to the rotor being in edgewise flight. When using a fixed wake model, the rotor blades are constantly immersed in the wake, as shown in Fig. 5.5a. With the relaxed wake model, the wake is driven away from the rotor disk as shown in Fig. 5.5b, which reduces the impact of the wake on the rotor blades and on the resulting forces and moments. The relaxed wake results will be the focus of the following discussion due to the significant impact of the wake shape on the resulting force and moment predictions.



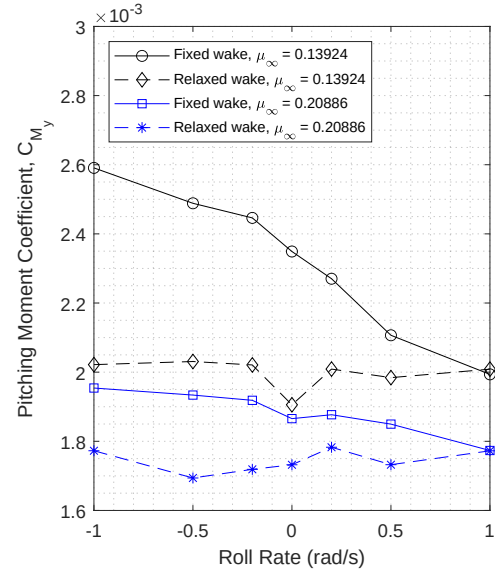
(a) Thrust coefficient versus roll rate.



(b) Torque coefficient versus roll rate.



(c) Rolling moment coefficient versus roll rate.



(d) Pitching moment coefficient versus roll rate.

Figure 5.4: Effects of roll rate for a T-MOTOR 18x6.1 rolling about its longitudinal axis.

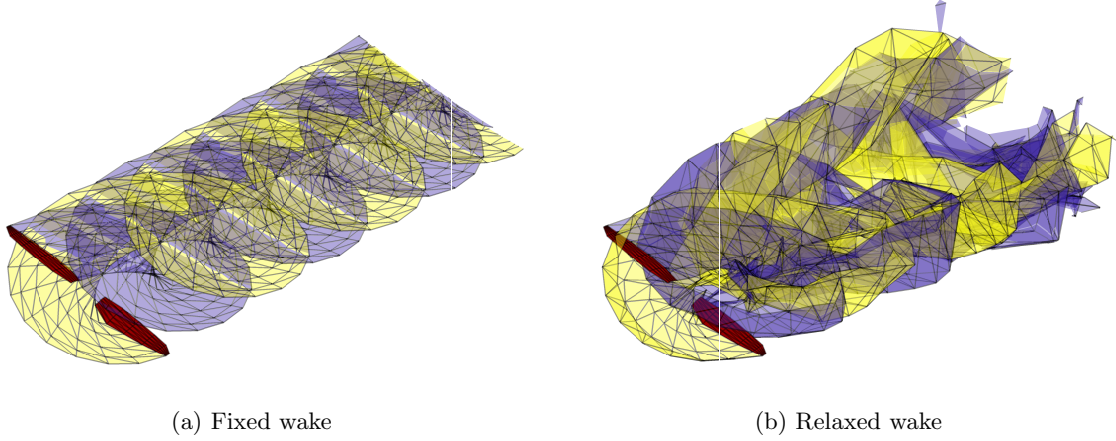


Figure 5.5: An example of a fixed and relaxed wake analysis for a rotor in edgewise flight ($\mu_\infty = 0.209$).

The time-averaged thrust coefficients produced by the relaxed wake analyses are relatively unchanging with roll rate, remaining within 3% of the steady-state thrust values for both freestream advance ratios. Similarly, the torque and pitching moment coefficients vary in the order of 2-3% of their steady-state values without a clear trend. For a freestream advance ratio of $\mu_\infty = 0.139$, the rolling moment coefficient varies from +5% to -14% relative to the steady-level case, for roll rates of -1 and 1 rad/s, respectively. This spread changes to +2% and -5% at a freestream advance ratio of $\mu_\infty = 0.209$, indicating that the rolling maneuver has a greater effect on the rolling moment coefficient at lower freestream advance ratios, where the roll rate is proportionally larger in comparison with the forward speed of the rotor disk. These results suggest that the change in roll moment due to a roll maneuver along the longitudinal axis of a rotor is not insignificant. This effect is likely an important consideration when analyzing the flight dynamics of a diamond-configuration quadrotor.

5.1.2 Rolling About an Offset Axis

To simulate the impact of a rolling maneuver on the rotors of a square-configuration quadrotor, or the outboard rotors of a diamond-configuration quadrotor, a second study was performed which rotates the rotor about an axis offset from the longitudinal axis. Figure 5.6 illustrates a top-down view of the system, with the origin being offset 0.25 m from the x-axis. As with the previous study, the rotor rotates counterclockwise and is subjected to fully-edgewise flow. The primary difference is that with an offset axis, a rolling maneuver now changes the axial velocity across the entire rotor in addition to the

rolling motion present in the previous study. A positive rolling maneuver about the x-axis in Fig. 5.6 will, for example, add additional axial velocity through the entire rotor plane due to this offset. This is illustrated in Fig. 5.7, which shows the kinematic velocity distribution across a rotor disk for a positive rolling motion about the offset axis.

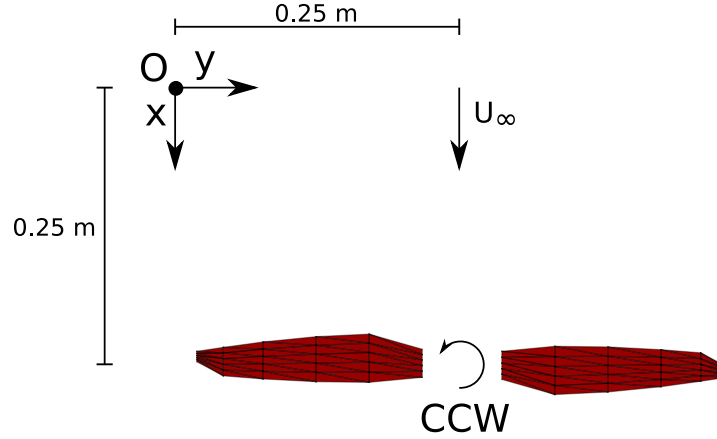


Figure 5.6: A top-down view of a rotor in edgewise flight with the roll axis offset from the rotor hub.

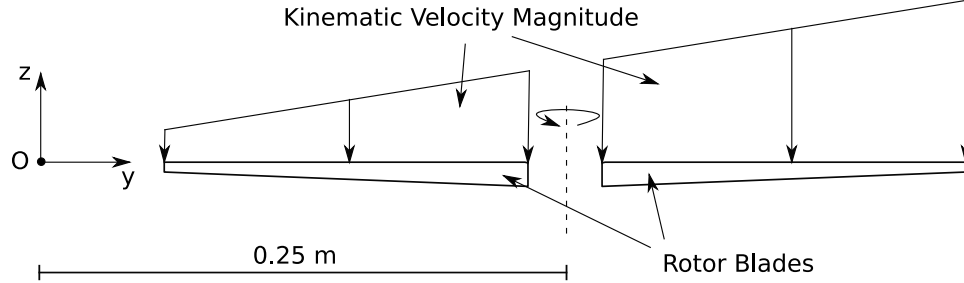


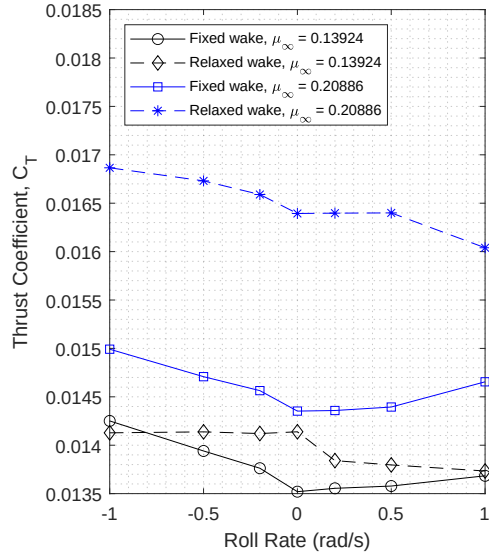
Figure 5.7: The kinematic velocity distribution across a rotor disk due to the roll axis being offset from the rotor hub.

The results from this study are presented in Fig. 5.8, which again presents a large disparity between the fixed and relaxed wake results due to the fact that the rotor is in fully edgewise flight. However, unlike the previous study, the relaxed wake thrust coefficients in Fig. 5.8a change with roll rate. As the roll rate increases from negative to positive, the thrust coefficient declines by approximately 6% for the $\mu_\infty = 0.209$ case, and approximately 3% for the $\mu_\infty = 0.139$ case. This drop in thrust is due to the changing axial flow through the rotor plane, which results from a rolling maneuver about an axis offset from the rotor hub. These results are supported by Fig. 4.16, which shows that the thrust coefficient

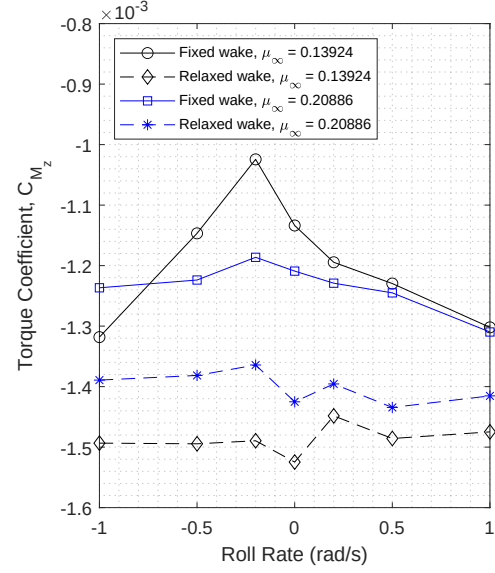
for the T-MOTOR 18x6.1 decreases for a given freestream advance ratio as the axial flow component is increased. Both the torque and rolling moments predicted by the relaxed wake analyses are relatively constant across the range of roll rates. The relaxed wake pitching moment coefficients, presented in Fig. 5.8d, decrease as the rolling moment increases, as a result of the reduced thrust output of the rotor.

5.1.3 Summary

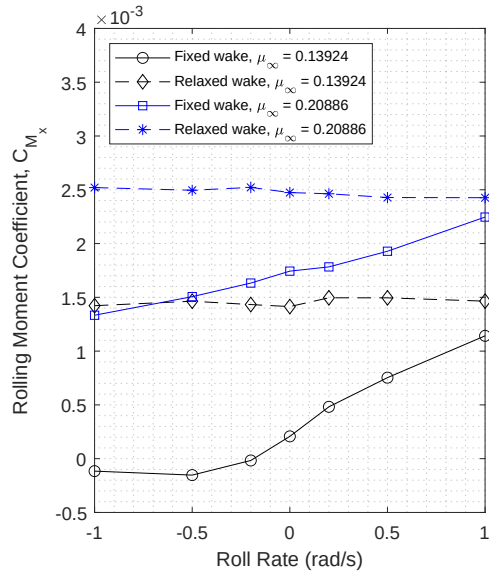
A T-MOTOR 18x6.1 was subjected to rolling maneuvers using the DDE method to investigate the effects on the rotor thrust and moment coefficients. Due to the rotor being analyzed in edgewise flight, large variations were seen between the fixed and relaxed wake thrust and moment coefficients, highlighting the importance of a relaxed wake model when analyzing such systems. With the rotor rolling about its longitudinal axis, there was little variation in the reported thrust, torque, and pitching moment coefficients for the relaxed wake results. The roll moment coefficient varied in magnitude by 2-14% relative to the steady-state value, indicating that a rolling maneuver can impact the roll moment in a significant way. With the rotor rotating about an axis offset from the longitudinal axis, the additional changes in axial velocity across the rotor due to this offset impacted the thrust and pitching moment coefficients, altering them in the order of 5-10% across the range of roll rates. These results indicate that the change in axial velocity due to a rotation about an offset axis does play a quantifiable role in the thrust and pitching moments produced by the rotor, and may have to be considered when modelling the flight dynamics of multirotor vehicles.



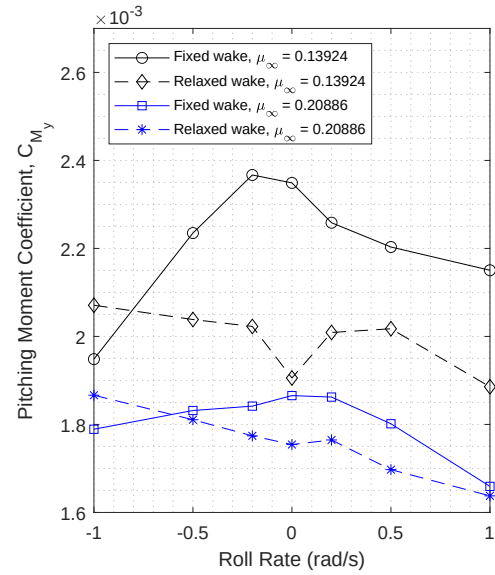
(a) Thrust coefficient versus roll rate.



(b) Power coefficient versus roll rate.



(c) Rolling moment coefficient versus roll rate.



(d) Pitching moment coefficient versus roll rate.

Figure 5.8: Effects of roll rate for a T-MOTOR 18x6.1 rolling about an offset point.

5.2 Transient Unsteady Rotor Analysis

The thrust acting on a rotor in forward flight may vary with time due to advancing and retreating blade effects acting on the individual rotor blades. While conventional helicopters compensate for these effects with hinged blades, small multirotor vehicles typically use rigid rotors. Without compensation, the thrust output of the rotor is cyclical, with the amplitude being linked to the forward speed and angle of the vehicle, as well as the rotational velocity of the rotor. Transient thrust has potential implications for both the control systems and the structural design of the rotor blades and central chassis due to fatigue. Despite these potential implications, quantification of the time-dependent thrust on rigid rotors at various tip-plane angles of attack is notably lacking from the literature. The purpose of this study is to investigate and quantify the transient thrust response of small rigid rotors in forward flight under various operating conditions through numerical predictions of unsteady blade and wake effects.

5.2.1 Transient Analysis of Rigid Rotors in Forward Flight

Using the DDE method, the T-MOTOR 18x6.1 and APC 12x5.5MR rotors introduced in Sections 4.3 and 4.4, respectively, were analyzed at various normalized freestream velocities at tip-path plane angles of $\alpha_{tpp} = 0^\circ$ and 15° , resulting in four unique datasets. An overview of the time-averaged thrust coefficients plotted against normalized freestream velocity is given in Fig. 5.9. For both rotors, the rate of change of the thrust coefficient with normalized freestream velocity is higher at a tip-path plane angle of $\alpha_{tpp} = 0^\circ$. For both tip-path plane angles, the APC rotor has a higher predicted thrust coefficient than the T-MOTOR across the range of normalized freestream velocities.

The thrust coefficients versus azimuth location for each of these four datasets are plotted in Fig. 5.10. In most cases, the peak rotor thrust appears around $\psi = 90 - 110^\circ$ and $270 - 290^\circ$, which corresponds to a blade passing the region of maximum advancement. The onset of this peak is delayed to higher azimuth angles as the normalized freestream velocity is decreased.

Most of the cases shown in Fig. 5.10, except for those shown in Fig. 5.10d, have small secondary peaks occurring roughly $20 - 50^\circ$ after the primary peaks. To explain the cause of these peaks, Fig. 5.11 illustrates the APC rotor at two azimuth locations corresponding to a primary and secondary peak for the case of $\mu_\infty = 0.25$ in Fig. 5.10b. Figure 5.11a is a top-down snapshot of the APC rotor during the primary peak at an azimuth location $\psi = 108^\circ$. At this azimuth location, the shed vortex of a previously

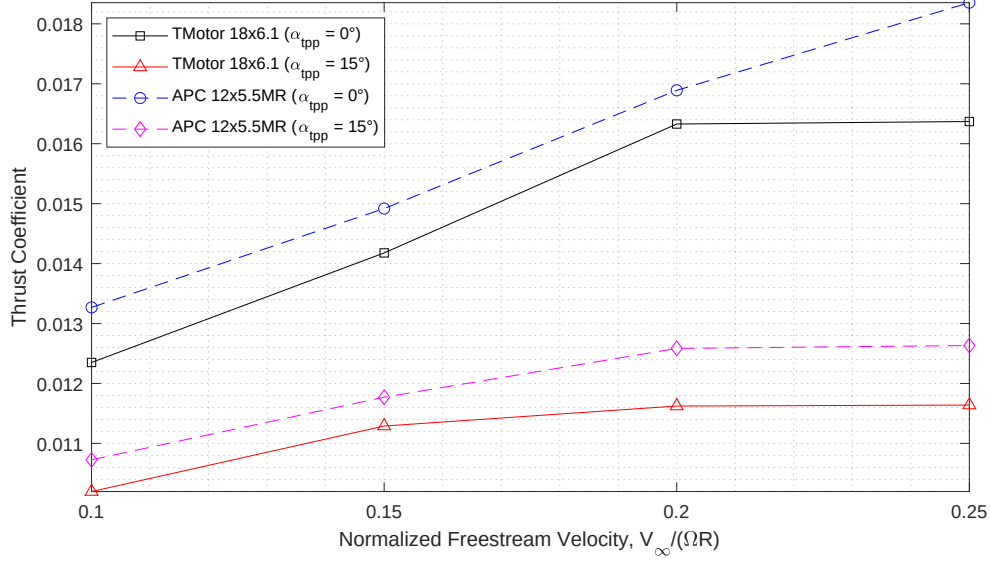
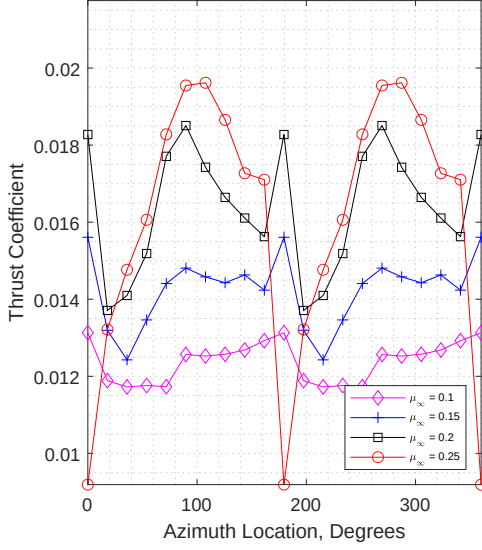


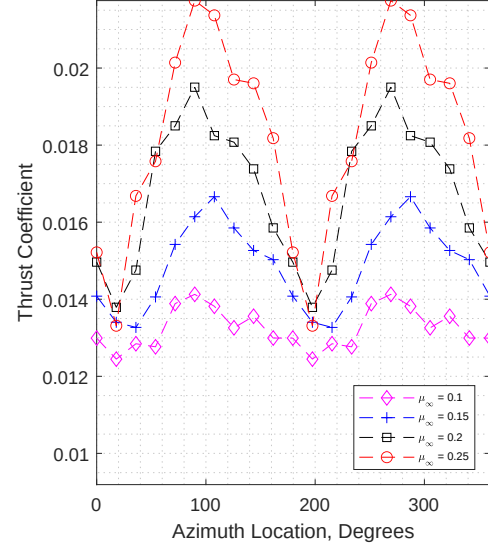
Figure 5.9: Thrust coefficient vs. normalized freestream velocity as predicted with the DDE method.

advancing blade passes over the advancing blade. As this tip-vortex moves inboard along the blade, the upwash outside of this tip-vortex increases the effective angle of attack of the blade sections along the outboard portion of the rotor blade. Similarly, at this azimuth location, the retreating blade is outside of the tip-vortex of a previous blade pass, which increases the effective angle of attack of the blade sections along the retreating blade, which are experiencing low relative velocities. The rotor and wake geometry corresponding to the secondary peaks of the case of $\mu_\infty = 0.25$ in Fig. 5.10b are displayed in Fig. 5.11b. This figure represents an azimuth location of $\psi = 126^\circ$. The main difference between Figs. 5.11a and 5.11b is that the retreating blade has entered into the tip-vortex of a previous blade pass, indicating that the primary and secondary peaks in the transient thrust for this case could be created by a negative effect acting on the retreating blade, caused by this interaction. As the retreating blade enters further into the downwash region, the thrust output of the rotor collapses, leading to the negative thrust peak at $\psi = 180 - 200^\circ$.

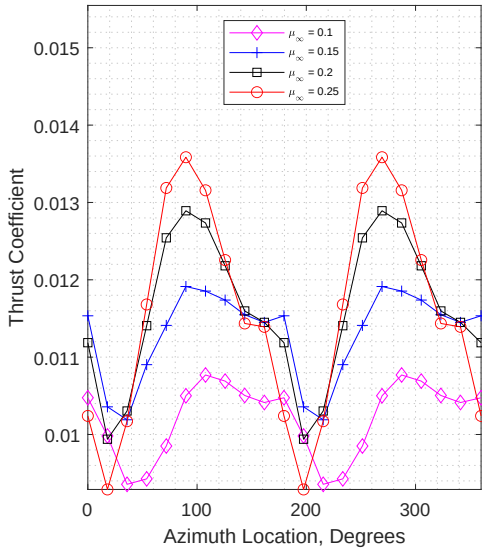
The secondary thrust peaks and troughs for the fully-edgewise T-MOTOR results presented in Fig. 5.10a indicate signs of a tip-vortex interaction occurring at $\psi = 0^\circ$ and 180° , with the effects growing more pronounced as the normalized freestream velocity and subsequent rotor thrust are increased. With



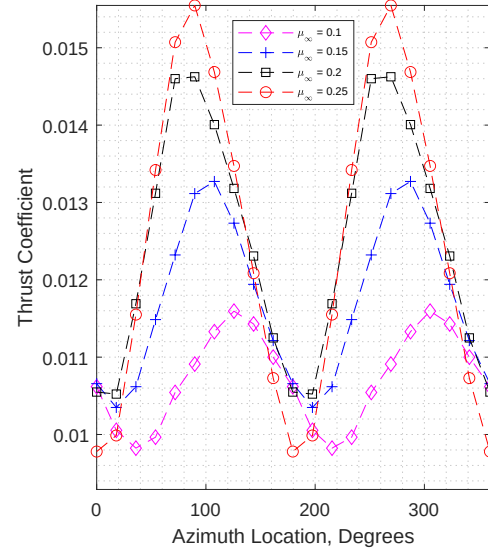
(a) Dataset 1: T-MOTOR 18x6.1 ($\alpha_{tpp} = 0^\circ$)



(b) Dataset 2: APC 12x5.5MR ($\alpha_{tpp} = 0^\circ$)

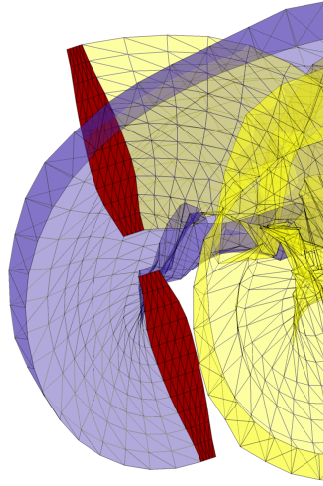


(c) Dataset 3: T-MOTOR 18x6.1 ($\alpha_{tpp} = 15^\circ$)

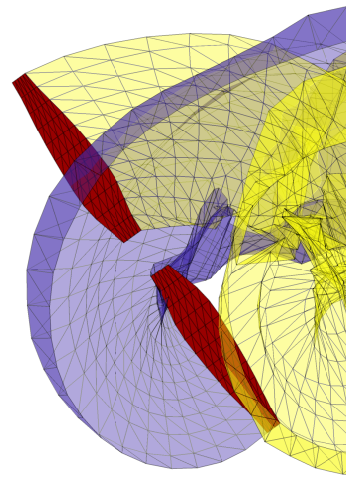


(d) Dataset 4: APC 12x5.5MR ($\alpha_{tpp} = 15^\circ$)

Figure 5.10: Thrust coefficient vs. azimuth location as predicted with the DDE method.



(a) $\psi = 108^\circ$



(b) $\psi = 126^\circ$

Figure 5.11: Rotor and wake visualization of the APC 12x5.5MR ($\alpha_{tpp} = 0^\circ$, $\mu_\infty = 0.25$) as modeled in the DDE method.

the blades pointed up and downstream at both $\psi = 0^\circ$ and 180° , these data points reflect when one of the two blades is pointed downstream and fully-engulfed in the rotor wake, as seen previously with the blade sectional thrust analysis presented in Section 4.4. In this region, the downstream blade is outputting comparatively low amounts of thrust while simultaneously encountering the shed tip vortices from previous blade passes. As the rotor thrust output is increased with larger normalized freestream velocities, the strength of these shed tip vortices increases and their effects on the rotor thrust become more severe. The differences seen between the fully-edgewise T-MOTOR and APC tip-vortex interactions are due to the wake propagation, which defines the location of these shed tip vortices and the resulting severity of their interactions with the rotor blades.

In order to identify further trends in the transient thrust data presented in Fig. 5.10, a normalized thrust coefficient is used:

$$\hat{C}_T = \frac{C_T}{\bar{C}_T} \quad (5.1)$$

where $\bar{C}_T$ represents the mean thrust coefficient as shown in Fig. 5.9. The amplitude of oscillation of this normalized thrust is then:

$$\hat{\lambda} = \frac{1}{2}(\hat{C}_{T,max} - \hat{C}_{T,min}) = \frac{C_{T,max} - C_{T,min}}{2\bar{C}_T} \quad (5.2)$$

Figure 5.12 shows this normalized thrust oscillation amplitude, $\hat{\lambda}$, plotted against the normalized freestream velocity of Eq. 4.4 for all four datasets. As expected, the normalized thrust oscillation amplitude increases with the normalized freestream velocity as advancing and retreating blade effects become increasingly severe. In the extreme, such as the cases at a normalized freestream velocity of $\mu_\infty = 0.25$, the normalized thrust oscillation amplitude of the rotors are in the range of 20%-30% of the mean rotor thrust output, indicating the rotor thrust output varies by $0.4\bar{C}_T$ - $0.6\bar{C}_T$ per revolution. The rotor blade design appears to impact the relationship between the normalized thrust oscillation amplitude and the normalized freestream velocity, as evidenced by the differences between the T-MOTOR and APC results, which become more pronounced with an increasing normalized freestream velocity. Observing these four datasets, there does not appear to be a discernible link between the normalized thrust oscillation amplitude and tip-path plane angles of $\alpha_{tp} = 0^\circ$ and 15° , though the normalized thrust oscillation amplitude will be zero for all normalized freestream velocities at the tip-path plane angle $\alpha_{tp} = 90^\circ$.

Understanding the magnitude of oscillations in the thrust of a rigid rotor provides added context

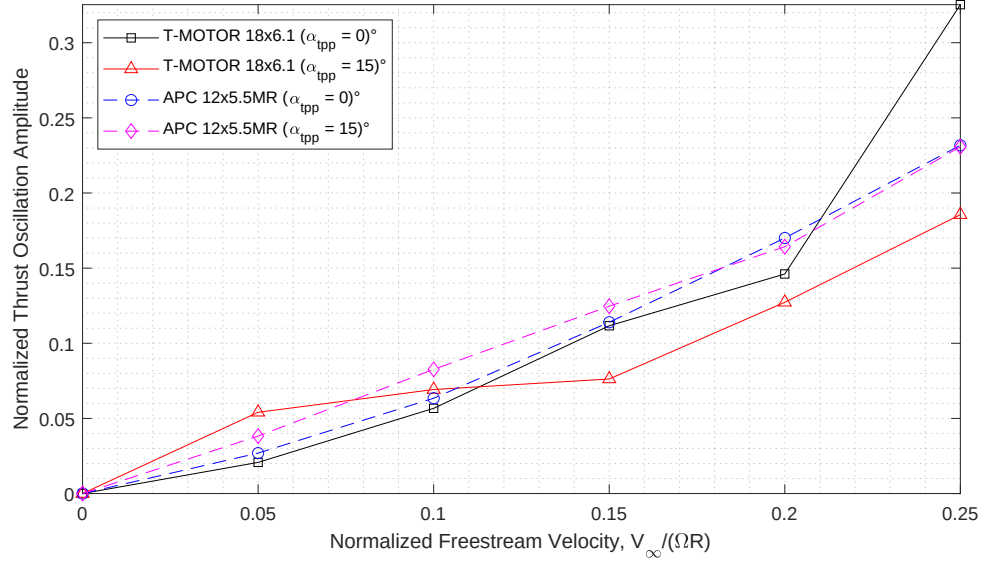
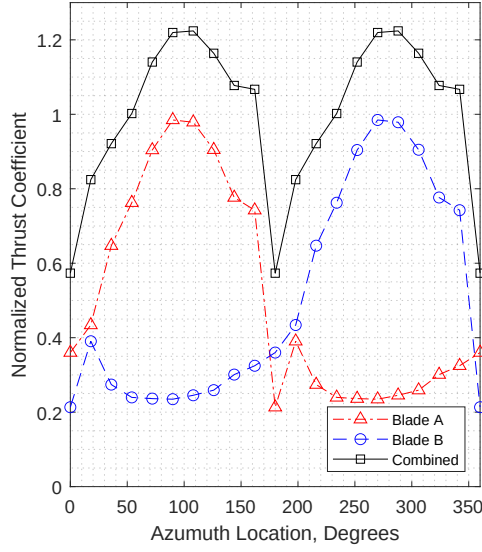


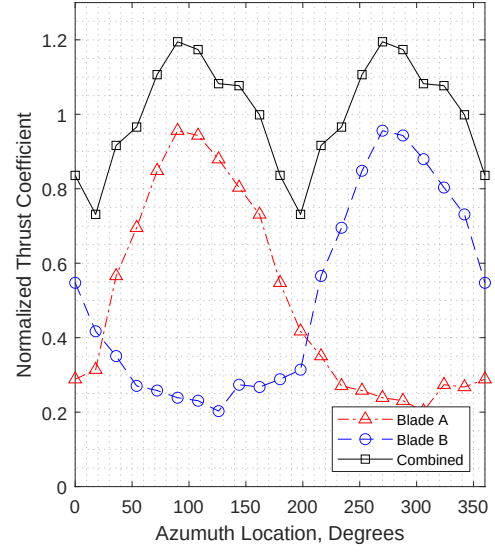
Figure 5.12: Normalized thrust oscillation amplitude vs. normalized freestream velocity as predicted with the DDE method.

for the rotor designer. The total thrust oscillation amplitude of the rotor in Fig. 5.12 only represents the force oscillations of both blades together and does not represent the thrust oscillation amplitude of a single blade. In order to quantify the individual blade contributions to the thrust coefficient, two distinct operating conditions were chosen for closer analysis. Figure 5.13 shows the normalized thrust coefficients of the rotors plotted against azimuth location, including individual blade contributions, for tip-path plane angles of $\alpha_{tpp} = 0^\circ$ and 15° and normalized freestream velocities of $\mu_\infty = 0.25$ and 0.1 , respectively. The thrust coefficients were normalized using Eq. 5.1 to allow a direct comparison.

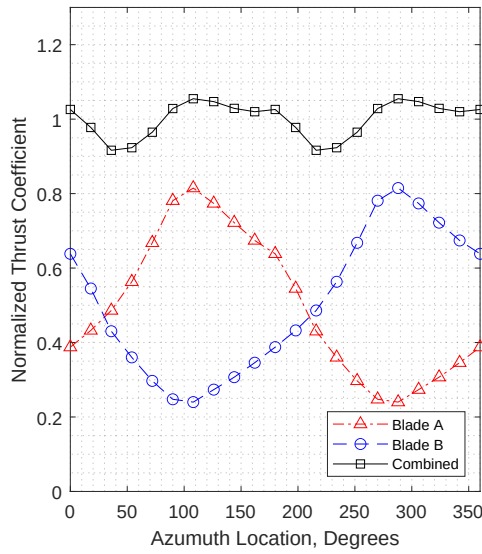
The individual blade contributions for the two fully-edgewise cases in Figs. 5.13a - 5.13b vary by approximately $0.7 - 0.9\bar{C}_T$ over a revolution. In both cases, the peak thrust output of the rotor aligns with the peak thrust output of the advancing blade, which occurs at azimuth locations of approximately 95° for Blade A and 275° for Blade B for both the T-MOTOR and APC rotors under this operating condition. The partially-edgewise cases in Figs. 5.13c - 5.13d both have normalized blade thrust contributions varying by approximately $0.6\bar{C}_T$. Under this specific operating condition, the thrust output of the retreating blade of the T-MOTOR rotor begins to recover faster than the advantage of the advancing blade is lost, leading to a slight double peak in the total thrust output of the rotor.



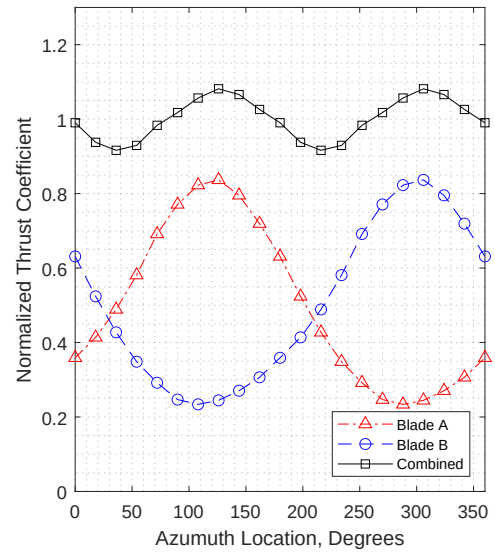
(a) T-MOTOR 18x6.1 ($\alpha_{tpp} = 0^\circ$, $\mu_\infty = 0.25$)



(b) APC 12x5.5MR ($\alpha_{tpp} = 0^\circ$, $\mu_\infty = 0.25$)



(c) T-MOTOR 18x6.1 ($\alpha_{tpp} = 15^\circ$, $\mu_\infty = 0.1$)



(d) APC 12x5.5MR ($\alpha_{tpp} = 15^\circ$, $\mu_\infty = 0.1$)

Figure 5.13: Individual blade contributions to thrust coefficient as predicted with the DDE method.

In order to generalize the results shown in Fig. 5.13, normalized individual blade thrust oscillation amplitudes were found for all datasets. These results are summarized in Fig. 5.14. Even at low normalized freestream velocities, the oscillation amplitudes for a single rotor blade are approximately $0.2\bar{C}_T$, seeing a total variation in thrust output of 40% of the mean rotor thrust. This oscillation amplitude increases to approximately $0.4\bar{C}_T$ as normalized freestream velocity increases, indicating that the thrust output of a single rotor blade may approach the mean thrust output of the entire rotor over one revolution under these conditions. Analysis of this nature should prove useful when considering aeroelastic, aeroacoustic, and structural properties during the design phase of rigid rotors.

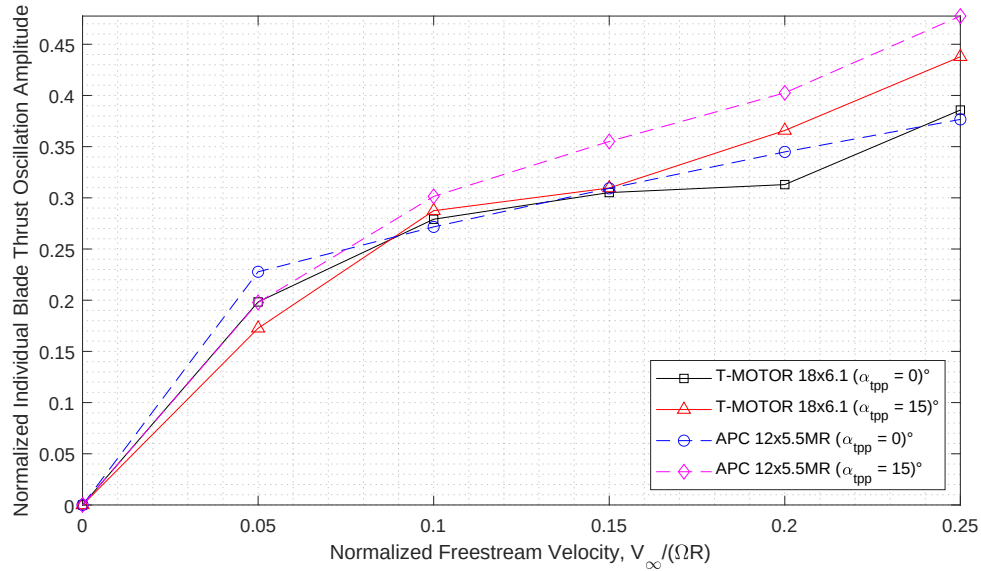


Figure 5.14: Normalized single blade thrust oscillation amplitude vs. normalized freestream velocity as predicted with the DDE method.

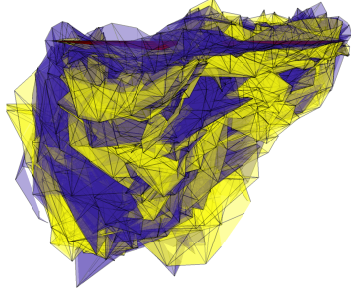
The relationship between the normalized individual blade thrust oscillation amplitude and the normalized freestream velocity does not appear to be linear. In addition to this, as with the total normalized thrust oscillation amplitude shown in 5.12, there does not appear to be a discernible pattern between the two tip-path plane angles for each rotor. The exceptions to this are the $\alpha_{tpp} = 15^\circ$ cases at high normalized freestream velocities, where the normalized individual blade thrust oscillation amplitude exceeds those of the fully-edgewise cases. This is likely due to how the mean rotor thrust changes with normalized freestream velocity. Figure 5.9 shows that as normalized freestream velocity increases, the thrust

output of the rotors at $\alpha_{tpp} = 0^\circ$ increase at a greater rate than at $\alpha_{tpp} = 15^\circ$. However, advancing and retreating blade effects become more severe as the normalized freestream velocity increases, which leads to higher normalized individual blade thrust oscillation amplitudes for $\alpha_{tpp} = 15^\circ$ as the mean thrust output of the rotor does not change significantly. Coupled with the fact that these oscillation values should approach zero as the tip-path plane angle approaches 90° , it is likely that $\alpha_{tpp} = 0 - 15^\circ$ represents the tip-path plane angle region which results in the highest normalized individual blade thrust oscillation amplitudes for these two rotors.

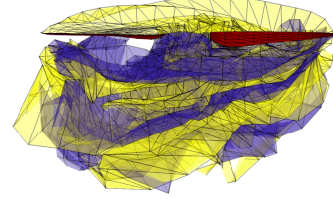
Another important takeaway from this analysis is an apparent asymmetry to the wake structure behind a rigid rotor in forward flight, which was noted by Misiorowski et al. (12). Figure 5.15 provides frontal views of two T-MOTOR 18x6.1 cases, with Fig. 5.15a representing the rotor in fully-edgewise flow at a high normalized freestream velocity, and Fig. 5.15b representing the rotor at a tip-path plane angle of $\alpha_{tpp} = 15^\circ$ at a more modest normalized freestream velocity. Due to the higher blade thrust oscillation amplitude present in the fully-edgewise case, the wake structure is highly asymmetric, with a large increase in downwash behind the advancing blade where the individual blade thrust output is the highest. The wake of the rotor in Fig. 5.15b, however, is more uniform, but still exhibits some degree of asymmetry. This asymmetric velocity field downstream of the rotor disk is in conflict with Glauert’s assumption that a rotor in high-speed forward flight has an induced downwash similar to that of a circular wing, as discussed by Leishman (42), implying that this assumption is not necessarily valid for rigid rotors. Knowing the wake structure behind a rigid rotor in forward flight may be of use to multicopter designers when considering rotor-wake interactions, such as determining the optimal rotation directions of the rotors or the best orientation of the vehicle for forward flight. Research of this nature has been conducted by Barcelos et al. (67) and Misiorowski et al. (12).

5.2.2 Summary

The purpose of this study was to quantify the transient thrust output of rigid rotors in forward flight. Two small rigid rotors were analyzed at various forward-flight operating conditions. Apart from the peaks and troughs in the transient thrust coefficient due to advancing and retreating blade velocity effects, wake effects acting on the rotor blades were also shown to affect the transient thrust coefficient under certain operating conditions. Additionally, the analysis indicates that the wake structure behind a rigid rotor in forward flight is asymmetric. The increased thrust output of the advancing blade leads



(a) $\alpha_{tpp} = 0^\circ$, $\mu_\infty = 0.30$



(b) $\alpha_{tpp} = 15^\circ$, $\mu_\infty = 0.111$

Figure 5.15: Frontal view of a T-MOTOR 18x6.1 and wake as modeled in the DDE method.

to a wake structure that is convected away faster from the rotor disk on the advancing side than on the retreating side.

A normalized thrust oscillation amplitude was introduced to quantify the oscillations in the transient thrust coefficient. For both of the rotors, the predicted transient thrust oscillation amplitudes varied from $0.05 - 0.3\bar{C}_T$ for tip-path plane angles of attack of 0° and 15° across a range of normalized freestream velocities, indicating that the transient thrust of a rotor may vary by approximately 60% of the mean rotor thrust per revolution. The predicted individual blade thrust oscillation amplitudes ranged from $0.15 - 0.45\bar{C}_T$, meaning that it is not uncommon for the thrust output of a single blade to vary by a value in the same order as the mean thrust of the rotor. Differences in the relationship between the normalized thrust oscillation amplitude and the normalized freestream velocity for the two rotors indicate that blade design plays an important role in the thrust oscillation of the rotor as the forward speed changes.

Chapter 6

Conclusions

The potential flow-based DDE method introduced in this research was designed to model complex, unsteady systems with relaxed wakes. This was accomplished using distributed doublets with a continuous strength distribution, which removes the potential for infinite induced velocities within the system. As a result, the relaxed wake model of the DDE method can be coupled with a fully-unsteady analysis framework without instabilities, whether interactions occur between the wake elements themselves or the aerodynamic surfaces and wake. Due to the higher-order distributed nature of the elements, the unsteady apparent mass approximation does not have stability constraints limiting the timestep size, as is the case with conventional UVLM or DVE-based methods.

The DDE method was evaluated using a series of four validation studies. A panel density study using planar elliptical wings showed that comparatively few elements were needed to match the classical lifting-line solution to an elliptical wing's span efficiency when using cosine spanwise element spacing. The unsteady capabilities of the DDE method were tested by modelling an airfoil subjected to a shape-edged gust. The results confirmed that the apparent mass approximation's stability is not constrained by timestep size, though the accuracy is related to the number of chordwise elements used to model the airfoil. The DDE method was then used to predict the time-averaged thrust of a small, rigid rotor at various freestream advance ratios and tip-path plane angles, agreeing well with experimentally-obtained data. As a final validation study, the transient sectional thrust of a rotor blade operating at a specific angle of attack and freestream advance ratio was predicted, with the results matching those obtained via a fully unsteady, transient CFD-based analysis.

The results of two studies using the DDE-based methods are provided in Chapter 5, both focusing on the unsteady analysis of rigid rotors. The first application used the DDE method to simulate the thrust and moments produced by a rotor in fully edgewise flight as it was subjected to various rolling maneuvers. When rolling about the longitudinal axis of the rotor, the maneuver affected the rolling moment coefficient by anywhere from 2-14%, depending on the roll rate and direction. The additional axial velocity from rolling about an axis offset from the longitudinal axis of the rotor affected the thrust and pitching moment produced in the order of 5-10% across a range of roll rates.

In the second study, the DDE method was used to predict the transient thrust output of two different rotors in forward flight, operating under a range of different normalized freestream velocities and tip-path plane angles. The rotors' transient thrust oscillated with an amplitude of up to 30% of the mean total thrust. The thrust output of the individual rotor blades was predicted to oscillate by up to 50% of the mean rotor thrust under certain conditions, underscoring the need for a fully unsteady framework when analyzing rotors in forward flight. Findings regarding transient oscillations offer additional context for structural and control systems designers, who currently rely predominantly on time-averaged rotor loads due to the inaccessible nature of this type of aerodynamic analysis.

Applying the DDE method to these unsteady systems opens up new possibilities in design analysis, exploration, and optimization. This was made possible through leveraging the advantages of distributed doublet sheets within a novel full-unsteady lifting surface framework. The DDE method offers clear advantages in labour and computational cost compared to CFD-based analysis, filling the capability gap in computational methods for today's complex and unconventional aircraft configurations.

6.1 Future Work

There are several extensions to the DDE method which will become the focus of future work. To further reduce the computational expense of the method, an adaptive panelling method could be implemented within the wake relaxation procedure, regionally varying the panel density based on the wake shape. This procedure could help maintain the geometric fidelity of the wake, particularly around the shed tip vortex, while mitigating the computational expense of unnecessary elements in the more planar regions of the wake.

The distributed doublet induction model outlined by Johnson (23) can be extended to include a

curved panel approximation, which allows the doublet sheets to be modelled as non-planar surfaces. The DDE method could be modified to incorporate this approximation, modelling the aerodynamic surfaces using non-planar quadrilateral elements. This capability would reduce the necessary number of surface elements while increasing the fidelity of the camber line modelling. The extent to which the inclusion of curved panels would benefit the DDE method is currently unknown and warrants investigation.

Further verification of the unsteady DDE force predictions will be conducted, particularly in comparison with classical theoretical unsteady aerodynamic solutions that incorporate the pitching and plunging of lifting surfaces. The suitability of the DDE method for aeroelastic and aeroacoustic predictions will be evaluated through verification studies, likely through the analysis of helicopter rotors where experimental data is readily available.

In terms of applications, the DDE method will be coupled with an optimization process to design rigid rotors for forward flight, leveraging the transient predictive capabilities demonstrated in Sections 4.4 and 5.2. An exploratory study into rigid rotor blade design could help designers understand the influence of the design parameters on the resulting thrust output under oscillating advancing-retreating conditions. In addition to rotor optimization, the DDE method will be applied to the analysis and design of integrated aircraft systems such as multirotor drones and urban air mobility vehicles, providing a low-cost alternative to high-fidelity CFD-based methods.

Appendix A

Induced Velocity Calculations

The following section outlines the velocity induction calculation for a distributed doublet element acting on an arbitrary point (ξ_p, η_p, ζ_p) in the element-local (ξ, η, ζ) coordinate system. This process is based in part on the doublet velocity evaluation outlined by Johnson (23). Recall from Section 3.1 that the velocity induced by a distributed doublet sheet of strength $\Gamma(\xi, \eta)$ is:

$$V = \iint_S \Gamma(\xi, \eta) \left[\frac{\hat{n}}{|r-s|^3} - 3\hat{n} \cdot (r-s) \frac{r-s}{|r-s|^5} \right] d\eta d\xi \quad (3.2)$$

This expression has to be solved for and expressed separately in terms of both the element strength coefficients, which are determined through the boundary conditions imposed on the system, and the influence coefficients, which are determined based on the shape of the element and the location of the point on which it is influencing. The reason the strength and influence coefficients must be separated when solving Eq. 3.2 is so that the influence coefficients can be used to solve for the strength coefficients of the element through the kinematic boundary conditions imposing flow tangency. Recall that the circulation of a distributed doublet element is defined in the element local reference frame as:

$$\Gamma = \frac{1}{2}A_1\eta^2 + A_2\eta + \frac{1}{2}B_1\xi^2 + B_2\xi + C_2\eta\xi + C_3 \quad (3.1)$$

where $A_1, A_2, B_1, B_2, C_2, C_3$ are the strength coefficients of the element. The velocity induced by a

distributed doublet element on an arbitrary point, represented by Eq. 3.2, must then be put in the following terms:

$$\begin{bmatrix} q_\xi \\ q_\eta \\ q_\zeta \end{bmatrix} = [D_i] \begin{bmatrix} A_1 \\ A_2 \\ B_1 \\ B_2 \\ C_2 \\ C_3 \end{bmatrix} \quad (\text{A.1})$$

where D_i represents a row vector containing the six influence coefficients. Only once the velocity induced by a DDE is put in terms of Eq. A.1 can the kinematic conditions be implemented.

Determining the six influence coefficients that comprise D_i is done through defining the influence coefficients using the following $\hat{J}_{(M,N)}$ terms, which are the result of decomposing Eq. 3.2 into manageable component integrals:

$$D_i = \begin{bmatrix} \frac{1}{2}\eta_p^2\hat{J}_{(1,1)} + \eta_p\hat{J}_{(1,2)} + \frac{1}{2}\hat{J}_{(1,3)} & \eta_p\hat{J}_{(1,1)} + \hat{J}_{(1,2)} & \frac{1}{2}\xi_p^2\hat{J}_{(1,1)} + \xi_p\hat{J}_{(2,1)} + \frac{1}{2}\hat{J}_{(3,1)} & \cdots \\ \xi_p\hat{J}_{(1,1)} + \hat{J}_{(2,1)} & \xi_p\eta_p\hat{J}_{(1,1)} + \xi_p\hat{J}_{(1,2)} + \eta_p\hat{J}_{(2,1)} + \hat{J}_{(2,2)} & \hat{J}_{(1,1)} \end{bmatrix} \quad (\text{A.2})$$

where the $\hat{J}_{(M,N)}$ terms are defined by Johnson (23) as:

$$\hat{J}_{(M,N)} = \frac{1}{4\pi} \begin{bmatrix} 3\zeta_p H_{(M+1,N,5)} \\ 3\zeta_p H_{(M,N+1,5)} \\ H_{(M,N,3)} - 3\zeta_p^2 H_{(M,N,5)} \end{bmatrix} \quad (\text{A.3})$$

Solving for the above expressions requires determining the $H_{(M,N,K)}$ terms, which represent the component integrals of Eq. 3.2. These $H_{(M,N,K)}$ terms take the form of:

$$H_{(M,N,K)} = \iint_{\Sigma} \frac{(\xi - \xi_p)^{M-1}(\eta - \eta_p)^{N-1}}{\sqrt{(\xi - \xi_p)^2 + (\eta - \eta_p)^2 + \zeta_p^2}} d\xi d\eta \quad (\text{A.4})$$

The method for solving these $H_{(M,N,K)}$ terms is specified by Johnson (23). The procedure for determining

the influence coefficients of a DDE element acting on an arbitrary point requires the solving of the $H_{(M,N,K)}$ and $\hat{J}_{(M,N)}$ terms necessary for solving Eq. A.2. These influence coefficients may then be used with the element strength coefficients along with Eq. A.1 to determine the velocity induced by the DDE on the point, or they may be used to specify the kinematic conditions in order to solve for the element strength coefficients.

Appendix B

Force Calculations

B.1 Freestream and Induced Lift and Side Force

Recall from Sections 3.4.1 and 3.4.2 that the freestream and induced forces are defined by the following equations respectively:

$$\hat{F}_f = \rho \iint_S \hat{U}_a(\xi, \eta) \times \hat{\omega}(\xi, \eta) \, d\eta \, d\xi \quad (3.4)$$

$$\hat{F}_i = \rho \iint_S \hat{W}_a(\xi, \eta) \times \hat{\omega}(\xi, \eta) \, d\eta \, d\xi \quad (3.7)$$

with the only difference being the velocity vector field used within the expression. Both $\hat{F}_f$ and $\hat{F}_i$ are then dotted with the lift and side force (or axial and radial force) direction unit vectors. The freestream force calculation uses a first-order approximation of the freestream velocity vector field across an element, denoted by $U_a(\xi, \eta)$ and defined by the expression:

$$\hat{U}_a(\xi, \eta) = (k_{\xi a}\xi + k_{\xi b}\eta + k_{\xi c})\hat{\xi} + (k_{\eta a}\xi + k_{\eta b}\eta + k_{\eta c})\hat{\eta} + (k_{\zeta a}\xi + k_{\zeta b}\eta + k_{\zeta c})\hat{\zeta} \quad (3.6)$$

where k represents a coefficient of this linear approximation. These k coefficients are found by determining the freestream velocity vector components at the three vertices of the element, and assuming a linear variation in the three components of this vector field across the surface of the element.

Similarly, the induced velocity vector field approximation that is used in the induced force expression, denoted by $\hat{W}_a(\xi, \eta)$, is a first-order approximation of the induced velocity vector field across the surface of an element. This is found in a similar manner to Eq. 3.6, using instead the induced velocity from all surface and wake elements in the system.

$$\hat{W}_a(\xi, \eta) = (g_{\xi a}\xi + g_{\xi b}\eta + g_{\xi c})\hat{\xi} + (g_{\eta a}\xi + g_{\eta b}\eta + g_{\eta c})\hat{\eta} + (g_{\zeta a}\xi + g_{\zeta b}\eta + g_{\zeta c})\hat{\zeta} \quad (\text{B.1})$$

The procedure for solving Eqs. 3.6 and B.1 is the same, with the only difference being the coefficients k or g used in the solution process, which represent the freestream and induced velocity vector field approximations. The solutions to Eqs. 3.7 and 3.6 are vector quantities with three components corresponding to the local (ξ, η, ζ) reference frame:

$$\hat{F}_f = \begin{bmatrix} \hat{F}_{f,\xi} \\ \hat{F}_{f,\eta} \\ \hat{F}_{f,\zeta} \end{bmatrix} \quad (\text{B.2})$$

$$\hat{F}_i = \begin{bmatrix} \hat{F}_{i,\xi} \\ \hat{F}_{i,\eta} \\ \hat{F}_{i,\zeta} \end{bmatrix} \quad (\text{B.3})$$

In the following sections, the solutions to the three components of Eq. B.2 are provided. As previously stated, this process is the same for the components of Eq. B.3, with the difference being the coefficients of the induced velocity vector field approximation are used rather than those of the freestream velocity vector field approximation.

B.1.1 Solving for $\hat{F}_{f,\xi}$

The ξ -component of the freestream force is calculated via the following procedure:

$$t_1 = (\rho k_{\zeta b})$$

$$t_2 = (C^2)$$

$$t_4 = (E^2)$$

$$t_{10} = \rho B_1$$

$$t_{12} = \rho C_2$$

$$t_{14} = k_{\zeta b} t_{10} + k_{\zeta a} t_{12}$$

$$t_{15} = -t_2 + t_4$$

$$t_{18} = \rho k_{\zeta a}$$

$$t_{19} = -C + E$$

$$t_{23} = \xi_1^2$$

$$t_{24} = t_{23}^2$$

$$t_{25} = \xi_3^2$$

$$t_{26} = t_{25}^2$$

$$t_{37} = \rho B_2$$

$$t_{40} = k_{\zeta c} t_{12} + k_{\zeta b} t_{37}$$

$$t_{46} = -2CD_{LE} + 2D_{TE}E$$

$$t_{51} = -k_{\zeta c} t_{10} - k_{\zeta a} t_{37}$$

$$t_{53} = -D_{LE} + D_{TE}$$

$$t_{62} = D_{LE}^2$$

$$t_{64} = D_{TE}^2$$

$$t_{73} = -t_{62} + t_{64}$$

$$t_{76} = \rho k_{\zeta c}$$

$$t_{88} = (\xi_3 - \xi_1)$$

$$\begin{aligned}
\hat{F}_{f,\xi} = & (-t_{24} + t_{26})(((-t_2 C + t_4 E) C_2 t_1)/3 + t_{15} t_{14}/2 + t_{19} B_1 t_{18})/4 + \\
& (-t_{23} \xi_1 + t_{25} \xi_3)((-3 D_{LE} t_2 + 3 t_4 D_{TE}) C_2 t_1)/3 + t_{15} t_{40}/2 + t_{46} t_{14}/2 - t_{19} t_{51} + t_{53} B_1 t_{18})/3 + \\
& (-t_{23} + t_{25})(((-3 t_{62} C + 3 E t_{64}) C_2 t_1)/3 + t_{46} t_{40}/2 + t_{73} t_{14}/2 + t_{19} B_2 t_{76} - t_{53} t_{51})/2 + \\
& (t_{88} (-t_{62} D_{LE} + t_{64} D_{TE}) C_2 t_1)/3 + t_{88} t_{73} t_{40}/2 + t_{88} t_{53} B_2 t_{76}
\end{aligned} \tag{B.4}$$

B.1.2 Solving for $\hat{F}_{f,\eta}$

The η -component of the freestream force is calculated via the following procedure:

$$t_1 = (\rho k_{\zeta b})$$

$$t_2 = (C^2)$$

$$t_4 = (E^2)$$

$$t_{10} = \rho A_1$$

$$t_{12} = \rho C_2$$

$$t_{14} = k_{\zeta a} t_{10} + k_{\zeta b} t_{12}$$

$$t_{15} = -t_2 + t_4$$

$$t_{18} = \rho k_{\zeta a}$$

$$t_{19} = -C + E$$

$$t_{23} = \xi_1^2$$

$$t_{24} = t_{23}^2$$

$$t_{25} = \xi_3^2$$

$$t_{26} = t_{25}^2$$

$$t_{38} = \rho A_2$$

$$t_{40} = k_{\zeta c} t_{10} + k_{\zeta b} t_{38}$$

$$t_{46} = -2CD_{LE} + 2D_{TE}E$$

$$t_{51} = k_{\zeta c} t_{12} + k_{\zeta a} t_{38}$$

$$t_{53} = -D_{LE} + D_{TE}$$

$$t_{62} = D_{LE}^2$$

$$t_{64} = D_{TE}^2$$

$$t_{73} = -t_{62} + t_{64}$$

$$t_{76} = \rho k_{\zeta c}$$

$$t_{88} = (\xi_3 - \xi_1)$$

$$\begin{aligned}
\hat{F}_{f,\eta} = & (-t_{24} + t_{26})(((-t_2 C + t_4 E) A_1 t_1)/3 + t_{15} t_{14}/2 + t_{19} C_2 t_{18})/4 + \\
& (-t_{23} \xi_1 + t_{25} \xi_3)((-3 D_{LE} t_2 + 3 t_4 D_{TE}) A_1 t_1)/3 + \\
& t_{15} t_{40}/0.2 e1 + t_{46} t_{14}/2 + t_{19} t_{51} + t_{53} C_2 t_{18})/3 + \\
& (-t_{23} + t_{25})(((-3 t_{62} C + 3 E t_{64}) A_1 t_1)/3 + t_{46} t_{40}/2 + t_{73} t_{14}/0.2 e1 + t_{19} A_2 t_{76} + t_{53} t_{51})/2 + \\
& (t_{88} (-t_{62} D_{LE} + t_{64} D_{TE}) A_1 t_1)/3 + t_{88} t_{73} t_{40}/2 + t_{88} t_{53} A_2 t_{76}
\end{aligned} \tag{B.5}$$

B.1.3 Solving for $\hat{F}_{f,\zeta}$

The ζ -component of the freestream force is calculated via the following procedure:

$$t_1 = \rho k_{\eta b}$$

$$t_3 = \rho k_{\xi b}$$

$$t_5 = (-A_1 t_1 - C_2 t_3)$$

$$t_6 = (C^2)$$

$$t_8 = (E^2)$$

$$t_{13} = \rho k_{\eta a}$$

$$t_{17} = \rho k_{\xi a}$$

$$t_{19} = -A_1 t_{13} - B_1 t_3 - C_2 t_1 - C_2 t_{17}$$

$$t_{20} = -t_6 + t_8$$

$$t_{23} = -C + E$$

$$t_{29} = \xi_1^2$$

$$t_{30} = t_{29}^2$$

$$t_{31} = \xi_3^2$$

$$t_{32} = t_{31}^2$$

$$t_{42} = \rho k_{\eta c}$$

$$t_{46} = \rho k_{\xi c}$$

$$t_{48} = -A_1 t_{42} - A_2 t_1 - B_2 t_3 - C_2 t_{46}$$

$$t_{54} = -2D_{LE}C + 2D_{TE}E$$

$$t_{59} = -B_1 t_{46} - B_2 t_{17}$$

$$t_{61} = -D_{LE} + D_{TE}$$

$$t_{66} = A_2 t_{13} + C_2 t_{42}$$

$$t_{76} = D_{LE}^2$$

$$t_{78} = D_{TE}^2$$

$$t_{86} = -t_{76} + t_{78}$$

$$t_{103} = (\xi_3 - \xi_1)$$

$$\begin{aligned}
\hat{F}_{f,\zeta} = & (-t_{30} + t_{32})(((-t_6 C + t_8 E)t_5)/3 + t_{20}t_{19}/2 - t_{23}B_1t_{17} - t_{23}C_2t_{13})/4 + \\
& (-t_{29}\xi_1 + t_{31}\xi_3)((-3t_6 D_{LE} + 3t_8 D_{TE})t_5)/3 + t_{20}t_{48}/2 + \\
& t_{54}t_{19}/2 + t_{23}t_{59} - t_{61}B_1t_{17} - t_{23}t_{66} - t_{61}C_2t_{13})/3 + \\
& (-t_{29} + t_{31})(((-3Ct_{76} + 3Et_{78})t_5)/3 + t_{54}t_{48}/2 + t_{86}t_{19}/2 - \\
& t_{23}B_2t_{46} + t_{61}t_{59} - t_{23}A_2t_{42} - t_{61}t_{66})/2 + \\
& (t_{103}(-t_{76}D_{LE} + t_{78}D_{TE})t_5)/3 + t_{103}t_{86}t_{48}/2 - t_{103}t_{61}B_2t_{46} - t_{103}t_{61}A_2t_{42}
\end{aligned} \tag{B.6}$$

B.2 Induced Drag Force

The following section outlines the induced drag calculation within the DDE method. This procedure is performed for the distributed doublet elements that lay on the trailing edge of the wing or rotor blade. To illustrate this, Fig. B.1 shows a top-down view of a generic rectangular wing composed of six elements, with the trailing edge of the wing highlighted in blue. The induced drag force is calculated through integrating the Kutta-Joukowski theorem formulation introduced in Eq. 3.8 across the trailing edge, which in this case is from vertices E-F-G-H. This is accomplished through integrating this expression in the local element reference frame across the appropriate edges of each DDE laying along the trailing edge of the lifting surface, which in this case are elements 4, 5, and 6. Due to how the local element reference frame of the elements are established, this edge is always defined by (ξ_1, η_1) and (ξ_3, η_3) for each trailing-edge DDE as illustrated in Fig. 3.2.

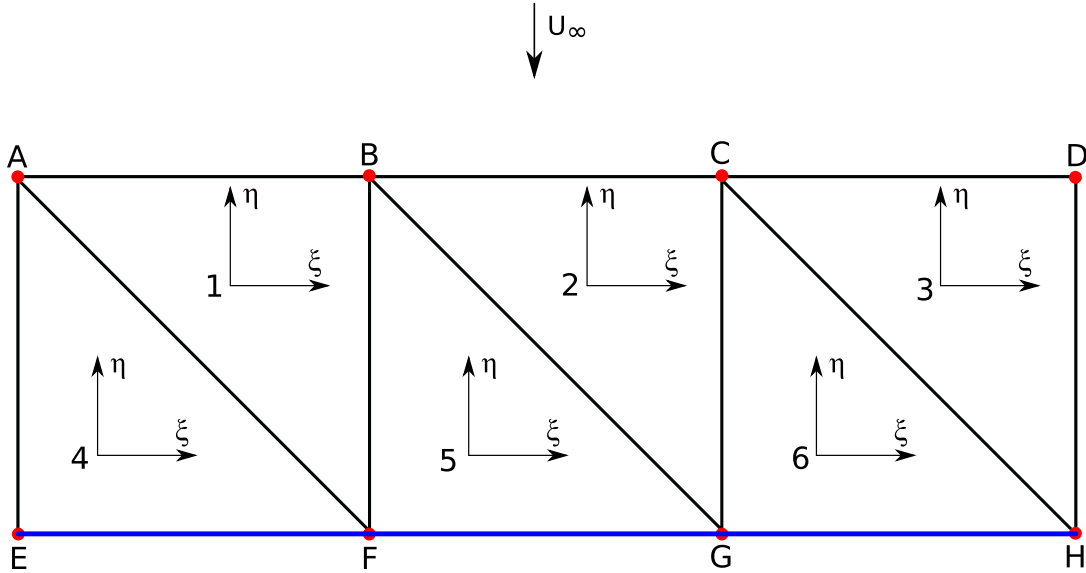


Figure B.1

The η -location of the trailing edge in terms of ξ within the location element reference frame of a DDE is:

$$\eta_{TE} = E\xi + D_{TE} \quad (\text{B.7})$$

where:

$$E = \frac{(\eta_3 - \eta_1)}{\xi_3 - \xi_1} \quad (\text{B.8})$$

$$D_{TE} = \eta_1 - \xi_1 E \quad (\text{B.9})$$

The circulation along the trailing edge of the element can then be defined in terms of ξ and the element strength coefficients by combining Eqs. 3.1 and B.7:

$$\Gamma_T = \frac{1}{2}(A_1 E^2 + 2C_2 E + B_1)\xi^2 + \frac{1}{2}((2A_1 E + 2C_2)D_T + 2EA_2 + 2B_2)\xi + \frac{1}{2}A_1 D_T^2 + A_2 D_T + C_3 \quad (\text{B.10})$$

With this definition of the trailing edge circulation strength, the resulting induced force acting on the trailing edge is given by the expression introduced in Eq. 3.8, restated here with the appropriate limits on the integration:

$$\hat{F}_{i,d} = \begin{bmatrix} \hat{F}_{i,d,\xi} \\ \hat{F}_{i,d,\eta} \\ \hat{F}_{i,d,\zeta} \end{bmatrix} = \rho \int_{\xi_1}^{\xi_3} \hat{W}_T(\xi) \times \hat{v}_T \Gamma_T(\xi) d\xi \quad (\text{B.11})$$

where $\hat{W}_T(\xi)$ is an approximation of the wake-induced velocity vector along the trailing edge of the element in terms of ξ , defined by Eq. 3.9. The term $\hat{v}_t$ is a unit vector representing the global spanwise direction of the lifting surface in the local (ξ, η, ζ) reference frame:

$$\hat{v}_t = (\xi_d, \eta_d, \zeta_d) \quad (\text{B.12})$$

The integration in Eq. B.11 is performed across the spanwise ξ -direction of the element. The resultant induced drag force, $\hat{F}_{i,d}$, is a vector quantity with three components corresponding to the local (ξ, η, ζ) reference frame. These three components are solved for individually, following the procedures in the subsequent three sections.

B.2.1 Solving for $\hat{F}_{i,d,\xi}$

The ξ -component of the induced drag force is calculated via the following procedure:

$$t_1 = (E^2)$$

$$t_9 = (\eta_d w_1 - v_1 \zeta_d)(t_1 A_1 + 2C_2 E + B_1)$$

$$t_{10} = \xi_1^2$$

$$t_{11} = (t_{10}^2)$$

$$t_{18} = A_1 D_{TE}$$

$$t_{20} = A_2 v_1$$

$$t_{21} = v_2 C_2$$

$$t_{31} = \zeta_d(A_1 v_2 t_1 + 2E(v_1 t_{18} + t_{20} + t_{21}) + 2D_{TE} C_2 v_1 + v_2 B_1 + 2B_2 v_1)$$

$$t_{35} = w_1 A_2$$

$$t_{36} = w_2 C_2$$

$$t_{46} = \eta_d(A_1 w_2 t_1 + 2E(w_1 t_{18} + t_{35} + t_{36}) + 2w_1 D_{TE} C_2 + 2w_1 B_2 + w_2 B_1)$$

$$t_{50} = (\xi_3^2)$$

$$t_{53} = t_{31} - t_{46}$$

$$t_{60} = v_2 A_2$$

$$t_{61} = v_3 C_2$$

$$t_{64} = D_{TE}^2$$

$$t_{65} = A_1 t_{64}$$

$$t_{77} = \zeta_d\left(\frac{4}{3}A_1 v_3 t_1 + \frac{8}{3}E(v_2 D_{TE} A_1 + t_{60} + t_{61}) + \frac{4}{3}v_1 t_{65} + \frac{8}{3}D_{TE}(t_{20} + t_{21}) + \frac{8}{3}v_2 B_2 + \frac{4}{3}v_3 B_1 + \frac{8}{3}C_3 v_1\right)$$

$$t_{81} = w_2 A_2$$

$$t_{82} = w_3 C_2$$

$$t_{96} = \frac{4}{3}\eta_d(A_1 w_3 t_1 + 2E(w_2 t_{18} + t_{81} + t_{82}) + A_1 t_{64} w_1 + 2D_{TE}(t_{35} + t_{36}) + 2w_1 C_3 + 2w_2 B_2 + w_3 B_1)$$

$$t_{99} = (t_{50} \xi_3)$$

$$t_{103} = t_{77} - t_{96}$$

$$t_{105} = t_{18} + A_2$$

$$t_{119} = \zeta_d(4Et_{105}v_3 + 2A_1t_{64}v_2 + 4D_{TE}(t_{60} + t_{61}) + 4v_2C_3 + 4v_3B_2)$$

$$t_{131} = 4\eta_d((Et_{105}w_3) + (A_1t_{64}w_2)/0.2e1 + (D_{TE}(t_{81} + t_{82})) + (w_2C_3) + (w_3B_2))$$

$$t_{134} = (t_{50}^2)$$

$$\begin{aligned} \hat{F}_{i,d,\xi} = & -\frac{\rho}{8}(\xi_1 - \xi_3)(-\frac{4}{5}t_{11}t_9 + t_{10}\xi_1(-\frac{4}{5}\xi_3t_9 + t_{31} - t_{46}) + t_{10}(-\frac{4}{5}t_{50}t_9 + (\xi_3t_{53}) + t_{77} - t_{96}) + \\ & \xi_1(-\frac{4}{5}t_{99}t_9 + (t_{50}t_{53}) + \xi_3t_{103} + t_{119} - t_{131}) - \frac{4}{5}t_{134}t_9 + (t_{99}t_{53}) + t_{50}t_{103} + \xi_3(t_{119} - t_{131}) + \\ & (4(-\eta_d w_3 + v_3 \zeta_d)(2A_2 D_{TE} + 2C_3 + t_{65}))) \end{aligned}$$

(B.13)

B.2.2 Solving for $\hat{F}_{i,d,\eta}$

The η -component of the induced drag force is calculated via the following procedure:

$$\begin{aligned}
t_3 &= (E^2) \\
t_{11} &= (-u_1\zeta_d + w_1\xi_d)(t_3A_1 + 2C_2E + B_1) \\
t_{12} &= \xi_1^2 \\
t_{13} &= (t_{12}^2) \\
t_{20} &= A_1D_{TE} \\
t_{22} &= A_2u_1 \\
t_{23} &= u_2C_2 \\
t_{33} &= \zeta_d(A_1u_2t_3 + 2E(u_1t_{20} + t_{22} + t_{23}) + 2C_2D_{TE}u_1 + u_2B_1 + 2B_2u_1) \\
t_{37} &= w_1A_2 \\
t_{38} &= w_2C_2 \\
t_{48} &= (A_1w_2t_3 + 2E(w_1t_{20} + t_{37} + t_{38}) + 2w_1D_{TE}C_2 + 2w_1B_2 + w_2B_1)\xi_d \\
t_{52} &= (\xi_3^2) \\
t_{55} &= t_{33} - t_{48} \\
t_{62} &= u_2A_2 \\
t_{63} &= u_3C_2 \\
t_{66} &= D_{TE}^2 \\
t_{67} &= A_1t_{66} \\
t_{79} &= \zeta_d\left(\frac{4}{3}A_1u_3t_3 + \frac{8}{3}E(u_2D_{TE}A_1 + t_{62} + t_{63}) + \frac{4}{3}u_1t_{67} + \frac{8}{3}D_{TE}(t_{22} + t_{23}) + \frac{8}{3}u_2B_2 + \frac{4}{3}u_3B_1 + \frac{8}{3}C_3u_1\right) \\
t_{83} &= w_2A_2 \\
t_{84} &= w_3C_2 \\
t_{98} &= \frac{4}{3}\xi_d(A_1w_3t_3 + 2E(w_2t_{20} + t_{83} + t_{84}) + A_1t_{66}w_1 + 2D_{TE}(t_{37} + t_{38}) + 2w_1C_3 + 2w_2B_2 + w_3B_1) \\
t_{101} &= (t_{52}\xi_3)
\end{aligned}$$

$$t_{105} = t_{79} - t_{98}$$

$$t_{107} = t_{20} + A_2$$

$$t_{121} = \zeta_d(4Et_{107}u_3 + 2A_1t_{66}u_2 + 4D_{TE}(t_{62} + t_{63}) + 4u_2C_3 + 4u_3B_2)$$

$$t_{133} = 4((Et_{107}w_3) + (A_1t_{66}w_2)/0.2e1 + (D_{TE}(t_{83} + t_{84})) + (w_2C_3) + (w_3B_2))\xi_d$$

$$t_{136} = (t_{52}^2)$$

$$\begin{aligned} \hat{F}_{i,d,\eta} = & \frac{\rho(\xi_1 - \xi_3)}{8} \left(-\frac{4}{5}t_{13}t_{11} + t_{12}\xi_1 \left(-\frac{4}{5}\xi_3t_{11} + t_{33} - t_{48} \right) + t_{12} \left(-\frac{4}{5}t_{52}t_{11} + (\xi_3t_{55}) + t_{79} - t_{98} \right) + \right. \\ & \xi_1 \left(-\frac{4}{5}t_{101}t_{11} + (t_{52}t_{55}) + \xi_3t_{105} + t_{121} - t_{133} \right) - \frac{4}{5}t_{136}t_{11} + (t_{101}t_{55}) + t_{52}t_{105} + \xi_3(t_{121} - t_{133}) + \\ & \left. (4(u_3\zeta_d - w_3\xi_d)(2A_2D_{TE} + 2C_3 + t_{67})) \right) \end{aligned} \tag{B.14}$$

B.2.3 Solving for $\hat{F}_{i,d,\zeta}$

The ζ -component of the induced drag force is calculated via the following procedure:

$$t_3 = (E^2)$$

$$t_{11} = (-\eta_d u_1 + v_1 \xi_d)(t_3 A_1 + 2C_2 E + B_1)$$

$$t_{12} = \xi_1^2$$

$$t_{13} = (t_{12}^2)$$

$$t_{20} = A_1 D_{TE}$$

$$t_{22} = A_2 u_1$$

$$t_{23} = u_2 C_2$$

$$t_{26} = D_{TE} C_2$$

$$t_{33} = \eta_d (A_1 u_2 t_3 + 2E(u_1 t_{20} + t_{22} + t_{23}) + 2u_1 t_{26} + u_2 B_1 + 2B_2 u_1)$$

$$t_{37} = A_2 v_1$$

$$t_{38} = v_2 C_2$$

$$t_{47} = (A_1 v_2 t_3 + 2E(v_1 t_{20} + t_{37} + t_{38}) + 2v_1 t_{26} + v_2 B_1 + 2B_2 v_1) \xi_d$$

$$t_{51} = (\xi_3^2)$$

$$t_{54} = t_{33} - t_{47}$$

$$t_{61} = u_2 A_2$$

$$t_{62} = u_3 C_2$$

$$t_{65} = D_{TE}^2$$

$$t_{66} = A_1 t_{65}$$

$$t_{78} = \eta_d \left(\frac{4}{3} A_1 u_3 t_3 + \frac{8}{3} E(u_2 D_{TE} A_1 + t_{61} + t_{62}) + \frac{4}{3} u_1 t_{66} + \frac{8}{3} D_{TE}(t_{22} + t_{23}) + \frac{8}{3} u_2 B_2 + \frac{4}{3} u_3 B_1 + \frac{8}{3} C_3 u_1 \right)$$

$$t_{83} = v_2 A_2$$

$$t_{84} = v_3 C_2$$

$$t_{97} = \frac{4}{3} \xi_d (A_1 v_3 t_3 + 2E(v_2 D_{TE} A_1 + t_{83} + t_{84}) + v_1 t_{66} + 2D_{TE}(t_{37} + t_{38}) + 2C_3 v_1 + 2v_2 B_2 + v_3 B_1)$$

$$t_{100} = (t_{51}\xi_3)$$

$$t_{104} = t_{78} - t_{97}$$

$$t_{106} = t_{20} + A_2$$

$$t_{120} = \eta_d(4Et_{106}u_3 + 2A_1t_{65}u_2 + 4D_{TE}(t_{61} + t_{62}) + 4u_2C_3 + 4u_3B_2)$$

$$t_{132} = 4((Et_{106}v_3) + (A_1t_{65}v_2)/0.2e1 + (D_{TE}(t_{83} + t_{84})) + (v_2C_3) + (v_3B_2))\xi_d$$

$$t_{135} = (t_{51}^2)$$

$$\begin{aligned} \hat{F}_{i,d,\zeta} = & -\frac{\rho(\xi_1 - \xi_3)}{8}(-\frac{4}{5}t_{13}t_{11} + t_{12}\xi_1(-\frac{4}{5}\xi_3t_{11} + t_{33} - t_{47}) + t_{12}(-\frac{4}{5}t_{51}t_{11} + (\xi_3t_{54}) + t_{78} - t_{97}) \\ & + \xi_1(-\frac{4}{5}t_{100}t_{11} + (t_{51}t_{54}) + \xi_3t_{104} + t_{120} - t_{132}) - \frac{4}{5}t_{135}t_{11} + (t_{100}t_{54}) + t_{51}t_{104} + \xi_3(t_{120} - t_{132}) + \\ & (4(\eta_d u_3 - v_3 \xi_d)(2A_2 D_{TE} + 2C_3 + t_{66}))) \end{aligned} \tag{B.15}$$

B.3 Apparent Mass

Recall from Section 2.1.4 the expression for apparent mass:

$$\hat{F} = \rho \frac{\partial}{\partial t} \left[\iint_S \Gamma(\xi, \eta) \, d\eta \, d\xi \right] \hat{i} \quad (3.10)$$

The integrated circulation term is found through integrating across both the spanwise and streamwise direction in the local reference frame:

$$\iint_S \Gamma(\xi, \eta) = \int_{\xi_1}^{\xi_3} \int_{\eta_{LE}}^{\eta_{TE}} \Gamma(\xi, \eta) \, d\eta \, d\xi \quad (B.16)$$

In the local element reference frame, the η -location of the leading edge of the element, defined by (ξ_2, η_2) and (ξ_3, η_3) as illustrated in Fig. 3.2, can be put in terms of ξ as:

$$\eta_{LE} = C\xi + D_{LE} \quad (B.17)$$

where:

$$C = \frac{(\eta_3 - \eta_2)}{\xi_3 - \xi_2} \quad (B.18)$$

$$D_{LE} = \eta_2 - \xi_2 C \quad (B.19)$$

Similarly, for the trailing edge of the element:

$$\eta_{TE} = E\xi + D_{TE} \quad (B.20)$$

where:

$$E = \frac{(\eta_3 - \eta_1)}{\xi_3 - \xi_1} \quad (B.21)$$

$$D_{TE} = \eta_1 - \xi_1 E \quad (B.22)$$

With these definitions, the integral in Eq. B.16 can be solved analytically using the follow procedure:

$$t_1 = (C^2)$$

$$t_8 = E^2$$

$$t_9 = (A_1 t_8)$$

$$t_{10} = (C_2 E)$$

$$t_{15} = A_1 t_1 C + 3t_1 C_2 + 3C B_1 - (t_9 + 3t_{10} + 3B_1)E$$

$$t_{16} = (\xi_1^2)$$

$$t_{24} = 4B_1$$

$$t_{26} = D_{LE}(4A_1 t_1 + 8C C_2 + t_{24})$$

$$t_{30} = D_{TE}(-4t_9 - 8t_{10} - t_{24})$$

$$t_{31} = C - E$$

$$t_{32} = C A_2$$

$$t_{33} = E A_2$$

$$t_{37} = 4(t_{32} + t_{33} + 2B_2)t_{31}$$

$$t_{40} = (\xi_3^2)$$

$$t_{42} = t_{26} + t_{30} + t_{37}$$

$$t_{46} = D_{LE}^2$$

$$t_{47} = 6t_{46}(A_1 C + C_2)$$

$$t_{49} = 12D_{LE}(t_{32} + B_2)$$

$$t_{52} = D_{TE}^2$$

$$t_{53} = 6t_{52}(-A_1 E - C_2)$$

$$t_{55} = 12D_{TE}(-t_{33} - B_2)$$

$$t_{57} = 12t_{31} C_3$$

$$\begin{aligned}
\int_{\xi_1}^{\xi_3} \int_{\eta_{LE}}^{\eta_{TE}} \Gamma(\xi, \eta) \, d\eta \, d\xi = \frac{1}{24} & ((\xi_1 - \xi_3)(t_{16}\xi_1 t_{15} + t_{16}(\xi_3 t_{15} + t_{26} + t_{30} + t_{37}) + \\
& \xi_1(t_{40}t_{15} + \xi_3 t_{42} + t_{47} + t_{49} + t_{53} + t_{55} + t_{57}) + t_{40}\xi_3 t_{15} + t_{40}t_{42} + \\
& \xi_3(t_{47} + t_{49} + t_{53} + t_{55} + t_{57}) + 4t_{46}D_{LE}A_1 + 12t_{46}A_2 + 24C_3D_{LE} - \\
& 4(A_1 t_{52} + 3A_2 D_{TE} + 6C_3)D_{TE})) \quad (B.23)
\end{aligned}$$

Equation B.23 is solved for every surface element within every timestep. After the timestepping procedure has finished, a finite differencing scheme is used to approximate the time rate of change of this integrated circulation for use in Eq. 3.10.

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